



SHACKLES

Crosby®

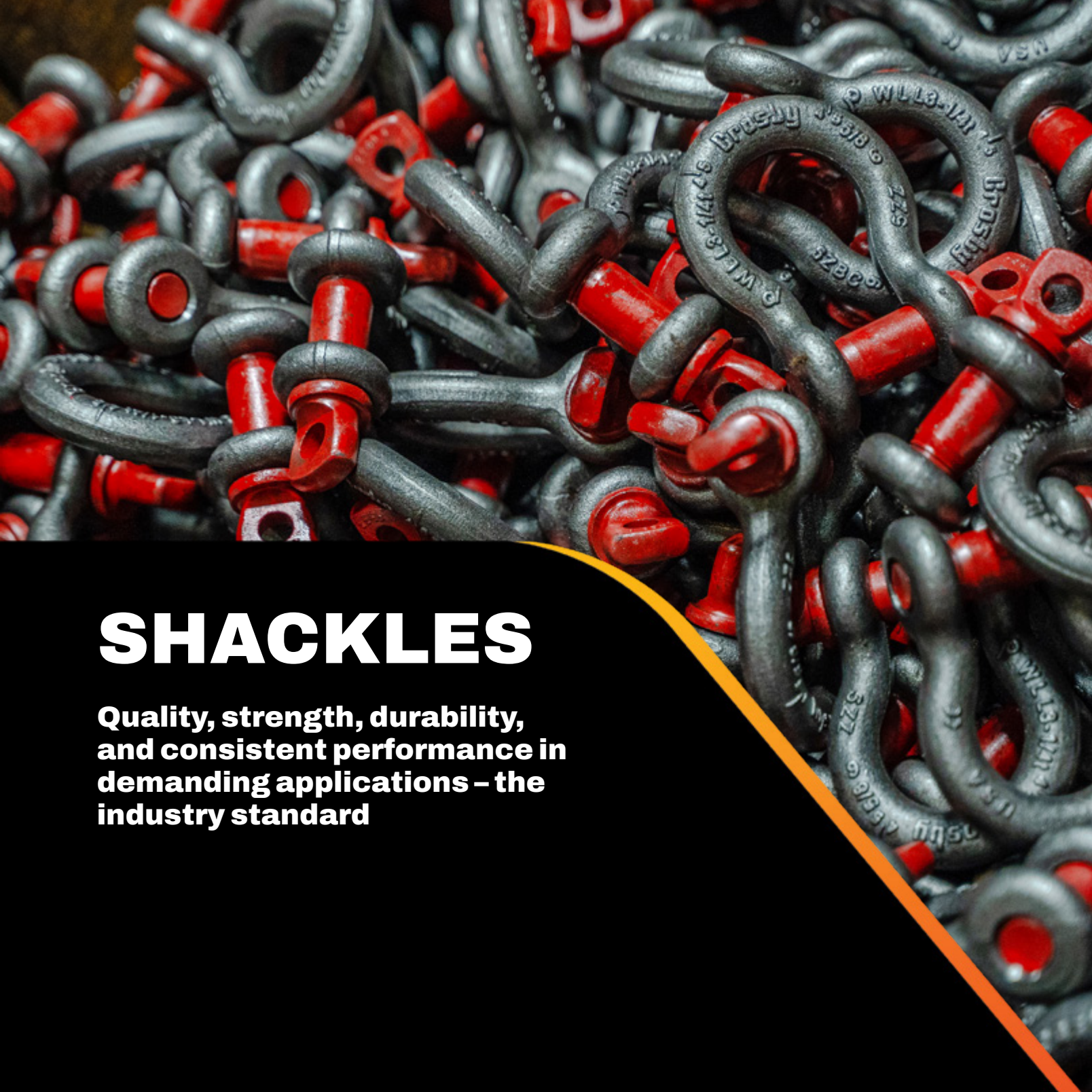


CM®

IMPORTED

CoreLifting.com





SHACKLES

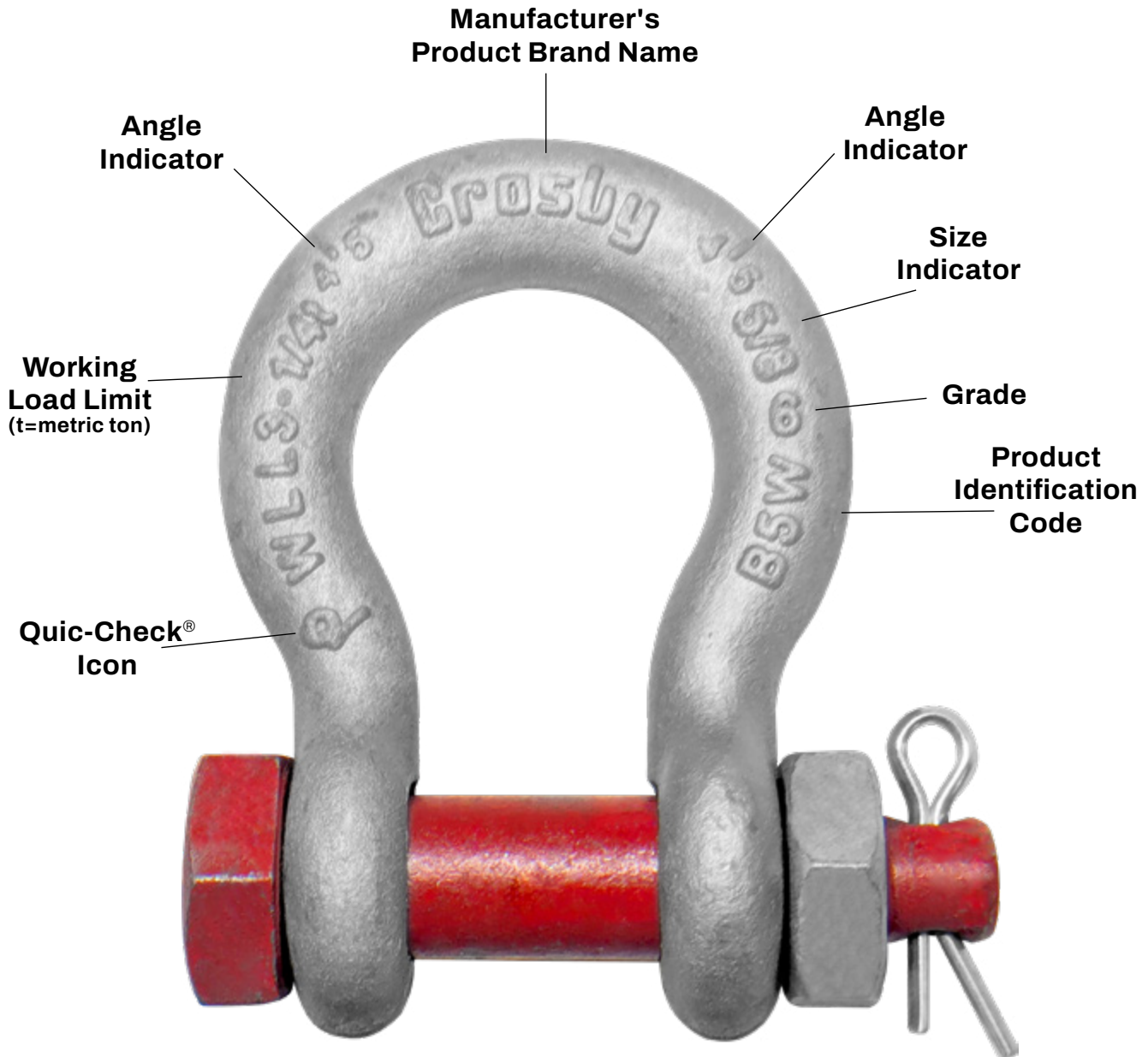
**Quality, strength, durability,
and consistent performance in
demanding applications – the
industry standard**

kitocrosby.com



VALUE LONG AFTER THE SALE

Crosby products are well known for quality, design, and safety features. It's important to know how to identify, interpret, and utilize the forged-in markings on your hardware to help ensure proper rigging for the life of the shackle, hook, or clip.



Watch our video training series on product identification



Shackle identification



Hook identification



Clip identification

[kitocrosby.com/identification](https://www.kitocrosby.com/identification)

SHACKLES

DESIGN

The theoretical reserve capability of carbon shackles should be at a minimum 5 to 1, and alloy shackles a minimum of 4 to 1. Known as the Design Factor, it is usually computed by dividing the catalog Ultimate Load by the Working Load Limit.

The Ultimate Load is the average load or force at which the product fails or no longer supports the load.

The Working Load Limit is the maximum mass or force which the product is authorized to support in general service. The Design Factor is generally expressed as a ratio such as 5 to 1, or 5:1.

Also important to the design of shackles is the selection of proper steel to support fatigue, ductility, and impact properties.

Questions to ask your rigging provider

What is the Working Load Limit and Design Factor for shackles?

Is deformation upon overloading a critical consideration in their design?

Do they jeopardize other properties by having high hardness in order to increase Working Load Limit or Design Factor?

Why choose Crosby

Crosby carbon shackles have the highest design factor (6 to 1) in the industry. All Crosby Design Factors are documented.

Kito Crosby purchases only special bar forging quality steel with cleanliness and guaranteed harden ability. All material chemistry is independently verified prior to manufacturing.

The design of Crosby shackles assures that strength, ductility, and fatigue properties are met.

CLOSED DIE FORGING

The proper performance of premium shackles depends on good manufacturing techniques that include proper forging and accurate machining. Closed-die forging of shackles assures clear lettering, superior grain flow, and consistent dimensional accuracy.

A closed-die forged bow allows for an increased cross section that, when coupled with quench and tempering, enhances strength and ductility.

Closed-die bow forgings combined with close tolerance pin holes assures good fatigue life. Close pin-to-hole tolerance has been proven to be critical for good fatigue life, particularly with screw pin shackles.

Questions to ask your rigging provider

Are their shackles closed-die forged with close tolerance pin holes?

Do their shackles have good fatigue life?

Do their shackles have a fatigue life that meets the new world standards?

Many forge bows utilize an open die forging process which allows for inconsistent dimensional accuracy and increased pin hole clearance, thus jeopardizing the fatigue life of the shackle in actual use.

Why choose Crosby

Each shackle is closed-die forged. Closed-die forging produces consistent dimensions. A closed-die forged bow allows for an increased cross section that, when coupled with quench and tempering, enhances strength and ductility.

Close tolerance holes and concentric pins with good surface finishes are provided by Kito Crosby and are proven to provide improved fatigue life in actual use.

Crosby shackles are fatigue rated as well as load rated. Close pin to hole tolerance has been proven to be critical for good fatigue life, particularly with screw pin shackles.

FATIGUE PROPERTIES

The mechanical properties of steel when a load is repeatedly applied is known as its fatigue strength. Fatigue testing determines the ability of a material to withstand repeated applications of a load. The load by itself may be too small to produce a failure. There are three factors involved when considering fatigue strength: the number of cycles at which a crack initiates, the number of cycles at which the crack starts to grow, and the number of cycles at which the fitting fails. One accepted method of fatigue rating fittings is to test them to 1-1/2 times the working load limit for 20,000 cycles, without failure. This standard test is accepted as indicating indefinite life when used within the working load limit under normal circumstances.

Questions to ask your rigging provider

Does the material selection process recognize fatigue properties?

Do they have an active program to design and test fatigue properties?

Is there a program in place to fatigue rate all load-bearing products that are used in critical applications?

Why choose Crosby

Kito Crosby has an active program to determine fatigue properties. Included in this program is the use of finite element design methods to predict possible weak areas, which in turn allows us to design in superior fatigue properties.

Kito Crosby specifies material of specific cleanliness and guaranteed hardenability which enhances fatigue. We design and manufacture products with fatigue in mind and ensure all load-bearing products used in critical applications being fatigue rated.

QUENCHED & TEMPERED

Quench and tempering assures the uniformity of performance and maximizes the properties of the steel. This means that each shackle meets its rated strength and has required ductility, toughness, impact, and fatigue properties.

The requirements of your job demand this reliability and consistency. This process develops a tough material that reduces the risk of brittle, catastrophic failure.

The shackle bow will deform if overloading occurs, giving warning before ultimate failure.

Questions to ask your rigging provider

Are their bows and pins quenched and tempered?

If not, are they willing to accept inferior impact toughness, product deformation, and the increased risk of inconsistency?

Why do many manufacturers not recommend non-heat-treated shackles for overhead lifting?

Why do some recommend quench and tempering for alloy but not carbon grades?

Many normalize the shackle bows. As a result, desired properties are not achieved. A few even provide bows in an 'as-forged' condition, resulting in the possibility of brittle failure.

Why choose Crosby

All Crosby shackle bows and pins are quenched and tempered, which enhances their performance under cold temperatures and adverse field conditions. Crosby carbon shackles are recommended for all critical applications including overhead lifting. Alloy shackles are recommended when specific dimensional requirements dictate a size that requires higher working load limits. Crosby shackles provide the tensile strength, ductility, impact, and fatigue properties that are essential if they are to perform time after time in adverse conditions.

These properties assure that the inspection criteria set forth by ANSI will effectively monitor the ability of the shackles to continue in service.

Watch our video on the Quench & Tempered process at kitocrosby.com/QT.



QUENCHED & TEMPERED

IDENTIFICATION & APPLICATION INFO

The proper application of shackles requires that the correct type and size of shackle be used. The shackle's Working Load Limit, its size, a traceability code, and the manufacturer's name should be clearly marked in the bow.

Traceability of the material chemistry and properties is essential for total confidence in the product. Material chemistry should be independently verified prior to manufacturing.

Questions to ask your rigging provider

Do they have an active traceability system used in manufacturing?

Is the material chemistry independently verified?

What training support is provided?

Why choose Crosby

We forge the Crosby name or "CG," the Working Load Limit, and the Product Identification Code (PIC) into each bow, and the Crosby name or "CG," and the PIC into each pin of its full line of screw pin, round pin, and bolt type anchor and chain shackles. Kito Crosby also provides training on the proper use of shackles.

Watch our training video on shackle identification at kitocrosby.com/identification.

CROSBY VALUE ADDED

- **Charpy impact properties:** Crosby shackles are quenched and tempered and have enhanced impact properties for greater toughness at all temperatures. If requested at the time of order, Kito Crosby can provide Charpy impact properties.
- **Fatigue properties:** Fatigue properties are available for 1/3 to 55 metric ton shackles. These Crosby shackles are fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- **Ductility properties:** Typical ductility properties are available for all sizes upon special request.
- **Hardness levels and material tensile strengths:** Typical values are available for all sizes of shackles, and actual values can be furnished if requested at the time of order.
- **Proof Testing:** If requested at the time of order, shackles can be proof tested with certificates.
- **Mag Certification:** If requested at the time of order, shackles can be magnetic particle inspected with certificates.
- **Certification:** Certification to world class standards is available upon special request at the time of order; American Bureau of Shipping, Lloyds Register of Shipping, Det Norske Veritas, American Petroleum Institute, RINA, Nuclear Regulatory Commission, and several other worldwide standards.
- **Applications:** **Round pin shackles** can be used in tie down, towing, suspension or lifting applications where the load is strictly applied in-line. **Screw pin shackles** can be used in any application where a round pin shackle is used. In addition, screw pin shackles can be used for applications involving side-loading circumstances. Reduced working load limits are required for side-loading applications. **Bolt type shackles** can be used in any application where round pin or screw pin shackles are used. In addition, they are recommended for permanent or long-term installations and where the load may slide on the shackle pin causing the pin to rotate.
- **Material analysis:** Kito Crosby can provide certified material (mill) analysis for each production lot, traceable by the Product Identification Code (PIC). Kito Crosby, through its own laboratory, verifies the analysis of each heat of steel. Kito Crosby purchases only **special bar** forging quality steel with specific cleanliness requirements and guaranteed hardenability.
- **Field inspection:** Written instructions for visual, magnaflux, and dye penetrant inspection of shackles are available from Kito Crosby. In addition, acceptance criteria and repair procedures for shackles are available.
- **QUIC-CHECK®:** Shackles incorporate two marking indicators forged into the shackle bow at 45° angles from vertical. These are utilized to quickly check the approximate angle of a two-legged hitch or check the angle of a single leg hitch. If the load is off vertical or side loaded a reduction in the Working Load Limit of the shackle is required.

G-213

Round pin anchor shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVA, Grade A, Class 1, except for those provisions required of the contractor.



G-209

Screw pin anchor shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVA, Grade A, Class 2, except for those provisions required of the contractor.



G-2130

Bolt type anchor shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVA, Grade A, Class 3, except for those provisions required of the contractor.



G-215

Round pin chain shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVB, Grade A, Class 1, except for those provisions required of the contractor.



G-210

Screw pin chain shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVB, Grade A, Class 2, except for those provisions required of the contractor.



G-2150

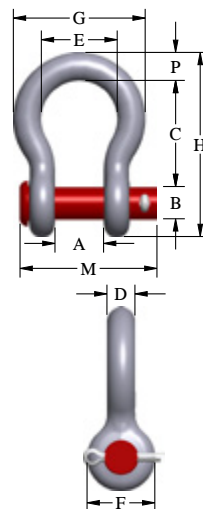
Bolt type chain shackles meet the performance requirements of Federal Specification RR-C-271G, Type IVB, Grade A, Class 3, except for those provisions required of the contractor.



G-213



- Forged, Quenched & Tempered, with alloy pins.
- Working Load Limit permanently shown on every shackle.
- Hot-dip galvanized.
- Sizes 3/8 inch and below are mechanically galvanized.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification. Charges for proof testing and certification available when requested at the time of order.
- Shackles are Quenched & Tempered and can meet DNV impact requirements of 42 Joules (31 ft-lb) at -20° C (-4° F).
- G-213 Round pin anchor shackles meet the performance requirements of Federal Specification RR-C-271H, Type IVA, Grade A, Class 1, except for those provisions required of the contractor.
- DO NOT SIDE LOAD ROUND PIN SHACKLES.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-213 Round Pin Anchor Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)												Replacement Pin Stock No.
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	M	P	
1/4	1/2	1018017	.13	.47	.06	.32	1.13	.06	.06	.78	.62	1.28	1.84	1.34	.25	-
5/16	3/4	1018035	.18	.53	.06	.39	1.22	.06	.06	.84	.75	1.46	2.09	1.63	.31	-
3/8	1	1018053	.29	.66	.06	.45	1.44	.13	.06	1.03	.92	1.79	2.50	1.86	.38	-
7/16	1 1/2	1018071	.38	.75	.06	.51	1.69	.13	.06	1.16	1.06	2.04	2.91	2.13	.44	-
1/2	2	1018099	.71	.81	.06	.64	1.88	.13	.06	1.31	1.18	2.31	3.28	2.38	.50	-
5/8	3 1/4	1018115	1.50	1.06	.06	.78	2.38	.13	.06	1.69	1.50	2.93	4.19	2.91	.69	-
3/4	4 3/4	1018133	2.32	1.25	.06	.88	2.81	.25	.06	2.00	1.81	3.50	4.97	3.44	.81	-
7/8	6 1/2	1018151	3.49	1.44	.06	1.03	3.31	.25	.06	2.28	2.10	4.04	5.83	3.82	.97	-
1	8 1/2	1018179	5.00	1.69	.06	1.16	3.75	.25	.06	2.69	2.38	4.69	6.56	4.53	1.06	-
1-1/8	9 1/2	1018197	6.97	1.81	.06	1.28	4.25	.25	.06	2.91	2.68	5.15	7.47	5.13	1.25	1082232
1-1/4	12	1018213	9.75	2.03	.06	1.41	4.69	.25	.06	3.26	3.00	5.76	8.26	5.50	1.38	1082250
1-3/8	13 1/2	1018231	13.25	2.25	.13	1.53	5.25	.25	.13	3.62	3.31	6.38	9.16	6.13	1.50	1082278
1-1/2	17	1018259	17.25	2.38	.13	1.66	5.75	.25	.13	3.88	3.62	6.94	10.00	6.50	1.62	1082296
1-3/4	25	1018277	29.46	2.88	.13	2.03	7.00	.25	.13	5.00	4.19	8.80	12.34	7.75	2.25	1082312
2	35	1018295	45.75	3.25	.13	2.28	7.75	.25	.13	5.75	4.81	10.15	13.69	8.75	2.40	1082330

6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit.

Load Rated

Fatigue Rated

QUIC-CHECK®

QT &

MAX TOUGH®

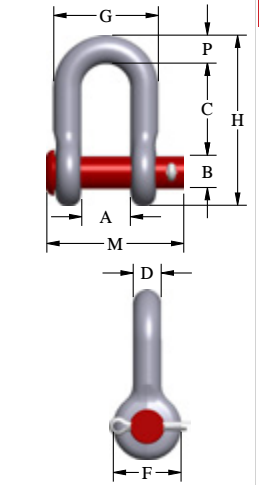
CE

APPLICATION AND WARNING INFORMATION
SECTION 17

G-215



- Forged, Quenched & Tempered, with alloy pins.
- Working Load Limit permanently shown on every shackle.
- Hot-dip galvanized.
- Sizes 3/8 inch and below are mechanically galvanized.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification. Charges for proof testing and certification available when requested at the time of order.
- Shackles are Quenched and Tempered and can meet DNV impact requirements of 42 Joules (31 ft-lb) at -20° C (-4° F).
- G-215 Round pin chain shackles meet the performance requirements of Federal Specification RR-C-271H, Type IVB, Grade A, Class 1, except for those provisions required of the contractor.
- DO NOT SIDE LOAD ROUND PIN SHACKLES.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-215 Round Pin Chain Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)											Replacement Pin Stock No.
				A	Tol. A ±	B	C	Tol. C ±	D	F	G	H	M	P	
1/4	1/2	1018810	.10	.47	.06	.32	.87	.06	.25	.62	.97	1.59	1.34	.25	-
5/16	3/4	1018838	.18	.53	.06	.39	1.04	.06	.31	.75	1.15	1.91	1.63	.31	-
3/8	1	1018856	.25	.66	.06	.45	1.25	.13	.38	.92	1.42	2.31	1.86	.38	-
7/16	1 1/2	1018874	.40	.75	.06	.51	1.45	.13	.44	1.06	1.63	2.67	2.13	.44	-
1/2	2	1018892	.50	.81	.06	.64	1.62	.13	.50	1.18	1.81	3.03	2.38	.50	-
5/8	3 1/4	1018918	1.21	1.06	.06	.78	2.00	.13	.63	1.50	2.32	3.76	2.91	.63	-
3/4	4 3/4	1018936	2.00	1.25	.06	.88	2.38	.25	.75	1.81	2.75	4.53	3.44	.81	-
7/8	6 1/2	1018954	3.28	1.44	.06	1.03	2.81	.25	.88	2.10	3.20	5.33	3.82	.97	-
1	8 1/2	1018972	4.75	1.69	.06	1.16	3.18	.25	1.00	2.38	3.69	5.94	4.53	1.00	-
1-1/8	9 1/2	1018990	6.30	1.81	.06	1.28	3.56	.25	1.13	2.68	4.07	6.78	5.13	1.25	1082232
1-1/4	12	1019016	9.00	2.03	.13	1.41	3.93	.25	1.25	3.00	4.53	7.50	5.50	1.38	1082250
1-3/8	13 1/2	1019034	12.00	2.25	.13	1.53	4.38	.25	1.38	3.31	5.01	8.28	6.13	1.50	1082278
1-1/2	17	1019052	16.15	2.38	.13	1.66	4.87	.25	1.50	3.62	5.38	9.05	6.50	1.62	1082296
1-3/4	25	1019070	26.00	2.88	.13	2.03	5.76	.25	1.75	4.19	6.38	10.97	7.75	2.12	1082312
2	35	1019098	43.25	3.25	.13	2.28	6.75	.25	2.10	5.00	7.25	12.74	8.75	2.36	1082330

6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit.



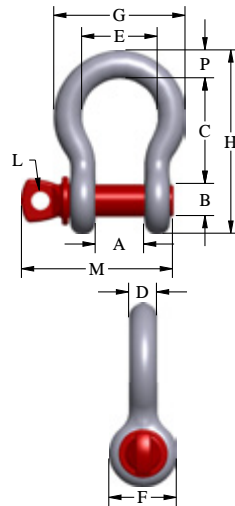
G-209



S-209



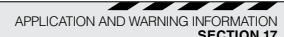
- Meets performance requirements of Grade 6 shackles.
- Forged, Quenched & Tempered, with alloy pins.
- Working Load Limit and Grade 6 permanently shown on every shackle.
- Hot-dip galvanized (G) or self colored (S).
- Sizes 3/8 inch and below are mechanically galvanized.
- Fatigue rated at 1-1/2 times the Working Load Limit at 20,000 cycles.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certifications. Proof testing and certification available when requested at the time of order, charges will apply.
- Approved for use at -40° F (-40° C) to 400° F (204° C).
- All 209 and 210 shackles can meet charpy requirements of 31 ft-lb (42 Joules) avg. at -4° F (-20° C) upon special request.
- Meets or exceeds all requirements of ASME B30.26.
- Type Approval certification in accordance with ABS Marine Vessel Rules, ABS Mobile Offshore Units, and ABS Guide for the Certification of Lifting Appliances available. Certificates available when requested at time of order and may include additional charges.
- G-209 Screw pin anchor shackles meet the performance requirements of Federal Specification RR-C-271, Type IVA, Grade A, Class 2, except for those provisions required of the contractor.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-209 / S-209 Screw Pin Anchor Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.		Weight Each (lb)	Dimensions (in)														Replacement Pin Stock No.
		G-209	S-209		A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	L	M	P		
3/16	1/3	1018357	—	.06	.38	.06	.25	.88	.06	.19	.60	.56	.98	1.47	.16	1.14	.19	-	
1/4	1/2	1018375	1018384	.10	.47	.06	.32	1.13	.06	.25	.78	.62	1.28	1.84	.19	1.43	.25	-	
5/16	3/4	1018393	1018400	.18	.53	.06	.38	1.21	.06	.31	.84	.75	1.46	2.09	.22	1.71	.31	-	
3/8	1	1018419	1018428	.31	.66	.06	.44	1.45	.13	.38	1.03	.92	1.79	2.50	.25	2.06	.38	-	
7/16	1 1/2	1018437	1018446	.38	.75	.06	.50	1.69	.13	.44	1.16	1.06	2.04	2.91	.31	2.37	.44	-	
1/2	2	1018455	1018464	.72	.81	.06	.63	1.88	.13	.50	1.31	1.18	2.31	3.28	.38	2.69	.50	-	
5/8	3 1/4	1018473	1018482	1.37	1.06	.06	.75	2.38	.13	.62	1.69	1.50	2.93	4.19	.44	3.34	.69	-	
3/4	4 3/4	1018491	1018507	2.35	1.25	.06	.88	2.81	.25	.75	2.00	1.81	3.50	4.97	.50	3.97	.81	-	
7/8	6 1/2	1018516	1018525	3.62	1.44	.06	1.00	3.31	.25	.88	2.28	2.10	4.04	5.83	.50	4.50	.97	-	
1	8 1/2	1018534	1018543	5.03	1.69	.06	1.16	3.76	.25	1.00	2.69	2.38	4.69	6.56	.56	5.13	1.06	-	
1-1/8	9 1/2	1018552	1018561	7.41	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	5.15	7.47	.63	5.97	1.25	1082599	
1-1/4	12	1018570	1018589	9.50	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	5.76	8.26	.69	6.50	1.38	1082615	
1-3/8	13 1/2	1018598	1018605	13.53	2.25	.13	1.56	5.22	.25	1.42	3.62	3.31	6.38	9.16	.75	6.93	1.50	1082633	
1-1/2	17	1018614	1018623	17.20	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	6.94	10.00	.81	7.43	1.62	1082651	
1-3/4	25	1018632	1018641	30.90	2.88	.13	2.03	7.00	.25	1.84	5.00	4.19	8.80	12.34	1.00	9.19	2.25	1082679	
2	35	1018650	1018669	45.00	3.25	.13	2.28	7.75	.25	2.08	5.75	4.81	10.15	13.68	1.13	10.36	2.40	1082697	
2-1/2	55	1018678	1018687	85.75	4.12	.25	2.78	10.51	.25	2.72	7.25	5.81	12.75	17.92	1.38	13.17	3.13	1082713	

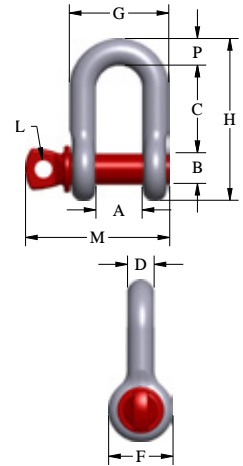
6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.



G-210



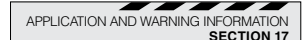
- Forged, Quenched & Tempered, with alloy pins.
- Working Load Limit and Grade 6 permanently shown on every shackle.
- Hot-dip galvanized.
- Sizes 3/8 inch and below are mechanically galvanized.
- Fatigue rated at 1-1/2 times the Working Load Limit at 20,000 cycles.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certifications. Proof testing and certification available when requested at the time of order, charges will apply.
- Approved for use at -40° F (-40° C) to 400° F (204° C).
- All 209 and 210 shackles can meet charpy requirements of 31 ft-lb (42 Joules) avg. at -4° F (-20° C) upon special request.
- Meets or exceeds all requirements of ASME B30.26.
- Type Approval certification in accordance with ABS Marine Vessel Rules, ABS Mobile Offshore Units, and ABS Guide for the Certification of Lifting Appliances available. Certificates available when requested at time of order and may include additional charges.
- G-210 Screw pin anchor shackles meet the performance requirements of Federal Specification RR-C-271, Type IVB, Grade A, Class 2, except for those provisions required of the contractor.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-210 Screw Pin Chain Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)												Replacement Pin Stock No.
				A	Tol. A ±	B	C	Tol. C ±	D	F	G	H	L	M	P	
1/4	1/2	1019150	.11	.47	.06	.32	.87	.06	.25	.62	.97	1.59	.19	1.43	.25	-
5/16	3/4	1019178	.17	.53	.06	.38	1.04	.06	.31	.75	1.15	1.91	.22	1.71	.31	-
3/8	1	1019196	.28	.66	.06	.44	1.25	.13	.38	.92	1.42	2.31	.25	2.06	.38	-
7/16	1 1/2	1019212	.43	.75	.06	.50	1.45	.13	.44	1.06	1.63	2.67	.31	2.37	.44	-
1/2	2	1019230	.65	.81	.06	.63	1.63	.13	.50	1.18	1.81	3.03	.38	2.69	.50	-
5/8	3 1/4	1019258	1.25	1.06	.06	.75	2.00	.13	.63	1.50	2.32	3.76	.44	3.34	.63	-
3/4	4 3/4	1019276	2.26	1.25	.06	.88	2.38	.25	.75	1.81	2.75	4.53	.50	3.97	.81	-
7/8	6 1/2	1019294	3.16	1.44	.06	1.00	2.81	.25	.88	2.10	3.20	5.33	.50	4.50	.97	-
1	8 1/2	1019310	4.75	1.69	.06	1.16	3.19	.25	1.00	2.38	3.69	5.94	.56	5.13	1.00	-
1-1/8	9 1/2	1019338	6.75	1.81	.06	1.25	3.56	.25	1.13	2.68	4.07	6.78	.63	5.97	1.25	1082599
1-1/4	12	1019356	9.06	2.03	.13	1.41	3.93	.25	1.25	3.00	4.53	7.50	.69	6.50	1.38	1082615
1-3/8	13 1/2	1019374	11.63	2.25	.13	1.56	4.36	.25	1.38	3.31	5.01	8.28	.75	6.93	1.50	1082633
1-1/2	17	1019392	15.95	2.38	.13	1.66	4.87	.25	1.50	3.62	5.38	9.05	.81	7.43	1.62	1082651
1-3/4	25	1019418	26.75	2.88	.13	2.03	5.76	.25	1.75	4.19	6.38	10.97	1.00	9.19	2.12	1082679
2	35	1019436	42.31	3.25	.13	2.28	6.75	.25	2.10	5.00	7.25	12.74	1.13	10.36	2.36	1082697
2-1/2	55	1019454	71.75	4.12	.25	2.78	8.00	.25	2.63	5.68	9.38	14.85	1.38	13.17	2.63	1082713

6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.



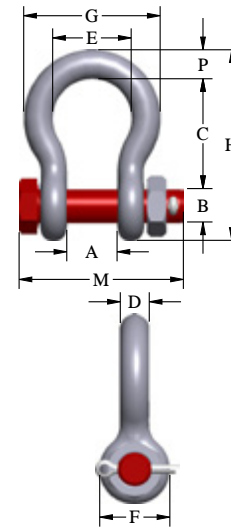
G-2130



S-2130



- Working Load Limit and Grade 6 permanently shown on every shackle.
- Forged, Quenched & Tempered, with alloy bolts.
- Hot-dip galvanized (G) or self colored (S). 85, 120, and 150-metric ton shackles are all hot-dip galvanized bows and the bolts are Dimetcoated® and painted red.
- Sizes 3/8 and below are mechanically galvanized.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit (1/3t - 55t).
- Approved for use at -40° F (-40° C) to 400° F (204° C).
- Meets or exceeds all requirements of ASME B30.26.
- Shackles 85 metric tons and larger are individually proof tested to 2.0 times the working load limit.
- Type Approval certification in accordance with ABS Marine Vessel Rules, ABS Mobile Offshore Units, and ABS Guide for the Certification of Lifting Appliances available. Certificates available when requested at time of order and may include additional charges.
- 3.1 Certification as standard available for charpy and statistical proof test from 3.25t up to 25 tons to DNV 2.7-1 and EN13889.
- Crosby 3.25t through 25t G-2130OC anchor shackles are type approved to DNV Certification Notes 2.7-1-Offshore Containers. These Crosby shackles are statistical proof and impact tested to 31 ft-lb (42 Joules) min. avg. at -4° F (-20° C). The tests are conducted by Crosby and 3.1 test certification is available upon request.
- All other 2130 shackles can meet charpy requirements of 31 ft-lb (42 Joules) avg at -4° F (-20° C) when requested at time of order.
- Meets the performance requirements of Federal Specification RR-C-271, Type IVA, Grade A, Class 3, except for those provisions required of the contractor.
- Look for the Red Pin®... the mark of genuine Crosby quality.

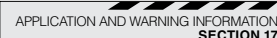


G-2130 / S-2130 Bolt Type Anchor Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.			Weight Each (lb)	Dimensions (in)												Replacement Bolt Stock No.
		G-2130	S-2130	G-2130OC		A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	M	P	
3/16	1/3 ‡	1019464	-	-	.06	.38	.06	.25	.88	.06	.19	.60	.56	.98	1.47	1.29	.19	-
1/4	1/2	1019466	-	-	.11	.47	.06	.32	1.13	.06	.25	.78	.62	1.28	1.84	1.56	.25	-
5/16	3/4	1019468	-	-	.22	.53	.06	.38	1.21	.06	.31	.84	.75	1.46	2.09	1.82	.31	-
3/8	1	1019470	-	-	.33	.66	.06	.44	1.45	.13	.38	1.03	.92	1.79	2.50	2.17	.38	-
7/16	1 1/2	1019471	-	-	.49	.75	.06	.50	1.69	.13	.44	1.16	1.06	2.04	2.91	2.51	.44	-
1/2	2	1019472	1019481	-	.79	.81	.06	.64	1.88	.13	.50	1.31	1.18	2.31	3.28	2.80	.50	-
5/8	3 1/4	1019490	1019506	1262013	1.68	1.06	.06	.77	2.38	.13	.62	1.69	1.50	2.93	4.19	3.56	.69	-
3/4	4 3/4	1019515	1019524	1262022	2.72	1.25	.06	.90	2.81	.25	.75	2.00	1.81	3.50	4.97	4.15	.81	-
7/8	6 1/2	1019533	1019542	1262031	3.95	1.44	.06	1.02	3.31	.25	.88	2.28	2.10	4.04	5.83	4.82	.97	-
1	8 1/2	1019551	1019560	1262040	5.66	1.69	.06	1.15	3.76	.25	1.00	2.69	2.38	4.69	6.56	5.39	1.06	-
1-1/8	9 1/2	1019579	1019588	1262059	8.27	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	5.15	7.47	5.90	1.25	1082839
1-1/4	12	1019597	1019604	1262068	11.71	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	5.76	8.26	6.69	1.38	1082857
1-3/8	13 1/2	1019613	1019622	1262077	15.83	2.25	.13	1.53	5.22	.25	1.42	3.62	3.31	6.38	9.16	7.21	1.50	1082875
1-1/2	17	1019631	1019640	1262086	19.00	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	6.94	10.00	7.73	1.62	1082893
1-3/4	25	1019659	1019668	1262095	33.91	2.88	.13	2.03	7.00	.25	1.84	5.00	4.19	8.80	12.34	9.63	2.25	1082919
2	35	1019677	1019686	-	52.25	3.25	.13	2.31	7.75	.25	2.08	5.75	4.81	10.15	13.69	10.77	2.40	1082937
2-1/2	55	1019695	1019702	-	98.25	4.12	.25	2.81	10.51	.25	2.72	7.25	5.81	12.75	17.92	13.53	3.13	1082955
3	† 85	1019711	-	-	154	5.00	.25	3.30	13.00	.25	3.12	7.88	6.50	14.62	21.50	15.13	3.63	1084449
3-1/2	† 120 ‡	1019739	-	-	265	5.25	.25	3.76	14.63	.25	3.63	9.00	8.00	17.02	24.88	17.00	4.38	1084452
4	† 150 ‡	1019757	-	-	338	5.50	.25	4.26	14.50	.25	4.00	10.00	9.00	18.00	25.68	18.00	4.56	1084456

6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications..

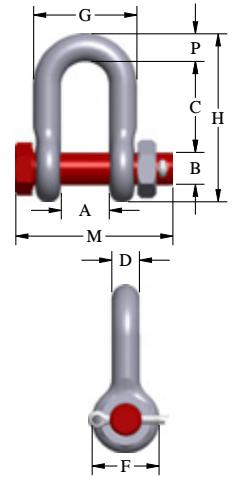
† Individually Proof Tested with certification. ‡ Furnished with eye bolts for handling.



G-2150



- Working Load Limit and Grade 6 permanently shown on every shackle.
- Forged, Quenched & Tempered, with alloy pins.
- Hot-dip galvanized. 85 ton shackles have hot-dip galvanized bows and the bolts are Dimetcoated® and painted red.
- Sizes 3/8 inch and below are mechanically galvanized.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit. (1/2t - 55t).
- Approved for use at -40° F (-40° C) to 400° F (204° C).
- Meets or exceeds all requirements of ASME B30.26.
- Sizes 1/2 - 25t meet the performance requirements of EN13889:2003.
- Shackles 55 metric tons and smaller can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification when requested at time of order.
- Type Approval certification in accordance with ABS Marine Vessel Rules, ABS Mobile Offshore Units, and ABS Guide for the Certification of Lifting Appliances available. Certificates available when requested at time of order and may include additional charges.
- Meets the performance requirements of Federal Specification RR-C-271, Type IVB, Grade A, Class 3, except for those provisions required of the contractor.
- All 2150 shackles can meet charpy requirements of 31 ft-lb (42 Joules) avg at -4° F (-20° C) upon special request.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-2150 Bolt Type Chain Shackles

Nominal Size (in)	Working Load Limit (t)*	Stock No.	Weight Each (lb)	Dimensions (in)											Replacement Bolt Stock No.
				A	Tol. A ±	B	C	Tol. C ±	D	F	G	H	M	P	
1/4	1/2	1019768	.13	.47	.06	.32	.97	.06	.25	.62	.97	1.59	1.56	.25	-
5/16	3/4	1019770	.23	.53	.06	.38	1.15	.06	.31	.75	1.15	1.91	1.82	.31	-
3/8	1	1019772	.33	.66	.06	.44	1.42	.13	.38	.92	1.42	2.31	2.17	.38	-
7/16	1 1/2	1019774	.49	.75	.06	.50	1.63	.13	.44	1.06	1.63	2.67	2.51	.44	-
1/2	2	1019775	.75	.81	.06	.64	1.81	.13	.50	1.18	1.81	3.03	2.80	.50	-
5/8	3 1/4	1019793	1.47	1.06	.06	.77	2.32	.13	.63	1.50	2.32	3.76	3.56	.63	-
3/4	4 3/4	1019819	2.52	1.25	.06	.90	2.75	.25	.75	1.81	2.75	4.53	4.15	.81	-
7/8	6 1/2	1019837	3.85	1.44	.06	1.02	3.20	.25	.88	2.10	3.20	5.33	4.82	.97	-
1	8 1/2	1019855	5.55	1.69	.06	1.15	3.69	.25	1.00	2.38	3.69	5.94	5.39	1.00	-
1-1/8	9 1/2	1019873	7.60	1.81	.06	1.25	4.07	.25	1.13	2.68	4.07	6.78	5.90	1.25	1082839
1-1/4	12	1019891	10.81	2.03	.06	1.41	4.53	.25	1.25	3.00	4.53	7.50	6.69	1.38	1082857
1-3/8	13 1/2	1019917	13.75	2.25	.13	1.53	5.01	.25	1.38	3.31	5.01	8.28	7.21	1.50	1082875
1-1/2	17	1019935	17.01	2.38	.13	1.66	5.38	.25	1.50	3.62	5.38	9.05	7.73	1.62	1082893
1-3/4	25	1019953	31.40	2.88	.13	2.03	6.38	.25	1.75	4.19	6.38	10.97	9.63	2.12	1082919
2	35	1019971	46.75	3.25	.13	2.31	7.25	.25	2.10	5.00	7.25	12.74	10.77	2.36	1082937
2-1/2	55	1019999	85.00	4.12	.25	2.81	9.38	.25	2.63	5.68	9.38	14.85	13.53	2.63	1082955
3	† 85	1020013	124.25	5.00	.25	3.30	11.00	.25	3.00	6.50	11.00	16.87	15.13	3.50	1084449

6:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.

† Individually Proof Tested with certification.

Load Rated

Fatigue Rated

TA
TYPE APPROVED

QUIC-CHECK®

Q&T
QUENCHED & TEMPERED

MAXTOUGH®

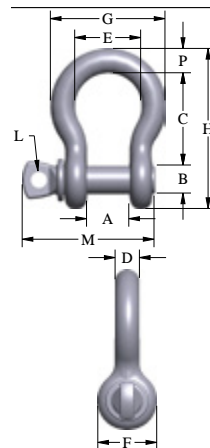
CE

APPLICATION AND WARNING INFORMATION
SECTION 17

G-209A
Grade 8



- Forged alloy steel, Quenched & Tempered, with alloy pins.
- Meets performance requirements of Grade 8 shackles.
- Working Load Limit permanently shown on every shackle.
- Hot-dip galvanized.
- Size 3/8 inch is mechanically galvanized.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification. Charges for proof testing and certification available when requested at the time of order.
- Approved for use at -40° C (-40° F) to 204° C (400° F).
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these shackles meet other critical performance requirements including impact properties and material traceability, not addressed by ASME B30.26.
- G-209A Screw pin anchor shackles meet the performance requirements of Federal Specification RR-C-271H, Type IVA, Grade B, Class 2, except for those provisions required of the contractor.



G-209A Alloy Screw Anchor Pin Shackles

Load Rated

QUICK-CHECK®

QT

CE

APPLICATION AND WARNING INFORMATION
SECTION 17

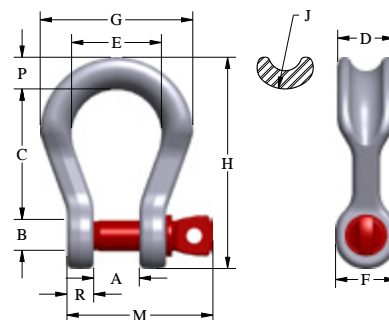
Nominal Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)													Replacement Screw Pin Stock No.
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	L	M	P	
3/8	2	1017450	.31	.66	.06	.44	1.45	.13	.38	1.03	.92	1.79	2.50	.25	2.06	.38	-
7/16	2.67	1017472	.38	.75	.06	.50	1.69	.13	.44	1.16	1.06	2.04	2.91	.31	2.37	.44	-
1/2	3.33	1017494	.63	.81	.06	.63	1.88	.13	.50	1.31	1.18	2.31	3.28	.38	2.69	.50	-
5/8	5	1017516	1.38	1.06	.06	.75	2.38	.13	.62	1.69	1.50	2.93	4.19	.44	3.34	.69	-
3/4	7	1017538	2.35	1.25	.06	.88	2.81	.25	.75	2.00	1.81	3.50	4.97	.50	3.97	.81	-
7/8	9.5	1017560	3.61	1.44	.06	1.00	3.31	.25	.88	2.28	2.10	4.04	5.83	.50	4.50	.97	-
1	12.5	1017582	5.32	1.69	.06	1.16	3.76	.25	1.00	2.69	2.38	4.69	6.56	.56	5.13	1.06	-
1-1/8	15	1017604	7.30	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	5.15	7.47	.63	5.97	1.25	1083867
1-1/4	18	1017626	9.88	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	5.76	8.26	.69	6.50	1.38	1083873
1-3/8	21	1017648	13.25	2.25	.13	1.56	5.22	.25	1.42	3.62	3.31	6.38	9.16	.75	6.93	1.50	1083885

4.5:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit (metric tons) and 2.2 times the Working Load Limit (short tons). For Working Load Limit reduction due to side loading applications, see Warnings & Applications.

G-2169



- Quenched & Tempered for maximum strength.
- Forged alloy steel.
- Available in galvanized finish.
- Can be individually proof tested and magnetic particle inspected upon request. Crosby certification available at time of order.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these shackles meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-2169 Alloy Screw Pin Wide Body Shackles

Load Rated

QT

APPLICATION AND WARNING INFORMATION
SECTION 17

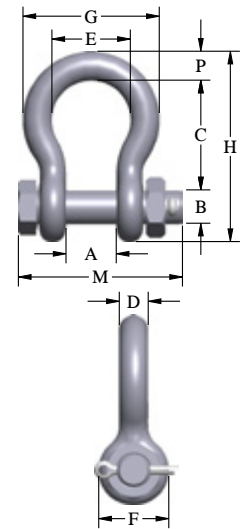
Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)														Replacement Screw Pin Stock No.
			A	B +/- .25	C	D +/- .02	E	F	G	H	J	K	L	M	P	R	
7	1021655	3.5	1.25	.88	3.56	1.60	2.50	1.82	4.10	5.87	1.25	1.25	.50	3.97	.96	.69	-
12 1/2	1021673	8.8	1.69	1.16	4.63	2.13	3.26	2.38	5.51	7.63	1.37	1.63	.56	5.13	1.25	.92	-
18	1021691	13	2.03	1.41	5.81	2.50	4.00	2.69	6.76	9.38	1.50	2.00	.69	6.25	1.41	1.16	1083873

5:1 Design Factor. Proof Load is 2 times the Working Load Limit.

G-2130A Grade 8



- Forged alloy steel, Quenched & Tempered, with bow and bolt.
- Meets or exceeds all requirements of Grade 8 shackles.
- Working Load Limit permanently shown on every shackle.
- Hot-dip galvanized.
- Meets or exceeds all requirements of ASME B30.26, including identification, ductility, design factor, proof load, and temperature requirements. Importantly, G-2130A meet other critical performance requirements, including impact properties, and material traceability not addressed by ASME B30.26.
- Shackles can be furnished proof tested with certificates to designated standards, such as ABS, DNV, Lloyds, or other certification when requested at time of order.
- Type Approval and certification in accordance with DNV 2.7-1 offshore containers.
- Shackles are Quenched & Tempered and meet DNV impact requirements of 42 Joules (31 ft-lb) at -40° C (-40° F).
- G-2130A Bolt Type Anchor shackles with thin head bolt – nut with cotter pin. Meets the performance requirements of Federal Specification RR-C-271H, Type IVA, Grade B, Class 3, except for those provisions required of the contractor.



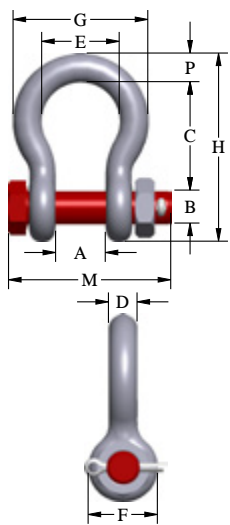
G-2130A Alloy Bolt Type Anchor Shackles Grade 8

Nominal Size (in)	Working Load Limit (t)*	Stock No.	Weight Each (lb)	Dimensions (in)											
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	H	L	M	N
1/2	2	1219472	.79	.81	.06	.64	1.88	.13	.50	1.31	1.18	3.28	2.31	2.80	.50
5/8	3.25	1219491	1.37	1.06	.06	.77	2.38	.25	.62	1.69	1.50	4.19	2.93	3.56	.69
3/4	4.75	1219516	2.71	1.25	.06	.90	2.81	.25	.75	2.00	1.81	4.97	3.50	4.15	.81
7/8	6.5	1219534	3.95	1.44	.06	1.02	3.31	.25	.88	2.28	2.10	5.83	4.04	4.82	.97
1	8.5	1219552	5.03	1.69	.06	1.15	3.76	.25	1.00	2.69	2.38	6.56	4.69	5.39	1.06
1-1/8	9.5	1219578	8.27	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	7.47	5.15	5.90	1.25
1-1/4	12	1219598	11.7	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	8.26	5.76	6.69	1.38
1-3/8	13.5	1219614	15.8	2.25	.13	1.53	5.22	.25	1.42	3.62	3.31	9.16	6.38	7.21	1.50
1-1/2	17	1219632	19.0	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	10.0	6.94	7.73	1.62

8:1 Design Factor. Maximum Proof Load is 2.5 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications..



G-2140

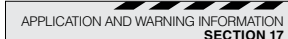


- Quenched & Tempered.
- Alloy bows, alloy bolts.
- Forged alloy steel 2 through 300 metric tons. Cast alloy steel 400 metric tons.
- Meets performance requirements of Grade 8 shackles.
- Working Load Limit is permanently shown on every shackle.
- 30, 40, 55, and 85 metric ton shackle bows are available galvanized (G) or self colored (S) with bolts that are galvanized and painted red.
- Size 3/8 inch is mechanically galvanized.
- 120, 150, 175 metric ton shackle bows are hot-dip galvanized; bolts are Dimetcoated and painted red.
- 200, 250, 300, 400 metric ton shackle bows are Dimetcoated; bolts are Dimetcoated and painted red.
- Approved for use at -40° C (-40° F) to 204° C (400° F).
- Shackles are Quenched & Tempered and can meet DNV impact requirements of 42 Joules (31 ft-lb) at -20° C (-4° F).
- Shackles 200 metric tons and larger are provided as follows:
 - Serialized bolt and bow
 - Material certification (chemical)
 - Magnetic particle inspected.
 - Certification must be requested at time of order.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. 2140 shackles meet other critical performance requirements including impact properties and material traceability, not addressed by ASME B30.26.
- Type Approval certification in accordance with ABS 2016 Steel Vessel Rules and 2016 ABS Guide for Certification of Lifting Appliances. Certificates are available when requested at time of order and may include additional charges.
- G-2140 meets the performance requirements of Federal Specification RR-C-271H, Type IVA, Grade B, Class 3, except for those provisions required of the contractor. For additional information, see Warnings & Applications.
- Look for the Red Pin®... the mark of genuine Crosby quality.

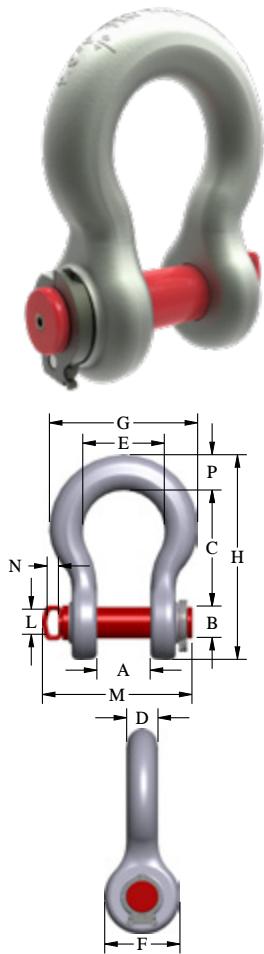
G-2140 Alloy Bolt Type Anchor Shackles

Nominal Shackle Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)														Replacement Bolt Stock No. *
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	L	M	N	P	
3/8	2	1021015	.33	.66	.06	.44	1.45	.13	.38	1.03	.92	1.79	2.50	-	2.17	-	.38	-
7/16	2 2/3	1021020	.49	.75	.06	.50	1.69	.13	.44	1.16	1.06	2.04	2.91	-	2.51	-	.44	-
1/2	3 1/2	1021029	.79	.81	.06	.64	1.88	.13	.50	1.31	1.18	2.31	3.28	-	2.80	-	.50	-
5/8	5	1021038	1.68	1.06	.06	.77	2.38	.13	.62	1.69	1.50	2.93	4.19	-	3.56	-	.69	-
3/4	7	1021047	2.72	1.25	.06	.90	2.81	.25	.75	2.00	1.81	3.50	4.97	-	4.15	-	.81	-
7/8	9 1/2	1021056	3.95	1.44	.06	1.02	3.31	.25	.88	2.28	2.10	4.04	5.83	-	4.82	-	.97	-
1	12 1/2	1021065	5.66	1.69	.06	1.15	3.76	.25	1.00	2.69	2.38	4.69	6.56	-	5.39	-	1.06	-
1-1/8	15	1021074	8.27	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	5.15	7.47	-	5.90	-	1.25	1084382
1-1/4	18	1021083	11.7	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	5.76	8.26	-	6.69	-	1.38	1084391
1-3/8	21	1021092	15.8	2.25	.13	1.53	5.22	.25	1.42	3.62	3.31	6.38	9.16	-	7.21	-	1.50	1084400
1-1/2	30	1021110	18.8	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	6.94	10.00	-	7.73	-	1.62	1084409
1-3/4	40	1021138	33.8	2.88	.13	2.03	7.00	.25	1.84	5.00	4.19	8.80	12.34	-	9.63	-	2.25	1084418
2	55	1021156	49.9	3.25	.13	2.31	7.75	.25	2.08	5.75	4.81	10.15	13.69	-	10.77	-	2.40	1084427
2-1/2	85	1021174	103	4.12	.25	2.81	10.51	.25	2.72	7.25	5.81	12.75	17.92	-	13.53	-	3.13	1084436
3	120	1021192	162	5.00	.25	3.30	13.00	.25	3.12	7.88	6.50	14.62	21.50	-	15.13	-	3.63	1084445
3-1/2	† 150	1021218	268	5.25	.25	3.76	14.63	.25	3.63	9.00	8.00	17.02	24.88	4.00	17.00	1.80	4.38	1084454
4	† 175	1021236	318	5.50	.25	4.26	14.50	.25	4.00	10.00	9.00	18.00	25.68	4.00	18.00	1.80	4.56	1084463
4-3/4	† 200	1021234	461	7.25	.25	4.76	15.18	.25	4.75	11.00	10.50	20.30	28.03	4.00	24.04	1.80	5.00	1087552
5	† 250	1021243	608	8.50	.25	5.01	18.50	.25	5.00	13.00	12.00	23.63	32.63	4.00	24.87	1.80	5.63	1087561
6	† 300	1021252	797	8.38	.25	6.01	18.72	.25	5.88	13.00	13.00	24.76	34.28	4.00	26.22	1.80	6.06	1087570
7*	† 400	1021478	1289	8.25	.25	7.01	22.50	.25	6.00	13.00	14.00	26.00	40.25	4.00	29.66	1.80	7.25	1020068

4.5:1 Design Factor for sizes 2 through 21 metric tons, 5.4:1 Design Factor for sizes 30 through 175 metric tons, 4:1 Design Factor for 200 through 400 metric tons. Maximum Proof Load is 2 times the Working Load Limit. *Cast alloy steel. †Furnished with round head bolts with a handle. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.



G-2140E



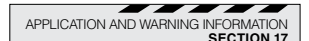
- Quenched & Tempered.
- Alloy bows, alloy bolts.
- Meets performance requirements of Grade 8 shackles.
- Working Load Limit is permanently shown on every shackle.
- 200, 250, and 300 metric ton shackle bows are Dimetcoated®; Pins are Dimetcoated and painted red.
- Approved for use at -40° F (-40° C) to 400° F (204° C).
- Shackles are Quenched & Tempered and can meet DNV impact requirements of 31 ft-lb (42 Joules) at -4° F (-20° C).
- All sizes are individually proof tested to 2.0 times the Working Load Limit.
- Shackles are provided as follows:
 - Serialized bolt and bow
 - Material certification (chemical)
 - Magnetic particle inspected
 - Certification must be requested at time of order
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these shackles meet other critical performance requirements including impact properties and material traceability, not addressed by ASME B30.26.
- Type Approval certification in accordance with ABS 2016 Steel Vessel Rules and 2016 ABS Guide for Certification of Lifting Appliances. Certificates available when requested at time of order and may include additional charges.
- G-2140E meets the performance requirements of Federal Specification RR-C-271H, Type IVA, Grade B, Class 3, except for those provisions required of the contractor.
- Look for the Red Pin®... the mark of genuine Crosby quality.

G-2140E Alloy Easy-Loc Shackles

Nominal Shackle Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)													
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	L	M	N	P
4-3/4	200	1021475	458	7.25	.25	4.76	15.18	.25	4.75	11	10.50	20.30	28.03	4	23.01	1.8	5.00
5	250	1021484	597	8.50	.25	5.01	18.50	.25	5.00	13	12.00	23.63	32.63	4	23.84	1.8	5.63
6	300	1021493	791	8.38	.25	6.01	18.72	.25	5.88	13	13.00	24.76	34.28	4	25.01	1.8	6.06

Nominal Shackle Size (in)	Working Load Limit (t)	Stock No.	Replacement Bolt Stock No.	Replacement EZ-Loc Collar Stock No.
4-3/4	200	1021475	1087603	1025837
5	250	1021484	1087612	1025837
6	300	1021493	1087621	1025855

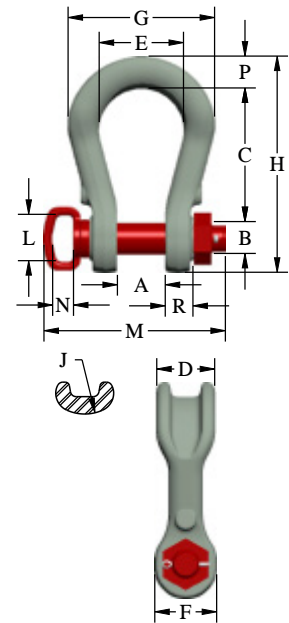
4:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.



G-2160



- Increase in shackle bow radius provides minimum 58% gain in sling bearing surface and eliminates need for a thimble.
- Increases usable sling strength a minimum of 15% and greatly improves life of wire rope slings.
- Can be used to connect synthetic web slings, synthetic round slings or wire rope slings.
- All sizes Quenched & Tempered for maximum strength.
- Forged alloy steel from 7 through 300 metric tons.
- Cast alloy steel from 400 through 2000 metric tons.
- Proof tested as follows:
 - 7 through 75 metric tons and 200 through 300 metric tons: 2 x WLL
 - 125 metric tons: 1.6 x WLL
 - 400 metric tons to 800t : 2 x WLL
 - 1000 metric tons to 2000t: 1.5 x WLL
- All ratings are in metric tons, embossed on side of bow.
- G-2160, (7 through 55t), are hot-dip galvanized and pins are painted red.
- G-2160 (75t and larger), bows are furnished Dimetcoated; Pins are Dimetcoated, then painted red.
- Approved for use at -40° C (-40° F) to 204° C (400° F).
- Bow and bolt are certified to meet Charpy impact testing of 42 Joules (31 ft-lb) min. avg. at -20° C (-4° F).
- All 2160 shackles are individually proof tested and magnetic particle inspected. Crosby certification available at time of order.
- Shackles requiring ABS, Lloyds and other certifications are available upon special request and must be specified at time of order.
- Type approved and certification to DNV Rules for Certification of Lifting Appliances, and are produced in accordance with DNV MSA requirements. Databook is provided that includes required documents.
 - Serialization / Identification
 - Material Testing (physical / chemical / Charpy)
 - Proof Testing
- Look for the Red Pin®... the mark of genuine Crosby quality.
- All Crosby 2160 shackles are capable of inline bow to bow assembly.



G-2160 Wide Body Shackles

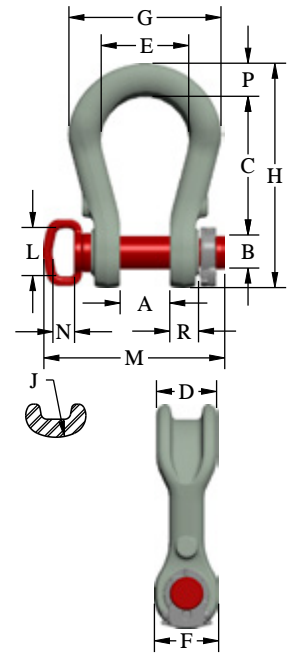
Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)															Effective Body Diameter	Replacement Bolt Stock No.
			A	B	C	D	E	F	G	H	J	L	M	N	P	R			
7	1021256	4.0	1.25	.88	3.56	1.60	2.50	1.82	4.10	5.87	1.25	—	4.14	—	.96	.69	2.1	—	
12 1/2	1021265	8.8	1.69	1.13	4.63	2.13	3.26	2.38	5.51	7.63	1.37	—	5.38	—	1.25	.92	2.4	—	
18	1021274	14.9	2.03	1.38	5.81	2.50	4.00	2.69	6.76	9.38	1.50	—	6.69	—	1.41	1.16	2.8	1083349	
30	1021276	26.3	2.74	1.63	6.94	3.13	5.00	3.50	8.50	11.38	2.50	—	8.10	—	1.75	1.38	4.1	1089522	
40	1021278	46.8	3.38	2.00	8.06	3.75	6.00	4.00	10.62	13.62	1.75	—	10.18	—	2.31	1.69	3.6	1089523	
55	1021287	68.0	3.25	2.25	9.36	4.50	7.00	4.63	12.26	15.63	2.00	—	10.36	—	2.63	2.00	4.3	1083375	
75	1022101	112	4.13	2.75	11.53	5.00	7.28	5.34	12.64	18.66	3.75	4	15.04	1.8	2.75	2.39	6.3	1089524	
125	1022110	193	5.12	3.15	14.37	5.91	8.66	6.50	15.47	23.00	3.75	4	18.32	1.8	3.50	3.10	6.8	1089533	
200	1022118	420	5.91	4.12	18.91	8.56	10.84	8.41	20.27	30.44	5.25	4	19.35	1.8	4.94	3.39	9.5	1089542	
300	1022127	805	7.38	5.25	23.63	10.38	12.62	10.50	23.93	37.66	6.13	4	22.61	1.8	5.78	4.30	11.4	1089551	
400	1021336	1180	8.66	6.30	24.41	14.17	14.76	12.36	26.30	41.81	7.20	—	30.71	—	8.07	5.91	14.3	1511032	
500	1021345	1754	9.84	7.09	26.97	17.13	17.90	13.39	31.19	46.85	8.74	—	33.98	—	9.65	6.46	17.3	1511033	
600	1021354	2210	10.83	7.87	29.13	18.70	19.26	14.76	32.13	50.87	9.57	—	36.93	—	10.43	7.17	18.9	1511034	
800	1021363	3384	13.46	9.06	32.28	21.10	21.81	17.32	35.59	57.56	10.71	—	42.44	—	12.09	8.35	21.3	1511037	
1000	1021381	4988	15.91	10.63	33.46	24.02	25.20	19.69	43.94	62.36	13.31	—	46.73	—	13.74	9.17	25.3	1511021	
1250	1021390	6802	17.40	11.81	36.54	26.57	27.87	22.44	49.57	68.86	14.61	—	52.05	—	15.20	9.80	27.9	1511027	
1550	1021399	8382	18.50	12.60	39.76	28.35	29.72	24.21	51.73	74.45	15.63	—	56.18	—	16.26	11.42	29.8	1511028	
2000	1021408	11868	20.47	15.16	44.88	30.31	31.89	26.77	55.59	83.46	16.97	—	62.87	—	17.60	12.70	32.1	1511029	

5:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit on 7 through 300 metric tons (except for 125 metric tons which is proof tested to 1.6 times the Working Load Limit). Maximum customer Proof Load is 1.5 times the Working Load Limit on 400 through 2000 metric tons (available on customer request at time of order 2xWLL).

G-2160E



- Increase in shackle bow radius provides minimum 58% gain in sling bearing surface and eliminates need for a thimble.
- Increases usable sling strength a minimum of 15% and greatly improves life of wire rope slings.
- Can be used to connect synthetic web slings, synthetic round slings or wire rope slings.
- All sizes Quenched & Tempered for maximum strength.
- Forged alloy steel from 75 through 300 metric tons.
- Cast alloy steel from 400 through 600 metric tons.
- Proof tested as follows:
 - 75 metric tons and 200-300 metric tons: 2 x WLL
 - 125 metric tons: 1.6 x WLL
 - 400-600 metric tons: 2 x WLL
- All ratings are in metric tons, embossed on side of bow.
- G-2160E, (75t and larger), bows are furnished Dimetcoated, and pins are Dimetcoated, then painted red.
- Approved for use at -40° C (-40° F) to 204 degrees C (400° F).
- Bow and bolt are certified to meet Charpy impact testing of 42 Joules (31 ft-lb) min. avg. at -20° C (-4 degrees F).
- All 2160E shackles are individually proof tested and magnetic particle inspected. Crosby certification available at time of order.
- Shackles requiring ABS, Lloyds and other certifications are available upon special request and must be specified at time of order.
- Shackles have DNV Type Approval to Rules for Certification of Lifting Appliances, and are produced in accordance with DNV MSA requirements. Databook is provided that includes required documents.
 - Serialization / Identification
 - Material Testing (physical / chemical / Charpy)
 - Proof Testing
- Look for the Red Pin®... the mark of genuine Crosby quality.



G-2160E Easy-Loc Wide Body Shackles

Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)													Effective Body Diameter
			A	B	C	D	E	F	G	H	J	M	N	P	R	
75	1021500	110	4.13	2.75	11.54	5.00	7.28	5.34	12.64	18.66	3.75	15.04	1.8	2.75	2.39	6.30
125	1021509	190	5.12	3.15	14.37	5.91	8.66	6.50	15.47	23.00	3.75	17.70	1.8	3.50	3.10	6.80
200	1021518	408	5.91	4.12	18.91	8.56	10.84	8.41	20.27	30.44	5.25	19.35	1.8	4.94	3.39	9.50
300	1021527	787	7.38	5.25	23.63	10.38	12.62	10.50	23.93	37.51	6.13	22.61	1.8	5.78	4.30	11.40
400	1021337	1156	8.66	6.30	24.41	14.17	14.76	12.36	26.22	41.81	7.20	30.31	-	8.07	5.91	14.27
500	1021346	1718	9.84	7.09	26.97	17.13	17.90	13.39	31.19	46.85	8.74	32.87	-	9.65	6.46	17.29
600	1021355	2159	10.83	7.87	29.13	18.70	19.26	14.76	32.13	50.87	9.57	36.93	-	10.43	7.17	18.91

Working Load Limit (t)	Stock No.	Replacement Bolt Stock No.	Replacement EZ-LOC Collar Stock No.
75	1021500	1089705	1025800
125	1021509	1089714	1025819
200	1021518	1089723	1025828
300	1021527	1089732	1025837
400	1021337	1511039	1025855
500	1021346	1511040	1025864
600	1021355	1511041	1025864

5:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit on 75 through 300 metric tons (except for 125 metric tons which is proof tested to 1.6 times the Working Load Limit). Maximum customer Proof Load is 1.5 times the Working Load Limit on 400 through 600 metric tons (available on customer request at time of order 2xWLL).

Shackle Bolt Securement

MADE EASY

The patented Easy-Loc V2™ shackle bolt securement system will change the way you make your critical lifts.



Open collar



Push collar onto bolt



Close collar

Wide opening ergonomic grip provides easy access for all hand sizes

316 stainless steel design resists corrosion

The new Easy-Loc V2™ can be retrofitted on all original Crosby Easy-Loc® Shackles

No cotter pin or tools required

- No cotter pins or tools required, reducing install/release time up to 90%.
- Meets all industry standards.
- Up to 60% lighter than conventional nut and cotter pin design.

Crosby®

kitocrosby.com



Watch video: kitocrosby.com/easy-loc

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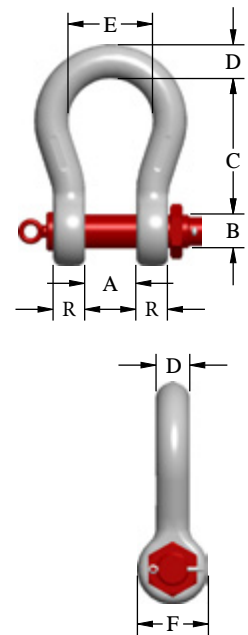
S-2135 / S-2145



- Trusted Crosby quality.
- Embossed Angle Indicators included.
- Meets performance requirements of Grade 8 shackles.
- 5:1 Design Factor.
- Individually proof loaded to 2 times the Working Load Limit.
- S-2135 and S-2145 shackles are available with aluminum paint and are not galvanized.
- Operating temperature range -20° C (-4° F) to 200° C (392° F) for S-2135 and S-2145.
- Material inspection certificate Type 3.1 according to EN 10204.
- DNV type approved: DNVGL-ST-0377 and DNVGL-ST-0378.
- Meets performance requirements of federal specification RR-C-271H, except for those provisions required of the contractor.
- Meets or exceeds all requirements of ASME B30.26.
- S-2135CT COLD TUFF® available in 85t with an operating temperature range of -60°C (-76°F) up to +200°C (392°F).
- DNV witness proof testing available on request for all sizes and models.
- Magnetic Particle Inspection available on request for all sizes and models.
- Look for the Red Pin®... the mark of genuine Crosby quality.

SHACKLES

1



S-2135 / S-2145 Bolt Type Anchor Shackles

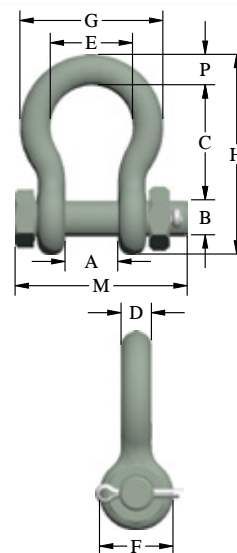
Frame Size (in)	Working Load Limit (t)	Stock No	Weight (lb)	Dimensions (in)							
				A	+/- 0	B	C	D	R	E	F
S-2135											
3	85	1205009	172	5.00	+ .25 / - 0	3.27	12.99	3.35	3.15	7.48	6.38
3 1/2	120	1205018	254	5.67	+ .25 / - 0	3.74	14.96	3.74	3.50	9.37	7.87
4	150	1205027	357	6.50	+ .33 / - 0	4.25	15.16	4.13	3.94	10.83	9.06
4 3/4	200	1205036	529	7.09	+ .33 / - 0	4.92	17.72	4.72	4.33	11.02	10.63
5	250	1205045	675	8.07	+ .40 / - 0	5.51	20.47	5.12	4.53	12.01	11.42
6	300	1205054	811	8.07	+ .40 / - 0	5.91	20.87	5.51	4.72	12.01	12.40
7	400	1205063	1327	9.06	+ .40 / - 0	6.89	22.64	6.30	6.30	12.80	14.37
7 1/4	500	1205234	1620	9.84	+ .47 / - 0	7.28	25.59	7.09	6.30	13.78	15.16
8	600	1205243	2136	10.83	+ .50 / - 0	8.07	25.59	7.87	7.28	14.76	16.93
8 1/2	800	1205261	2438	11.81	+ .60 / - 0	8.66	25.59	8.27	7.87	15.75	17.72
9 1/2	1000	1205270	3254	13.39	+ .66 / - 0	9.45	27.56	9.45	9.45	16.54	19.69
10	1250	1205279	4310	14.17	+ .75 / - 0	10.63	29.53	10.24	8.86	17.72	22.44
11	1500	1205288	5130	14.17	+ .75 / - 0	11.42	31.50	11.02	8.86	17.72	24.02
S-2135 CT											
3	85	1205099	172	5.00	+ .25 / - 0	3.27	12.99	3.35	3.15	7.48	6.38
S-2145											
3	120	1205072	172	5.00	+ .25 / - 0	3.27	12.99	3.35	80	7.48	6.38
3 1/2	150	1205081	254	5.67	+ .25 / - 0	3.74	14.96	3.74	89	9.37	7.87
4	175	1205090	357	6.50	+ .33 / - 0	4.25	15.16	4.13	100	10.83	9.06

Maximum Proof Load is 2.0 times the Working Load Limit.

G-2130CT / G-2140CT



- Forged, Quenched & Tempered, with alloy bolt.
 - G-2130CT - carbon steel
 - G-2140CT - alloy steel
- Working Load Limit permanently shown on every shackle.
- Individually serialized with certification.
- Fatigue Rated (G-2130CT only).
- All sizes are individually proof tested to 2.0 times the Working Load Limit.
- Finish is inorganic zinc primer.
- Bow and bolt are certified to meet Charpy impact testing of 42 Joules (31 ft-lb) min. avg. at -20° C (-4° F).
- Individually magnetic particle inspected with certification.
- Type Approval and certification in accordance with DNV 2.7-1 Offshore Containers, and Rules for Certification of Lifting Appliances, DNV-OS-E101 and are produced in accordance with DNV MSA requirements, including required documents.
- DNV certified minimum design temperature -4° F. May be used at -50° F (-45° C) in non DNV applications.
- Meets the performance requirements of Federal Specification RR-C-271H Type IVA:
 - G-2130CT - Grade A, Class 3, except for those provisions required of the contractor.
 - G-2140CT - Grade B, Class 3, except for those provisions required of the contractor.



G-2130CT COLD TUFF® Bolt Type Anchor Shackles

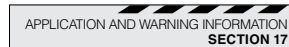
Nominal Shackle Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)											
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	H	G	M	P
3/4	4 3/4	1260568	2.72	1.25	.06	.90	2.81	.25	.75	2.00	1.81	4.97	3.50	4.15	.81
7/8	6 1/2	1260577	3.87	1.44	.06	1.02	3.31	.25	.88	2.28	2.10	5.83	4.04	4.82	.97
1	8 1/2	1260586	5.66	1.69	.06	1.15	3.76	.25	1.00	2.69	2.38	6.56	4.69	5.39	1.06
1-1/8	9 1/2	1260595	8.26	1.81	.06	1.25	4.27	.25	1.16	2.91	2.68	7.47	5.15	5.90	1.25
1-1/4	12	1260604	11.71	2.03	.06	1.41	4.69	.25	1.29	3.26	3.00	8.26	5.76	6.69	1.38
1-3/8	13 1/2	1260613	15.1	2.25	.13	1.53	5.22	.25	1.42	3.62	3.31	9.16	6.38	7.21	1.50
1-1/2	17	1260622	20.8	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	10.00	6.94	7.73	1.62
1-3/4	25	1260633	33.9	2.88	.13	2.03	7.00	.25	1.84	5.00	4.19	12.34	8.80	9.63	2.25

5.4:1 Design Factor. Maximum Proof Load is 2 times the Working Load Limit. For Working Load Limit reduction due to side loading applications, see Warnings & Applications.

G-2140CT COLD TUFF® Alloy Bolt Type Anchor Shackles

Nominal Shackle Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)											
				A	A ±	B	C	C ±	D	E	F	H	G	M	P
1-1/2	30	1260801	20.8	2.38	.13	1.66	5.76	.25	1.53	3.88	3.62	10.00	6.94	7.73	1.62
1-3/4	40	1260812	33.9	2.88	.13	2.03	7.00	.25	1.84	5.00	4.19	12.34	8.80	9.63	2.25
2	55	1260823	52.0	3.25	.13	2.31	7.75	.25	2.08	5.75	4.81	13.69	10.15	10.77	2.40
2-1/2	85	1260834	96.0	4.12	.25	2.81	10.51	.25	2.72	7.25	5.81	17.92	12.75	13.53	3.13
3	120	1260843	178.0	5.00	.25	3.30	13.00	.25	3.12	7.88	6.50	21.50	14.62	15.13	3.63
3-1/2	† 150	1260852	265.0	5.25	.25	3.76	14.63	.25	3.63	9.00	8.00	24.88	17.02	17.00	4.38
4	† 175	1260861	338.0	5.50	.25	4.26	14.50	.25	4.00	10.00	9.00	25.68	18.00	18.00	4.56
4-3/4	† 200	1260870	450.0	7.25	.25	4.76	15.18	.25	4.75	11.00	10.50	28.03	20.30	24.04	5.00
5	† 250	1260889	600.0	8.50	.25	5.01	18.50	.25	5.00	13.00	12.00	32.63	23.63	24.87	5.63

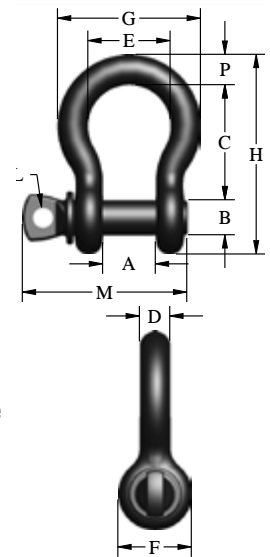
5.4:1 Design Factor on 30t through 175 metric tons. 4:1 Design Factor on 200 metric tons and larger. Maximum Proof Load is 2 times the Working Load Limit for all sizes.



S-209T



- Flat black baked on powder coat finish.
- Forged, Quenched & Tempered, with alloy pins.
- Working Load Limit and Grade 6 permanently shown on every shackle.
- Fatigue rated to 20,000 cycles at 1-1/2 times the Working Load Limit.
- Industry leading 6 to 1 Design Factor.
- Screw pin anchor shackles meet the performance requirement of Federal Specification RR-C-271H, Type IVA, Grade A, Class 2, except for those provisions required of the contractor.
- Meets the performance requirements of EN 13889.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these shackles meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.



S-209T Theatrical Shackles

Nominal Size (in)	Working Load Limit (t)	Stock No.	Weight Each (lb)	Dimensions (in)												
				A	Tol. A ±	B	C	Tol. C ±	D	E	F	G	H	L	M	P
3/8	1	1018706	.31	.66	.06	.44	1.45	.13	.38	1.03	.92	1.79	2.50	.25	2.06	.38
7/16	1 1/2	1018724	.38	.75	.06	.50	1.69	.13	.44	1.16	1.06	2.04	2.91	.31	2.37	.44
1/2	2	1018742	.72	.81	.06	.63	1.88	.13	.50	1.31	1.18	2.31	3.28	.38	2.69	.50
5/8	3 1/4	1018760	1.37	1.06	.06	.75	2.38	.13	.62	1.69	1.50	2.93	4.19	.44	3.34	.69
3/4	4 3/4	1018778	2.35	1.25	.06	.88	2.81	.25	.75	2.00	1.81	3.50	4.97	.50	3.97	.81

Maximum Proof Load is 2.0 times the Working Load Limit.

Load Rated

Fatigue Rated

QUIC-CHECK®

QT & TEMPERED

MAXTOUGH®

APPLICATION AND WARNING INFORMATION
SECTION 17



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A new podcast for lifting professionals



kitocrosby.com/podcast



QUIC-TAG™

Crosby®

Easy-to-secure Crosby RFID tags

Industry standards require periodic performance inspections to make sure lifting equipment is performing to specified levels.

The Crosby QUIC-TAG™ makes the inspection process more efficient, and its unique design can be retrofitted on numerous products.

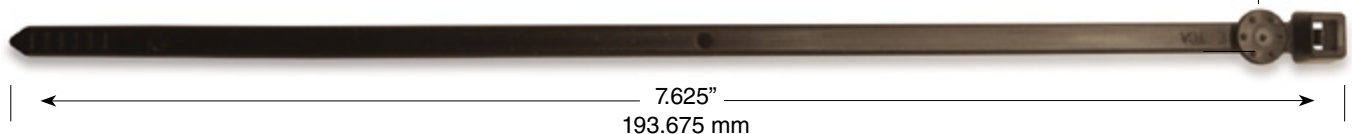
- Easy, fast, and secure attachment
- Engineered for extreme durability and strength with a low profile design
- Resistant to harsh environmental conditions including exposure to UV rays, water chemical exposure and temperatures up to 185°F (85°C)
- Compatible with the Crosby QUIC-CHECK® Inspection and Identification System
- 13.5 MHz operating frequency
- The most cost effective RFID tag offered by Kito Crosby

QUIC-CHECK®



RFID chip

Shown actual size:



Application Information

RIGGING PRACTICE SHACKLES

Screw pin shall be fully engaged. If designed for a cotter pin, it shall be used and maintained. Applied load should be centered in the bow to prevent side loading. Multiple sling legs should not be applied to the pin. If side loaded, the rated load shall be reduced according to Table 1 on the following page.

Screw Pin Shackles Pin Security



MOUSE SCREW PIN WHEN USED IN LONG-TERM OR HIGH-VIBRATION APPLICATIONS.
Mouse or Mousing (screw pin shackle) is a secondary securement method used to secure screw pin from rotation or loosening. Annealed iron wire is looped through hole in collar of pin and around adjacent leg of shackle body with wire ends securely twisted together.

Shackles



ROUND PIN
Do not side load, do not use as a collector ring, always use cotter pin.



SCREW PIN
Use when picking and placing a load, tighten pin prior to each lift.



BOLT-TYPE
Use in permanent or long-term installations, always use nut and cotter.

Connection of Slings to Shackles

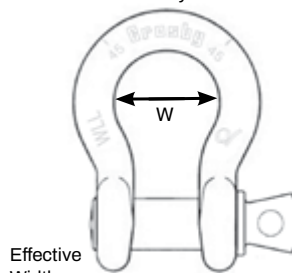


Diameter of shackle must be greater than wire rope diameter if no thimble in eye.



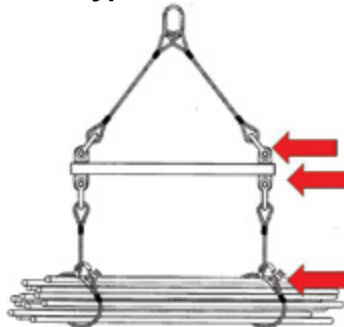
Shackle must be large enough to avoid pinching of synthetic slings.

Note that the effective width of the curved surface is only 75% of width.



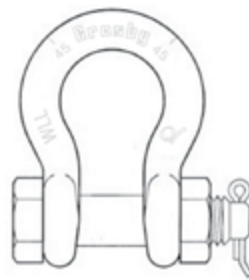
Effective Width = $.75 \times W$

Bolt-Type Shackles

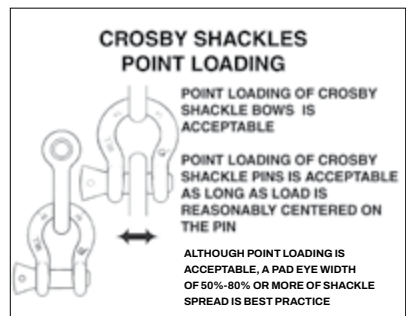
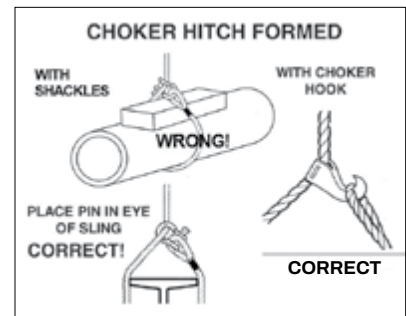
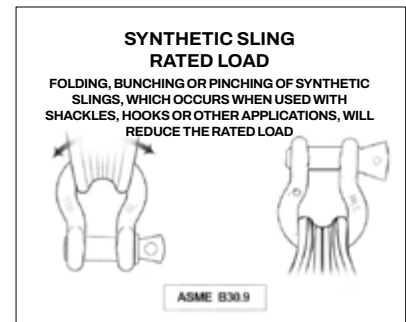
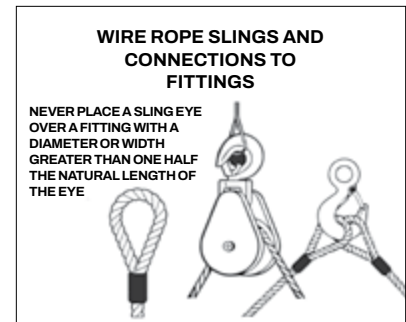
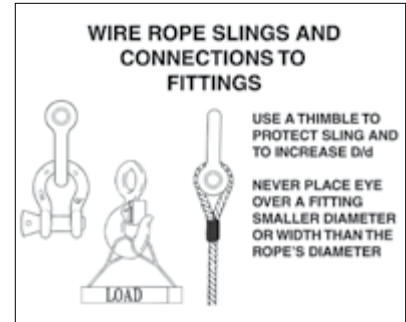


Use Bolt-Type Shackle for permanent or long-term connection.

Use Screw Pin Shackle for temporary connection.



Not necessary to tighten nut. Always use cotter pin.



Installation Guidelines

1. Extended prong cotter pins should be inserted into hole until the head is tangent to the bolt/pin, and oriented so the axis of the eye is parallel to the shank of the bolt/pin.
2. The prongs are to be bent in opposite directions around the bolt or pin as shown in Figure 1 below.
3. After installation, the cotter pin prongs should wrap around the bolt or pin by at least 60 degrees opposite directions of bolt or pin diameter.
4. The prongs may be bent with pliers or by gently tapping with a hammer. *Note: Avoid bending the prongs over sharp radii which may promote breakage. If a prong breaks off or becomes damaged during installation, replace the cotter pin.
5. The ends of the prongs may be curled to form a small loop to reduce the potential for snagging or puncture wounds.

Cotter Pin Sizes For Crosby Shackles

213 & 215 SHACKLES	
SHACKLE SIZE	COTTER PIN SIZE
1/4"	3/32 x 3/4"
5/16"	3/32 x 1"
3/8"	1/8 x 1"
7/16"	1/8 x 1"
1/2"	1/8 x 1"
5/8"	3/16 x 1 1/4"
3/4"	3/16 x 1 1/4"
7/8"	5/16 x 1 1/2"
1"	5/16 x 1 3/4"
1 1/8"	5/16 x 1 3/4"
1 1/4"	5/16 x 2"
1 3/8"	5/16" x 2 1/4"
1 1/2"	5/16" x 2 1/4"
1 3/4"	5/16" x 2 3/4"
2"	3/8 x 3"

2130 & 2150 SHACKLES	
SHACKLE SIZE	COTTER PIN SIZE
3/16"	3/32 x 3/4"
1/4"	3/32 x 3/4"
5/16"	3/32 x 1"
3/8"	1/8 x 1"
7/16"	1/8 x 1"
1/2"	1/8 x 1"
5/8"	3/16 x 1 1/4"
3/4"	3/16 x 1 1/4"
7/8"	1/4 x 1 1/2"
1"	1/4 x 1 3/4"
1 1/8"	1/4 x 1 3/4"
1 1/4"	1/4 x 2"
1 3/8"	5/16 x 2 1/4"
1 1/2"	5/16 x 2 1/4"
1 3/4"	5/16 x 2 3/4"
2"	3/8 x 3"
2 1/2"	7/16 x 4"
3"	3/8 x 4 1/2"
3 1/2"	3/8 x 4 1/2"
4"	3/8 x 4 1/2"

2140 SHACKLES	
SHACKLE SIZE	COTTER PIN SIZE
1 1/2"	5/16" x 2 1/4"
1 3/4"	5/16" x 2 3/4"
2"	3/8" x 3"
2 1/2"	7/16" x 4"
3"	3/8" x 4 1/2"
3 1/2"	3/8" x 4 1/2"
4"	3/8" x 4 1/2"
4 3/4"	3/8" x 7"
5"	3/8" x 8"
6"	3/8" x 8 1/2"
7"	3/8" x 10 1/2"
7 1/2"	3/8" x 10 1/2"
8"	3/8" x 13 1/2"

2160 SHACKLES	
SHACKLE WLL (t)	COTTER PIN SIZE
7	3/16" x 1 1/4"
12-1/2	1/4" x 1 3/4"
18	1/4" x 2"
30	5/16 x 2 1/4"
40	5/16" x 2 3/4"
55	3/8" x 3"
75	3/8" x 3"
125	3/8" x 4"
200	1/2" x 5 1/4"
300	5/8" x 6"
400	5/8" x 8"
500	3/4" x 9"
600	3/4" x 10"
700	3/4" x 11"
800	3/4 x 13" R3
900	3/4" x 13"
1000	3/4" x 14"
1250	3/4" x 15"
1500R3	3/4" x 17"
2000	5/8" x 18 3/4"
4"	3/8 x 4 1/2"

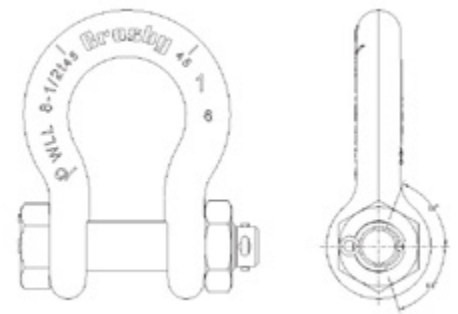


Figure 1
Cotter pin installation in a 1" bolt type shackle.

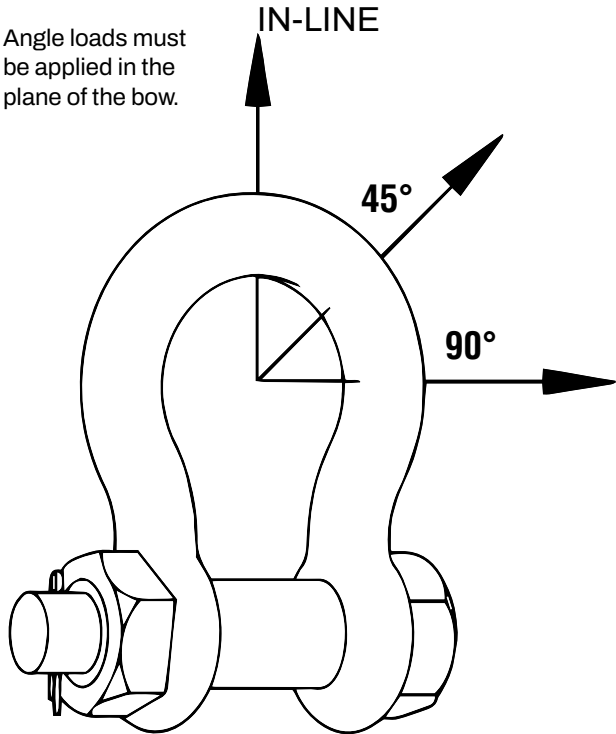
Application Information

Point Loading of Crosby® Shackles

It has been determined that all Crosby shackles can be point-to-point loaded to the Working Load Limit without bending of the pin/bolt. This loading can be bow-to-bow, bow-to-pin, or pin-to-pin (if there is not interference between the diameter of the shackle ears). However, caution should be given to maintain the load at the center of the span by spacers so the load will not slide over to one side, and overload that ear. See “Off Center Loading Of Crosby® Screw Pin & Bolt Type Shackles – 3/16” to 3” Sizes”.

Angular Loading Of Crosby® Screw Pin & Bolt Type Shackles

Crosby has made representative tests with smaller size shackles with the load applied at 90 degrees to the normal plane of loading (ie. in-line). The test results indicated that in order to maintain a proof load of 2 times the Working Load Limit (2 x WLL), the Working Load Limit should be reduced to 50% (ie. one-half the catalog working load rating). **DO NOT SIDE LOAD G/S-213 OR G/S-215 ROUND PIN SHACKLES.** Calculations based on the above test indicates the Working Load Limit should be reduced as shown below for loads applied at various angles to the normal plane of loading:



SIDE LOADED RATING REDUCTION
TABLE FOR 3/16” - 3” (120 METRIC TONS)

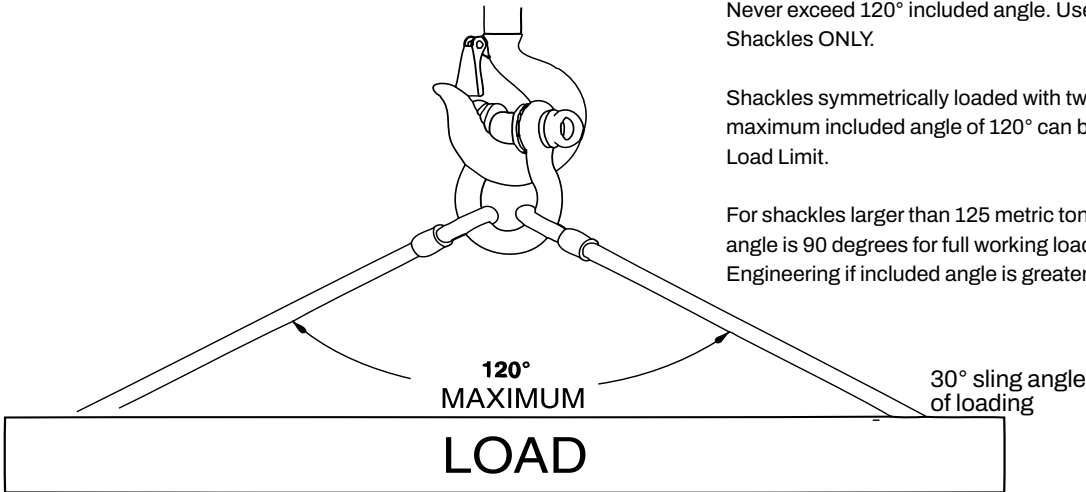
Table 1	
Side Loading Reduction Chart for Screw Pin and Bolt Type Shackles Only+	
Angle of Side Load from Vertical In-Line of Shackle	Adjusted Working Load Limit
0° - 10° In-Line*	100% of Rated Working Load Limit
11° - 20° from In-Line*	85% of Rated Working Load Limit
21° - 30° from In-Line*	75% of Rated Working Load Limit
31° - 45° from In-Line*	70% of Rated Working Load Limit
46° - 55° from In-Line*	60% of Rated Working Load Limit
56° - 70° from In-Line*	55% of Rated Working Load Limit
71° - 90° from In-Line*	50% of Rated Working Load Limit

+ In-Line load is applied perpendicular to pin. * DO NOT SIDE LOAD ROUND PIN SHACKLE.

Table 1	
SHACKLE SIZE GREATER THAN 3” ANGLE FROM IN-LINE (DEGREES) REDUCTION IN WLL	
0° - 5° In-Line*	0% of Rated Working Load Limit
6° - 10° from In-Line*	15% of Rated Working Load Limit
>10° from In-Line*	ANALYSIS REQ'D.

For shackles larger than 125 metric tons, where the angle of the side load is greater than 5 degrees, contact Crosby Engineering.

INCLUDED ANGLE - SHACKLES



Never exceed 120° included angle. Use Bolt Type and Screw Pin Shackles ONLY.

Shackles symmetrically loaded with two leg slings having a maximum included angle of 120° can be utilized to full Working Load Limit.

For shackles larger than 125 metric tons, the maximum included angle is 90 degrees for full working load limit. Contact Crosby Engineering if included angle is greater than 90 degrees.

For shackles larger than 125 metric tons, the maximum included angle is 90 degrees for full working load limit. Contact Crosby Engineering if included angle is greater than 90 degrees.

Application Information

Round Pin Shackles



Round Pin Shackles can be used in tie down, towing, suspension or lifting applications where the load is strictly applied in-line. Round pin shackles should never be used in rigging applications to gather multiple sling legs, or where side loading conditions may occur.

Bolt-Type Shackles



Bolt-Type Shackles can be used in any application where round pin or screw pin shackles are used. In addition, they are recommended for permanent or long term installations and where the load may slide on the shackle pin causing the pin to rotate. The bolt-type shackle's secondary securement system, utilizing a nut and cotter, eliminates the requirement to tighten nut before each lift or movement of load.

Screw Pin Shackles



Screw Pin Shackles are used in Pick and Place* applications. For permanent or long-term installations, Crosby recommends the use of bolt type shackles.

If you choose to disregard Crosby's recommendation, the screw pin shall be secured from rotation or loosening .

Screw pin shackles can be used for applications involving side-loading circumstances. Reduced working load limits are required for side-loading applications. While in service, do not allow the screw pin to be rotated by a live line, such as a choker application.

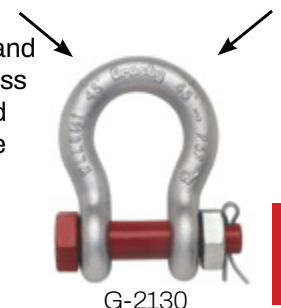
* Pick and Place application: Pick (move) a load and place as required. Tighten screw pin before each pick.

QUIC-CHECK® All Crosby Shackles, with the exception of 2160, 2169, 2170, 252 and 253 styles incorporate markings forged into the product that address an easy to use QUIC-CHECK® feature. Angle indicators are forged into the shackle bow at 45 degree** angles from vertical. These are utilized on screw pin and bolt type shackles to quickly check the



approximate angle of a two-legged hitch, or quickly check the angle of a single leg hitch when the shackle pin is secured and the pull of the load is off vertical (side loaded), thus requiring a reduction in the working load limit of the shackle.

** Round Pin Shackles utilize the 45 degree QUIC-CHECK® indicators to ensure load is applied strictly in-line.



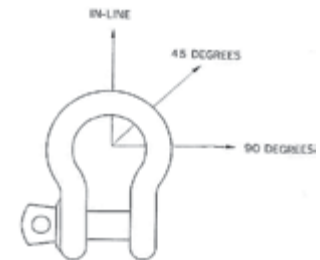
G-2130

Technical Information

2006/42/EC highlights the responsibility of the manufacturer, distributor and end user of lifting gear. Gunnebo Industries shackles are specified, monitored and documented in compliance with the most stringent requirements for the product concerned. A certified ISO 9001:2008 to 9001:2015 system is an evidence of our quality standard. See website or user instructions for assembly instructions. Meets listed current specifications and standards at time of publication of this catalog.

Instructions For Safe Use

1. The user is obliged to keep a valid Test Certificate for any shackle being used in a lifting operation.
2. Before use each shackle should be inspected to ensure that:
 - all markings in the body and the pin of the shackle are legible and in compliance with the relevant Test Certificate.
 - the shackle pin is of the correct type.
 - the body and pin are not distorted or unduly worn.
 - The body and pin are free from nicks, cracks, grooves and corrosion.
 - If there is any doubt with regards to the above criteria being met, the shackle should not be used for a lifting operation.
3. It is important to ensure that the pin is safely locked after assembly. For repeated lifting between inspections of the gear, it is recommended to use a safety bolt type shackle with nut and split-pin - the user must ensure that the split-pin is fitted, to prevent the nut from unscrewing during use.
4. Incorrect seating of a pin may be due to a bent pin, damaged threads or misalignment of the holes. Do not use the shackle under these circumstances, but refer the matter to a competent person (i.e. dealer, manufacturer)
5. Shackles should be fitted to the load in a manner that allows the shackle body to take the load in a straight line along its centerline to avoid undue bending stresses which will reduce the load capacity of the shackle. When using shackles in conjunction with multi-leg slings, due consideration should be given to the effect of the angle between the sling legs. When a shackle is used to secure the top block of a set of block and tackle the load on this shackle is increased by the value of the hoisting effect.
6. To avoid eccentric loading of the shackle it is recommended to center load on pin as far as possible over the total length of the pin or to use loose spacers.
7. Never modify, repair or reshape a shackle by welding, heating or bending as this will affect the nominal WLL.
8. Never heat treat a shackle as this may affect the WLL.



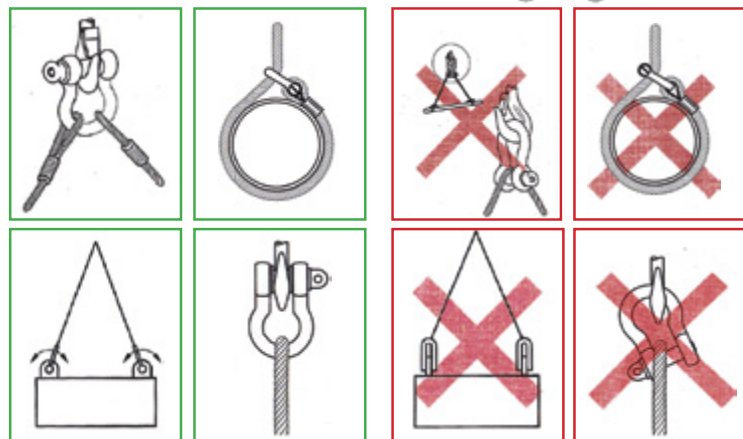
Side loads should be avoided as the products are not designed for this purpose. If side loads cannot be avoided, the following reduction factors must be taken into account:

Reduction for side loading

Load angle	New Working Load Limit
0°	100% of original WLL
45°	70% of original WLL
90°	50% of original WLL

Avoid applications where, due to load movement, the shackle pin can rotate

Shackle must be loaded in straight direction



Temperature

If extreme temperature situations are applicable, the following load reductions must be taken into account.

Reduction for elevated temperatures

Temperature:	New Working Load Limit
-20 - 200° C	100% of original Working Load Limit
200 - 300° C	90% of original Working Load Limit
300 - 400° C	75% of original Working Load Limit
> 400° C	not allowed



Product Catalog

Strength Through Innovation Since 1764



Feel confident in every situation

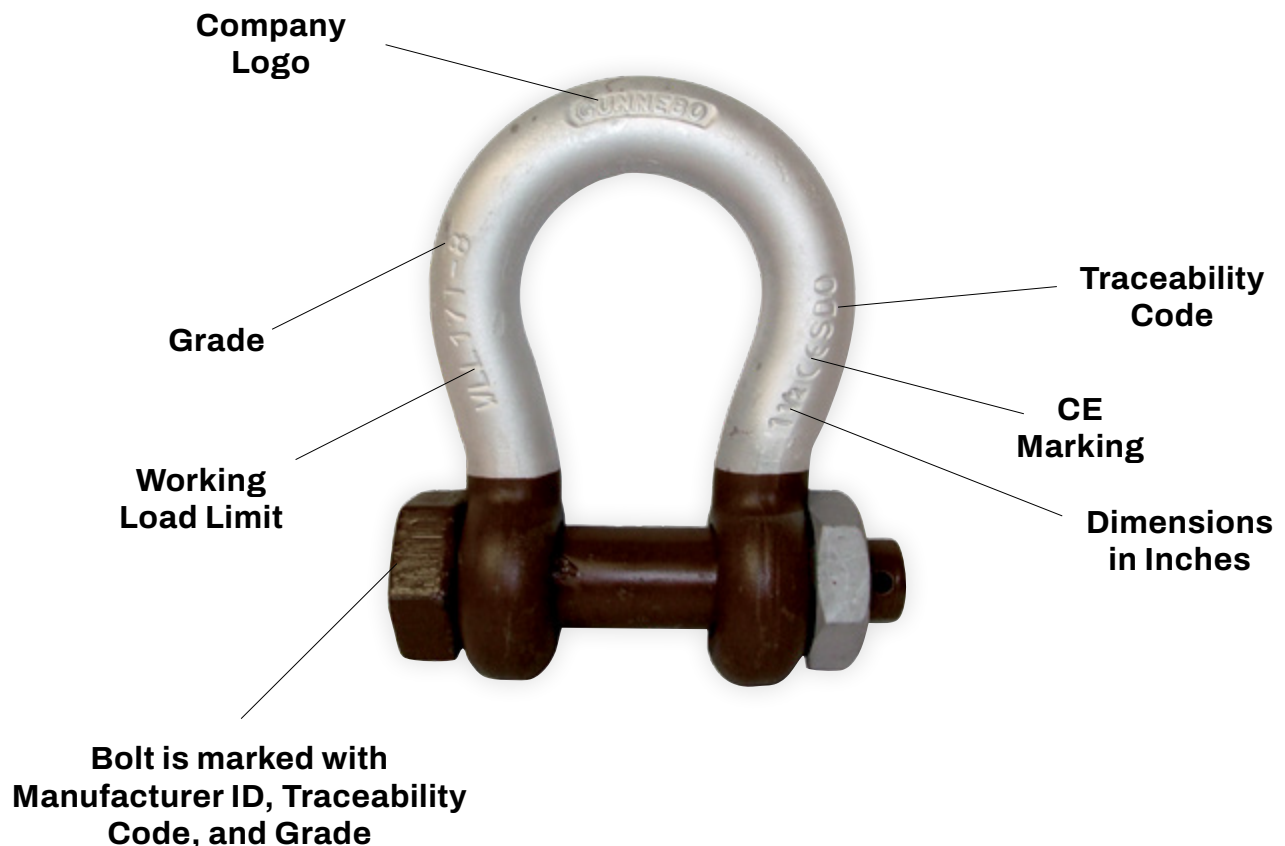
Gunnebo Industries shackles are made from a range of steel qualities, including acid proof stainless steel and high-grade alloy steel to comply with the most stringent specifications. Our factories comprise all facilities and systems for the manufacturing and control of a top-quality product. This includes tool design, an advanced tool shop, forging, heat treatment, machining, hot-dip galvanizing and quality control.

We offer a range of DNV 2.7-1 Type Approved lifting shackles for offshore containers, developed for the tough conditions of the offshore industry, where safety must be of the highest priority at all times. The heat treatment of these products ensures the proper ductility and strength to sustain shock loads which may be imposed when the container is lifted from the deck of a vessel.

Make sure you have the original

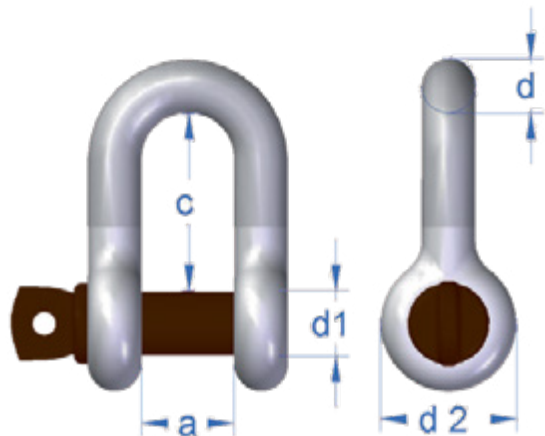
- High quality shackles acc. EN 13889 and US Fed. Spec RR-C. 271 (Grade A and Grade B)
- Consistent product quality
- Long experience of shackle production using modern manufacturing methods
- Local availability of expertise

To ensure you have a genuine Gunnebo Industries shackle, it should be marked as below:

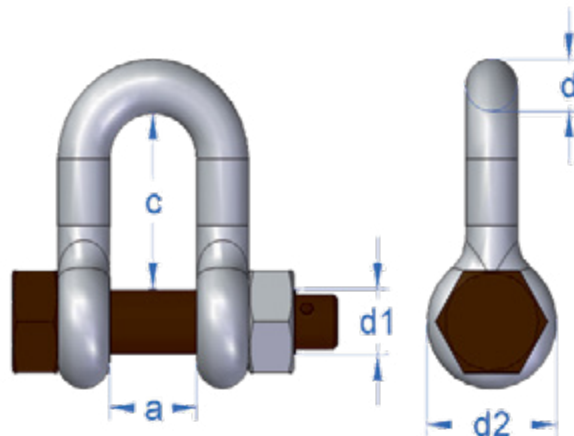


Dee Shackle No 834 and No 835

- Standard: DNV 2.7-1 Type Approved, EN 13889 and US Federal Spec. RR-C-271
- Material: High tensile carbon steel, Quenched & Tempered, Grade 6
- Finish: All parts hot-dip galvanized, pin brown painted on top of galvanized.
- Design Factor: 6:1
- Documentation: Test certificate and traceable raw material / inspection certificate acc. EN 10204 - 3.1.DNVGL-ST-E271-2.7-1 and E273-2.7-3 Type Approval Certification.
- Temperature: - 4°F to 392°F



Shackle No 834 with screw pin



Shackle No 835 with safety bolt



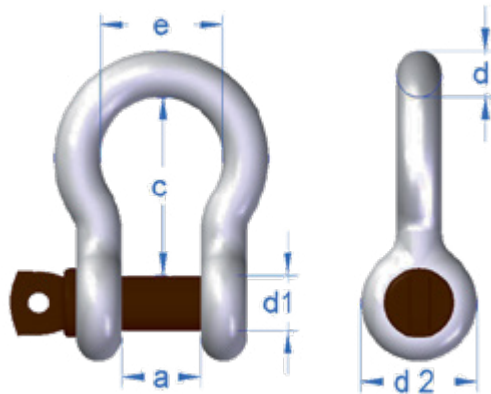
834 Screw Pin Stock No.	835 Safety Bolt Stock No.	WLL (t) 6:1	Pin d1 (in)	Nominal Size d		Inner Width a* (in)	Inner Length c* (in)	Eye Outer d2 (in)	834 Screw Pin Weight (lb)	835 Safety Bolt Weight (lb)
				(mm)	(in)					
A083416	A083516	3.25	0.75	16	5/8"	1.06	2.01	1.57	1.21	1.32
A083419	A083519	4.75	0.87	19	3/4"	1.22	2.36	1.89	2.20	2.43
A083422	A083522	6.5	0.98	22	7/8"	1.46	2.80	2.05	2.87	3.31
A083425	A083525	8.5	1.10	25	1"	1.69	3.19	2.36	4.19	4.85
A083428	A083528	9.5	1.26	28	1 1/8"	1.81	3.54	2.52	6.17	6.83
A083432	A083532	12.0	1.38	32	1 1/4"	2.05	3.94	2.83	7.937	9.26
A083435	A083535	13.5	1.50	35	1 3/8"	2.24	4.37	2.99	10.1	12.4
A083438	A083538	17.0	1.65	38	1 1/2"	2.36	4.80	3.31	14.3	16.5
A083445	A083545	25.0	1.97	45	1 3/4"	2.91	5.87	4.13	25.3	28.7

* Forging tolerance: +/- 5% on inside width/length.

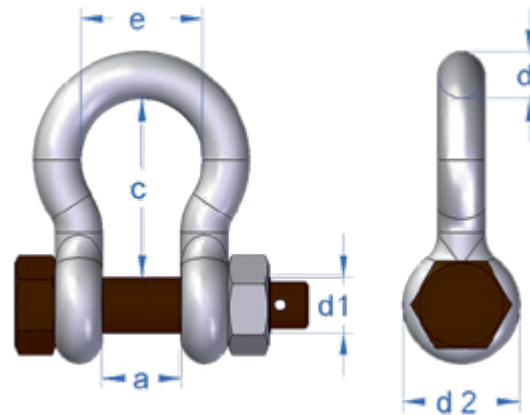
Split pin included

Bow Shackle No 854 and No 855

- Standard:** DNV 2.7-1 Type Approved, EN 13889 and US Federal Spec. RR-C-271
- Material:** High tensile carbon steel, Quenched & Tempered, Grade 6
- Finish:** All parts hot-dip galvanized, brown painted bolts on top of galvanized.
- Design Factor:** 6:1
- Documentation:** Test certificate and traceable raw material / inspection certificate acc. EN 10204 - 3.1.
DNVGL-ST-E271-2.7-1 and E273-2.7-3 Type Approval Certification.
- Temperature:** - 40°C to 200°C



Shackle No 854 with screw pin



Shackle No 855 with safety bolt



854 Screw Pin Stock No.	855 Safety Bolt Stock No.	WLL (t) 6:1	Pin d1 (in)	Nominal Size d		Inner Width a* (in)	Inner Length c* (in)	Bow Width e (in)	Eye Outer d2 (in)	854 Screw Pin Weight (lb)	855 Safety Bolt Weight (lb)
				(mm)	(in)						
A085413	A085513	2.0	0.63	13	1/2"	0.83	1.85	1.30	1.30	0.82	0.93
A085416	A085516	3.25	0.75	16	5/8"	1.06	2.36	1.65	1.57	1.43	1.54
A085419	A085519	4.75	0.87	19	3/4"	1.22	2.80	1.93	1.89	2.43	2.65
A085422	A085522	6.5	0.98	22	7/8"	1.46	3.31	2.36	2.05	3.31	3.75
A085425	A085525	8.5	1.10	25	1"	1.69	3.74	2.68	2.36	4.87	5.69
A085428	A085528	9.5	1.26	28	1 1/8"	1.81	4.25	2.91	2.52	6.83	7.50
A085432	A085532	12.0	1.38	32	1 1/4"	2.05	4.69	3.27	2.83	9.26	10.6
A085435	A085535	13.5	1.50	35	1 3/8"	2.24	5.20	3.50	2.99	13.2	15.4
A085438	A085538	17.0	1.65	38	1 1/2"	2.36	5.75	3.86	3.31	17.6	19.8
A085445	A085545	25.0	1.97	45	1 3/4"	2.91	7.01	5.00	4.13	29.7	33.0
A085452	A085552	35.0	2.24	50	2"	3.27	7.76	5.43	4.41	41.8	46.2
A085464	A085564	55.0	2.76	65	2 1/2"	4.13	10.2	7.09	5.71	83.7	85.9

* Forging tolerance: +/- 5% on inside width/length.

Split pin included

Arctic Shackle No 856

Bow shackle with safety bolt

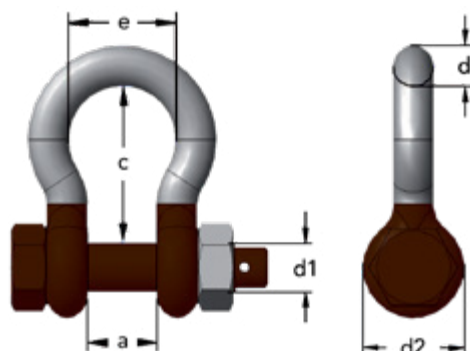


Unique benefits with the Arctic Shackle

Adverse weather and rough sea conditions in combination with extremely low temperatures, as often encountered for instance in the North Sea, places tough requirements on the products used. 856 Arctic shackles are specially designed for these conditions. The Arctic Shackle is type approved to DNV 2.7-1 Offshore containers and meets the impact requirements of 42 J at – 40 degrees °C.

The Arctic Shackle is a grade 8 shackle with all parts hot-dip galvanized, including the safety bolt, and has the characteristic brown color marking.

- Standard:** DNV 2.7-1, US Federal Spec. RR.C-271 and EN-13889
- Material:** Special alloy steel, Quenched & Tempered, Grade 8
- Finish:** All parts hot-dip galvanized + brown color marking
- Design Factor:** As specified in the table below
- Documentation:** Test certificate and traceable raw material / inspection certificate acc. EN 10204 - 3.1. DNVGL-ST-E271-2.7-1 and E273-2.7-3 Type Approval Certification.
- Temperature:** - 40°C to 200°C



Stock No.	WLL (t)	Design Factor	Pin d1 (in)	Nominal Size d		Inner Width a (in)	Inner Length c (in)	Eye Outer e (in)	Bow Width d2 (in)	Weight (lb)
				(mm)	(in)					
A085613	2.0	8.00	0.62	13	1/2"	0.82	1.85	1.29	1.29	0.92
A085616	3.25	8.00	0.74	16	5/8"	1.06	2.36	1.65	1.57	1.54
A085619	4.75	8.00	0.86	19	3/4"	1.22	2.79	1.92	1.88	2.64
A085622	6.5	7.85	0.98	22	7/8"	1.45	3.30	2.36	2.04	3.74
A085625	8.5	7.25	1.10	25	1"	1.69	3.74	2.67	2.36	5.51
A085628	9.5	6.94	1.25	28	1 1/8"	1.81	4.25	2.91	2.51	7.49
A085632	12.0	6.40	1.37	32	1 1/4"	2.04	4.68	3.26	2.83	10.5
A085635	13.5	6.10	1.49	35	1 3/8"	2.24	5.19	3.50	2.99	15.4
A085638	17.0	6.00	1.65	38	1 1/2"	2.36	5.74	3.85	3.30	19.8
A085645	25.0	6.00	1.96	45	1 3/4"	2.91	7.00	5.00	4.13	33.0
A085652	35.0	6.00	2.24	50	2"	3.26	7.75	5.43	4.56	46.2
A085664	55.0	6.00	2.75	65	2 1/2"	4.13	10.2	7.08	5.70	85.9

Split pin included

Super Shackle No 858

Bow shackle with safety bolt

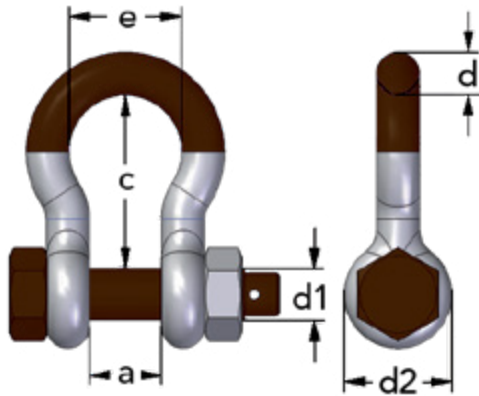


Unique Benefits with The Super Shackle

In certain situations a demand for extra Working Load Limit occurs in others the lifting environment has limited space for the lifting application. The 858 Super Shackle enables a higher working load limit for the same nominal size.

The Super shackle meets the US Federal Specification RR.C-271. It is a grade 8 shackle and has all parts hot dipped galvanized, including the safety bolt.

Standard:	US Federal Spec. RR.C-271 Type IVA Class 3, Grade B
Material:	High tensile steel. Quenched & Tempered, Grade 8
Finish:	All parts hot-dip galvanized + brown color marking
Design Factor:	5:1
Documentation:	Test certificate and traceable 3.1 certificate
Temperature:	-40°C to 200°C



Stock No.	WLL (t) 5:1	Pin d1 (in)	Nominal Size d		Inner Width a (in)	Inner Length c (in)	Bow Width e (in)	Eye Outer d2 (in)	Weight (lb)
			(mm)	(in)					
A085813	3.3	0.62	13	1/2"	0.82	2.00	1.29	1.29	0.88
A085816	5.0	0.74	16	5/8"	1.06	2.36	1.65	1.57	1.54
A085819	7.0	0.86	19	3/4"	1.22	2.79	1.92	1.88	2.64
A085822	9.5	0.98	22	7/8"	1.45	3.30	2.36	2.04	3.74
A085825	12.5	1.10	25	1"	1.69	3.74	2.67	2.36	5.51
A085828	15.0	1.25	28	1 1/8"	1.81	4.25	2.91	2.51	7.49
A085832	18.0	1.37	32	1 1/4"	2.04	4.68	3.26	2.83	10.5
A085835	21.0	1.49	35	1 3/8"	2.24	5.19	3.50	2.99	15.4
A085838	30.0	1.65	38	1 1/2"	2.36	5.74	3.85	3.30	19.4
A085845	40.0	1.96	45	1 3/4"	2.91	7.00	5.00	4.13	33.0

Split pin included

GREEN PIN[®]

PRODUCT CATALOGUE

IMPERIAL



INTRODUCTION

General

In case you do not use the products yourself but are reselling these as part of a manufactured product, please take our general cautions and warnings into account and make these known to your customers as well. In any case, we do not accept any responsibility or liability, nor can we be held responsible for any misuse or damage with, by or at your customers due to negligent use.

Definitions

Material

Various raw materials are used for the production of shackles, hooks and other lifting devices, depending on the use of the finished product. The following raw materials may be used:

- Mild steel, untreated, grade 30;
- High tensile steel, untreated or normalized, grade 40;
- High tensile steel, quenched and tempered, grade 60;
- Alloy steel, quenched and tempered, grade 80;
- Alloy steel, quenched and tempered, grade 100;
- Stainless steel AISI316L, AISI316 or Duplex 1.4462, grade 50.

Load

Following terms are used to define a load:

- Working Load Limit or WLL: the maximum load the product is designed to sustain, in general use and in in-line lifting.
- Proof Load or PL: this is the load applied on proof testing the product. At this load the product may not show visual deformation. For information about the proof load applied, we refer to the separate paragraph on testing.
- Minimum Breaking Load or MBL: the minimum load at which the product may fail or no longer sustain the load. Where applicable the MBL is specified.
- Shock Load: a sudden impact of the load on the lifting product. Shock loads are to be avoided at all times since they increase the stress on the product significantly and may affect its product life.

The unit that is used in this catalogue to indicate WLL, PL and MBL is t, which stands for metric tonne.

Safety factor

This indicates the ratio between the MBL and the WLL. For example, the standard range of Green Pin® shackles has a the safety factor of 6 : 1. This means that the shackle may only fail to retain the load, when the load is in exceedance of at least 6 times its designed WLL. Green Pin® chain fittings generally have a safety factor of 4 : 1 (lifting eyes have a safety factor of 5 : 1).

Product dimensions

All product dimensions mentioned in this catalogue are nominal dimensions. Product design, materials and/or specifications may be changed without prior notification. You can find the most up to date information including CAD-drawings on greenpin.com/products.

Finish

Products can have the following finish:

- Self-coloured: the product is delivered in the condition as it has been forged or machined and has undergone no specific finish treatment.
- Electro-galvanized: the finished product is electro-galvanized according to the customary standards. The thickness of the galvanization is at least 5 µm.
- Hot dipped galvanized: the finished product is hot dipped galvanized according to the customary standards. The thickness of the coating is at least 70 µm.
- Painted: the finished product is painted in a specific colour.
- Polished: stainless steel products are polished.

Standard

These refer to the specific standards indicated for the product.

Temperature range

This indicates the temperature range at which the product can be used. Beyond the advised temperature range the WLL of a product may be affected.

Abbreviations

The following abbreviations are used in this catalogue:

Product class abbreviations (for example, G-4161)

C Carbon steel
A Alloy steel
R Stainless steel
S Self-coloured
P Painted
E Electro-galvanized
G Hot dipped galvanized

Product name abbreviations (for example, Green Pin® Bow Shackle BN)

BN	Bolt & Nut or safety bolt	HH	Hook-Hook
CL	Clevis	HK	Hook
CP	Cotter Pin	H-type	Horizontal
D	D-Handle	JJ	Jaw-Jaw
E	Eye	ROV	Remotely Operated Vehicle
EE	Eye-Eye	RT	Recessed Trigger
EJ	Eye-Jaw	S	Shackle
EH	Eye-Hook	S/S	Stainless Steel
F	Fishtail handle	SC	Screw Collar or Screw Pin
FN	Fixed Nut	SCL	Swivel Clevis
FP	Flush Pin	SE	Swivel Eye
GR5	Grade 50	SQ	Square headed Screw Pin
GR8	Grade 80	U-type	Universal (Horizontal and Vertical)
GR10	Grade 100	V-type	Vertical

Certificates

Our company is ISO certified by Lloyd's; currently we are ISO 9001-2015 certified. Depending on the type of product and certificate availability for a certain product, below mentioned certificates can be provided.

Type 2.1	2.1	Works certificate to EN 10204 Statement of compliance with the order.
Type 2.2	2.2	Works certificate to EN 10204 Statement of compliance with the order, stating the results of non-specific inspection.
Type 3.1	3.1	Inspection certificate to EN 10204 Statement of compliance with the order, stating the results of material specific inspection. This includes chemical composition and mechanical properties at component level.
Type MTC a	MTC ^a	Manufacturer test certificate Statement of compliance with the order, stating the results of proof load testing samples of a production batch. Products are not individually tested.
Type MTC b	MTC ^b	Manufacturer test certificate Statement of compliance with the order, stating the results of individual proof load testing.
Type LROS	LROS	Proofload Statement Statement of witness of proof load testing and visual examination by a surveyor from Lloyds Register, stating the results of individual proof load testing.
Type DNV PL	DNV PL	Proofload Statement Statement of witness of proof load testing and visual examination by a surveyor from DNV, stating the results of individual proof load testing. For P-6043, these shackles are proof load tested with presence of a DNV surveyor.
Type MPI a	MPI ^a	Non-destructive testing report Statement of compliance with the order, stating the results of Magnetic Particle Inspection (M.P.I.) in accordance with EN 10228-1 on samples of a production batch. Products are not individually tested.
Type MPI b	MPI ^b	Non-destructive testing report Statement of compliance with the order, stating the results of individual Magnetic Particle Inspection (M.P.I.) in accordance with EN 10228-1.

Table continues on next page

SHACKLES



Applications

Shackles are used in lifting operations and static systems as removable links to connect (steel) wire rope, chain and other fittings. Screw pin shackles are used mainly for non-permanent applications. Safety bolt and fixed nut shackles are used for long-term or permanent applications or where the load may slide on the pin causing rotation of the pin. Chain- or dee shackles are mainly used on one-leg systems whereas anchor- or bow shackles are mainly used on multi-leg systems.

Range

Green Pin® offers a wide range of bow and dee shackles for a variety of applications. The range stretches from WLL 0.33 t to 3000 t. This provides our customers with a very extensive range to choose a shackle that suits their application best. Most of the shackles are directly available from stock. Furthermore, shackles can be supplied to many standards such as the US Federal Specification RR-C-271, EN 13889, ISO 2415, British Standard 3032, DIN 82101 etc. Additionally we offer a wide range of general commercial shackles, which are not suitable for lifting but merely for fixing purposes. Van Beest offers a wide range of other shackles to complement the Green Pin® assortment.

Design

All Green Pin® shackles have a specific design for a specific application. Please find below some examples of highly functional designs, to optimize the use of the Green Pin® shackles in daily use:

- Green Pin Super® Shackles which are made from grade 80 steel. They are designed to be used in confined spaces. The higher material strength is used to reduce the physical dimensions of the product whilst maintaining its WLL and functionality;
- Green Pin Polar® Shackles are for use in extreme climatic conditions with material properties generally guaranteed up to temperatures of -76 °F;
- Green Pin Power Sling® Shackles are designed to provide a better bending efficiency for the sling. A larger radius increases the life span of the sling significantly;
- Another example of a functional design is a shackle pin with a square sunken hole. Because of the flat head there is less risk of the shackle getting caught in a net or a line.

How to recognize a genuine Green Pin® shackle?

- Marking in the shackle body and pin
(from dia 1/2 inch and up)
- Steel grade (for example 6)
- CE conformity code
- Diameter bow (inch)
- Green Pin® logo (GP)
- Holland marking
- Working Load Limit (WLL in metric tonnes)
- Traceability code (2 letters)
- Green powder coated pin



Green Pin® Shackles meet all requirements of the Machinery Directive 2006/42/EC and its latest amendments.

Finish

Shackles supplied by Green Pin® are either hot dipped galvanized, electro-galvanized, painted or self-coloured, depending on the type of shackle and its application. You can find the finish of each type of shackle in the product section further on.



Certification

Upon request at time of order, all load rated shackles can be supplied with any of the following documents or certificates:

Free of charge:

2.1 2.2 3.1 MTC^a DNV 2.7-1^a DNV 2.7-1^b DNV 0377 DNV 0378 DNV NS 9415 CE IIA CE IIB ABS PDA ABS MA

With additional charges:

MTC^a MPI^a MPI^b US^a US^b DNV CG3 BL

On request the proof load test certificates can be supplied surveyed by an official classification society, such as LROS, DNV, BV, ABS or any other officially certified inspection body. Please verify your certification requirements with Green Pin® at the time of order.

Green Pin® Bow Shackles, Green Pin® Dee Shackles and Green Pin Polar® Shackles are DNV type approved. These shackles carry two DNV type approval certificates that show compliance with:

- DNV-ST-E271-2.71 Offshore Containers
- EN 12079-2 Offshore containers and associated lifting sets
- EN 13889 Forged steel shackles for general lifting purposes
- IMO/MSC Circular 860
- US Federal Specification RR-C-271
- DNV ST-E273 Portable Offshore Units
- DNV Standard No. 0378 Offshore and Platform Lifting Appliances
- ASME B30.26
- ISO 2415

The Green Pin Power Sling® Shackles are DNV type approved. This DNV type approval certificate is in compliance with:

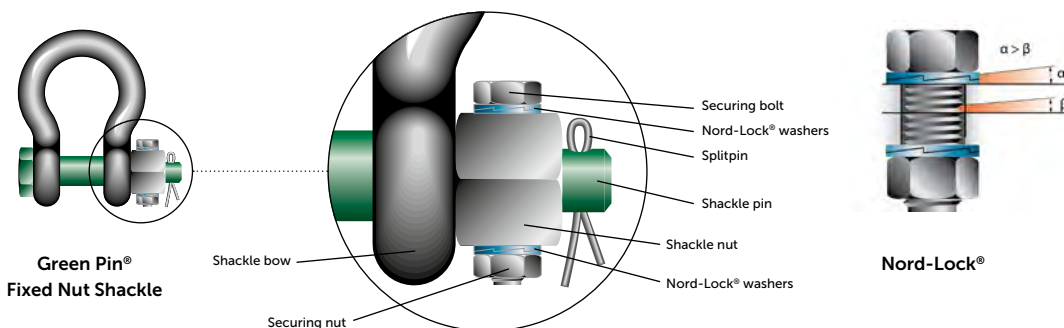
- DNV Standard for Certification No. 0377 Standard for Shipboard Lifting Appliances
- DNV Standard for Certification No. 0378 Offshore and Platform Lifting Appliances

Green Pin® Shackles G-4161, G-4163, G-4151, G-4153, G-5163, G-5261 and G-5263 are ABS Type Approved. The shackles have a Product Design Assessment Approval and a Manufacturer Assessment Approval Certificate. The shackles are type approved to be used as lifting gear or to be used as lifting device.

Type approval certificates can be found on greenpin.com.

Fixed Nut Shackles

Shackles can also be used in more permanent constructions. These can be subject to dynamic loads and/or extreme vibrations. In such applications there is a risk that, over time, the nut may start to move over the thread. We offer our range of Green Pin® Fixed Nut Shackles to avoid this risk. All Green Pin® shackles with bolt and nut can be equipped with an extra AISI 316 securing bolt that is assembled through the nut and shackle pin. This securing bolt is fastened with two sets of Nord-Lock® washers and a securing nut. This will keep the shackle nut in position. The Nord-Lock® wedge-locking washers lock when subjected to extreme vibration or dynamic loads.

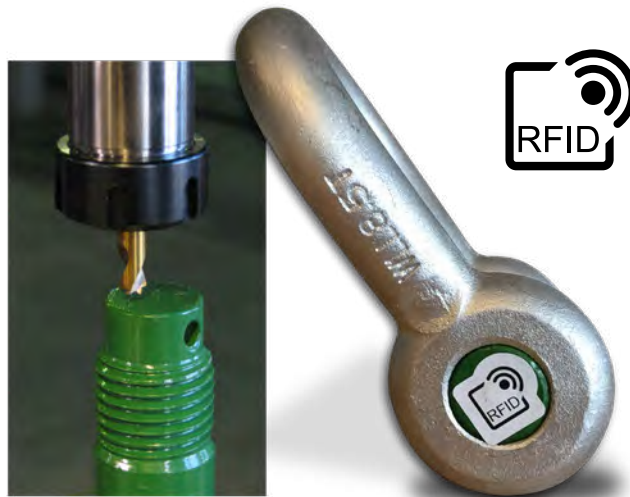


Green Pin® Shackles with RFID

All lifting equipment requires regular inspection. Tracking and filing reports on paper can be a time consuming task. Green Pin® offers a solution with an easily accessible RFID (Radio Frequency Identification) chip in our range of Green Pin® Shackles. This RFID chip responds to a radio-signal that is transmitted by a reader. Each chip has a unique number and this number links the individual shackle to a record in an inspection management system. The chips are impact resistant and durable and they are countersunk into the end of the shackle pin. The chips are NFC (Near Field Communication) compatible, allowing users to scan, identify and track the shackles with the latest generation of NFC compatible smartphones.

Green Pin® offers the option of RFID implementation in all Green Pin® shackles. For detailed technical information please go to www.greenpin.com/FAQ.

- **RF Protocol :** ISO 15693
- **Operating Frequency :** HF – 13.56 MHz



Testing

Generally load rated products are Proofload tested, and certificates can be supplied upon request. For specific information on certificates we refer to the separate paragraph on certification.

Green Pin® shackles are Proofload tested (bow-pin configuration) at the following loads:

working load limit	Green Pin® Bow or Dee Shackles Polar® Shackles Heavy Duty Shackles BigMouth® Bow Shackle proof load	Green Pin Super® Shackles proof load	Green Pin® Sling Shackles proof load	Green Pin Power Sling® Shackle proof load	Green Pin® Web Sling Shackle proof load	Green Pin BigMouth® Dee Shackle proof load
metric t	metric t	metric t	metric t	metric t	metric t	metric t
0.33	0.66					
0.5	1					
0.75	1.5					
1	2					
1.5	3					
2	4					
3.25	6.5				8.13	
3.3		6.6				
4.6						9.2
4.75	9.5				11.88	
5		10				
6.5	13				16.25	
7		14	14			
8.5	17				21.25	
8.6						17.2
9.5	19	19				
12	24					
12.5		25	25			
13.5	27					
15		30				
15.5						31
16	32					
17	34					
18		36	36			
21		42				
25	50					
30	60	60	60			
35	70					
40		80	80			
42.5	85					
55	110	110	110			
75	150		150			
85	170	170				
120	240	240				
125			250	250		
150	300	300	300	300		
175		350				
200	400		300	400		
250	500		375	500		
300	600		450	600		
400	600		532	800		
500	750		750	1000		
600	900		900	900		
700	1050		931	1050		
800	1200		1064	1200		
900	1350		1350	1350		
1000	1500		1330	1500		
1250	1875		1663	1875		
1500	2250					
1550			2061.5	2325		

Instructions for use

Select the correct type and WLL of the shackle for the particular application. If extreme circumstances or shock loading may occur, this must be taken into account when selecting the correct shackle. Please note that commercial shackles are not to be used for lifting applications.

Shackles should be inspected before use to ensure that:

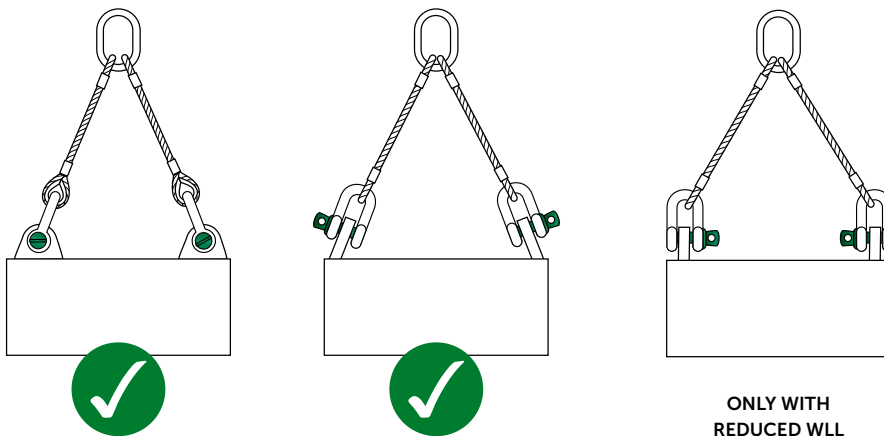
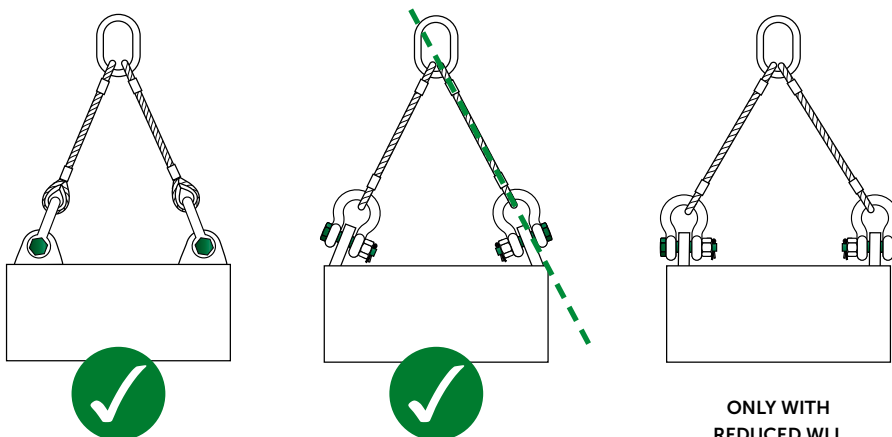
- all markings are legible;
- the body and pin are both of the same brand and type;
- the body and pin are both of the correct size;
- never use a safety bolt type shackle without using a securing pin;
- the pin, nut, cotter pin, or any other locking system cannot vibrate out of position;
- the threads of the pin and the body are undamaged;
- the body and the pin are not distorted or unduly worn;
- the body and pin are free from nicks, gouges, cracks and corrosion;
- shackles may not be heat treated as this may affect their WLL;
- never modify, repair or reshape a shackle by machining, welding, heating or bending as this will affect the WLL.

Assembly

Ensure that the pin is correctly screwed into the shackle eye: tighten it hand-tight, then secure it using a wrench or other suitable tool so that the collar of the pin is fully seated against the shackle eye. Ensure that the pin is of the correct length so that it penetrates the full depth of the threaded eye and the collar of the pin touches the surface of the shackle eye.

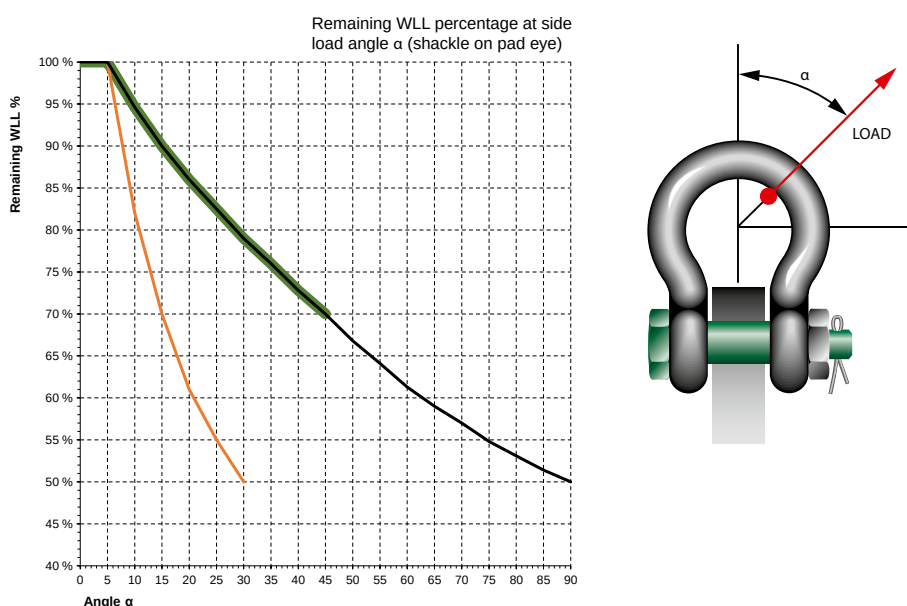
Incorrect positioning of the pin may be caused by a bent pin, too tight fitting thread or misalignment of the pin holes. Do not use the shackle under these circumstances. Never replace a shackle pin except with one of the same brand, type, make and size to ensure the shackle maintains its original WLL.

Make sure that the shackle is supporting the load correctly, i.e. along the axis of the shackle body centerline. Avoid bending loads, unstable loads and overloads.



Side loads

Side loads should be avoided, as the products are not designed for this purpose. If side loads cannot be avoided, the WLL of the shackle must be reduced:

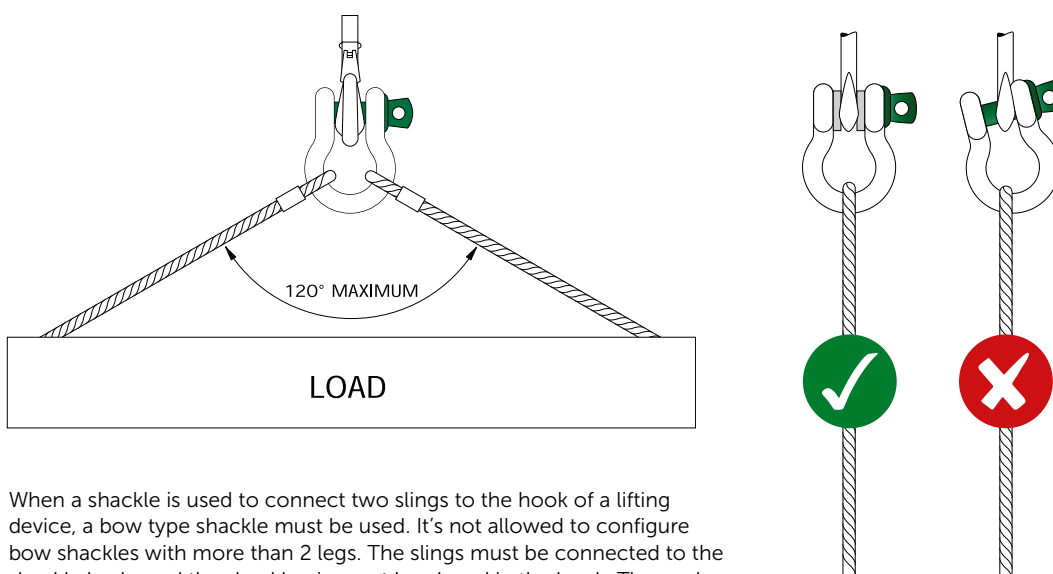


This black curve is valid for almost all Green Pin® shackles, except for ROV Shackles (P-5363 and P-5367) which are for in-line use only. The green curve is valid for Green Pin® Sling Shackles (P-6033, P-6013 and P-6065) and the orange curve is valid for the Green Pin Power Sling® Shackles (P-6043).

In-line lifting is considered to be a load perpendicular to the pin axis and in the plane of the bow. The load angles in the graph represent the deviating angles from in-line loading.

Maximum loading angle

When connecting shackles to multi-leg slings, consider the effect of the angle between the legs of the sling. As the angle increases, so does the load in the sling leg and consequently in any shackle attached to that leg.



When a shackle is used to connect two slings to the hook of a lifting device, a bow type shackle must be used. It's not allowed to configure bow shackles with more than 2 legs. The slings must be connected to the shackle body, and the shackle pin must be placed in the hook. The angle between the slings should not exceed 120°. If symmetrically loaded the shackle may be used to the full WLL.

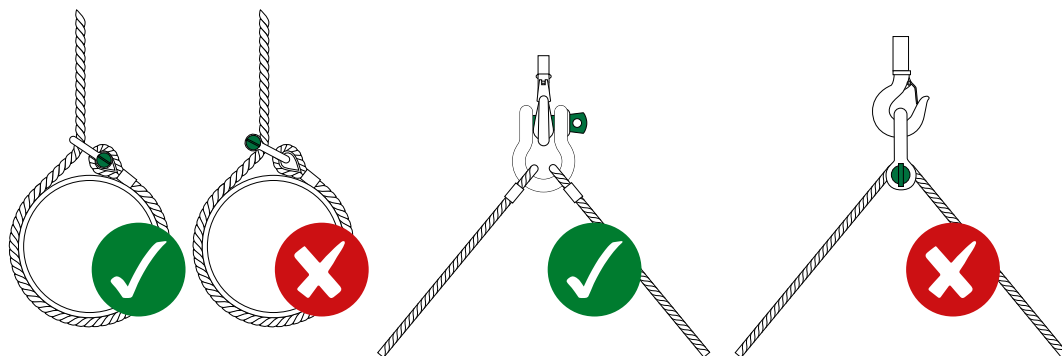
Avoiding eccentric loading

To avoid eccentric loading of the shackle a loose spacer may be used on either end of the shackle pin. Do not reduce the width between the shackle jaws by welding washers or spacers to the inside of the shackle eyes or by narrowing the jaws, as this will affect the WLL of the shackle.

When a shackle is attached to the top block of a set of wire rope blocks the load on this shackle is increased by the value of the hoisting effect.

Avoiding pin rotation

Avoid applications where the load moves over the shackle pin; the pin may rotate and possibly be unscrewed. If moving of the load cannot be avoided, or when the shackle is to be left in place for a prolonged period or where maximum pin security is required, use a shackle with a safety bolt, nut and cotter pin or a shackle with a fixed nut.

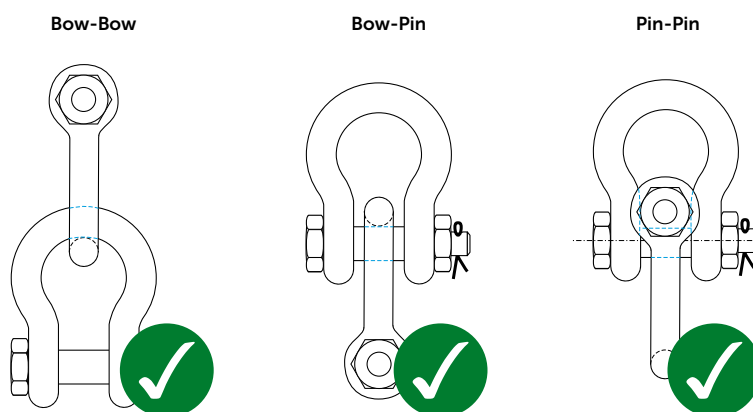


Chemicals

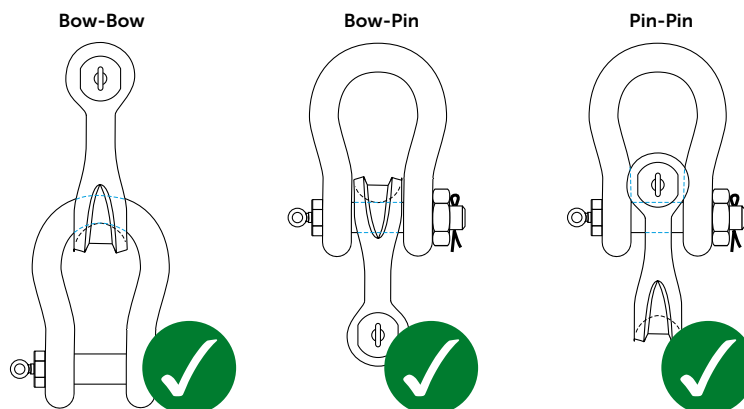
Shackles should not be immersed in acidic solutions or exposed to acidic fumes or other chemicals that are potentially harmful to the shackle.

Point loading

Shackles are used in lifting- and static systems as removable links to connect (steel) wire rope, chain and other fittings. Most of the times the load bearing component that connects to a shackle is of a rounded shape. Point loading of shackles during lifting operations is allowed. The maximum load of the configuration is limited by the component with the lowest WLL. Increasing the contact area by using bigger diameters and/or pad eyes can be an advantage. Sharp edges shall be avoided. Green Pin® shackles can also be used in below configurations. The maximum load of the configuration is limited by the component with the lowest WLL.

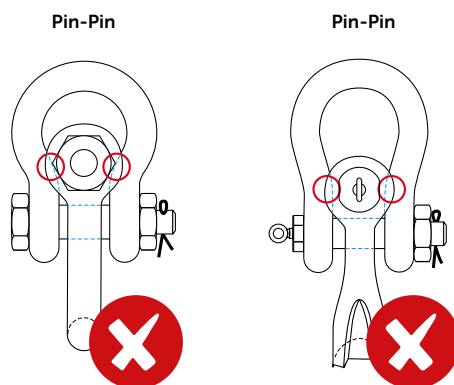


The crown of a Green Pin® Sling Shackle (P-6033) is wider than that of a standard shackle, thus creating a larger bearing surface. This improves the lifetime of the sling. Green Pin® Sling shackles can also be used in below configurations. The maximum load of the configuration is limited by the component with the lowest WLL. For information about point loading of the Green Pin Power Sling® Shackle (P-6043) please contact sales@vanbeest.eu.



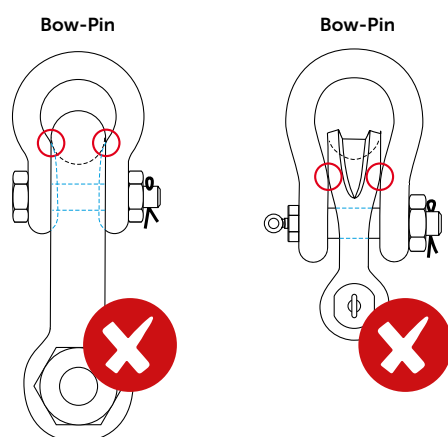
Pin- Pin configuration

When the shackle eyes touch and the pins do not bear properly, the configuration shall not be used.



Bow- Pin configuration

When the shackle body of the inner shackle touches the shackle eyes of the outer shackle and body and pin do not bear properly, the configuration shall not be used.



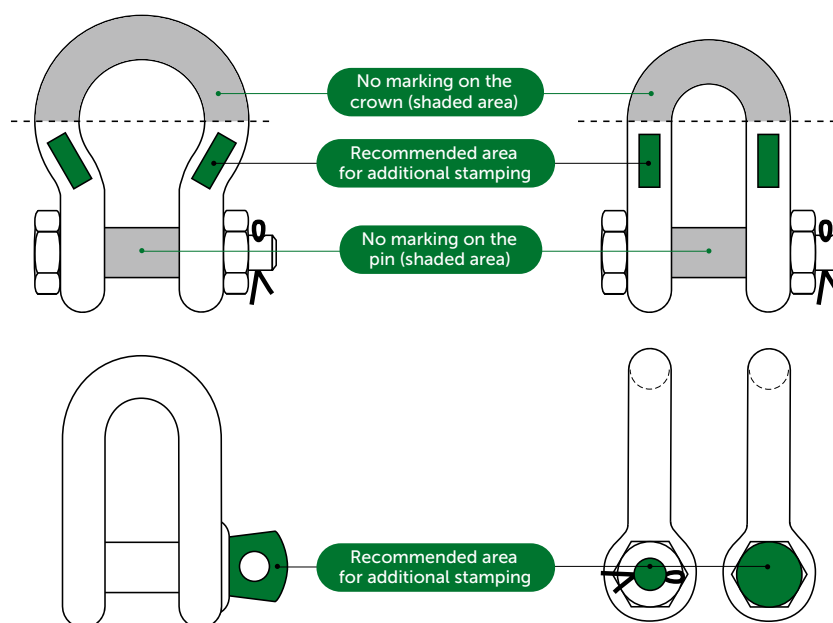
Contact Royal Van Beest, to check if a certain configuration is possible.

Additional markings

It is possible to add additional stamping on Green Pin® shackles, but make sure that you follow the recommendations below. If the recommendations are followed, the performance of the shackles is guaranteed.

- permanent identification marks or symbols are to be made by dot peen marking or with stamps having rounded profiles (low-stress stamps);
- laser markings are allowed as long as the heat of the laser does not influence in a negative way the material structure and properties. The laser marking must be legibly and indelibly marked in a place where the markings will not be removed by use;
- the number of marks on a shackle is to be kept to a minimum;
- the use of fractions and oblique strokes is to be avoided and a dot or hyphen is preferable to a dividing line;
- values of WLL are, generally, to be marked to one place of decimals (except for 0.25 and 0.75) up to 100 t and in integers thereafter. The word "tonnes" may be abbreviated to "t";
- recommended sizes of marks are
 - Diameter of the part to be marked > recommended size of the mark;
 - less than $\frac{1}{2}$ inch > $\frac{1}{8}$ inch;
 - $\frac{1}{2}$ inch to $1\frac{1}{32}$ inch > $\frac{3}{16}$ inch;
 - over $1\frac{1}{32}$ inch > $\frac{1}{4}$ inch.

Typical arrangements of marks can be found in the following illustrations.



Temperature

If extreme temperature situations occur, the following load reductions must be taken into account:

Temperature	Reduction for elevated temperatures New Working Load Limit
up to 392 °F	100 % of original Working Load Limit
392 - 572 °F	90 % of original Working Load Limit
572 - 752 °F	75 % of original Working Load Limit
> 752 °F	not allowed

The rating of shackles to EN 13889 assumes the absence of exceptionally hazardous conditions. Exceptionally hazardous conditions include offshore activities, the lifting of persons and the lifting of potentially dangerous loads such as molten metals, corrosive materials or fissile materials. In such cases a competent person should assess the degree of hazard and the WLL should be reduced accordingly.

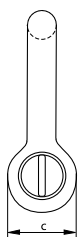
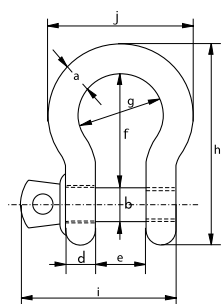
Inspection

Shackles must be regularly inspected in accordance with the safety standards and regulations given in the country of use. This is required because the products in use may be affected by wear, misuse, overloading etc. which may lead to deformation and alteration of the material structure. Inspection should take place at least every six months (follow the local rules in the country of use) and more frequently when the shackles are used in severe operating conditions.





G-4161



Green Pin® Bow Shackle SC

Standard bow shackle with screw collar pin

- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 2, grade A, from 2 t and upward these shackles comply with ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 0378 CE IIA ABS PDA ABS MA
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	lbs
0.33	³ / ₁₆	¹ / ₄	¹⁵ / ₃₂	³ / ₁₆	³ / ₈	⁷ / ₈	⁵ / ₈	1 ¹³ / ₃₂	1 ⁵ / ₃₂	1 ¹ / ₃₂	0.04
0.5	¹ / ₄	⁵ / ₁₆	⁵ / ₈	⁹ / ₃₂	¹⁵ / ₃₂	1 ⁵ / ₃₂	25/32	1 ⁷ / ₈	1 ¹ / ₂	1 ¹¹ / ₃₂	0.11
0.75	⁵ / ₁₆	³ / ₈	²⁵ / ₃₂	¹¹ / ₃₂	¹⁷ / ₃₂	1 ¹ / ₄	7/8	2 ⁷ / ₃₂	1 ²⁷ / ₃₂	1 ⁹ / ₁₆	0.22
1	³ / ₈	⁷ / ₁₆	²⁹ / ₃₂	¹³ / ₃₂	²¹ / ₃₂	1 ¹⁵ / ₃₂	1 ¹ / ₃₂	2 ¹⁷ / ₃₂	2 ¹ / ₈	1 ¹³ / ₁₆	0.31
1.5	⁷ / ₁₆	¹ / ₂	³¹ / ₃₂	⁷ / ₁₆	³ / ₄	1 ¹¹ / ₁₆	1 ⁵ / ₃₂	2 ⁷ / ₈	2 ¹¹ / ₃₂	2	0.42
2	¹ / ₂	⁵ / ₈	1 ¹¹ / ₃₂	¹ / ₂	⁷ / ₈	2	1 ¹ / ₄	3 ¹⁷ / ₃₂	2 ⁷ / ₈	2 ⁵ / ₁₆	0.79
3.25	⁵ / ₈	³ / ₄	1 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₁₆	2 ¹⁷ / ₃₂	1 ¹¹ / ₁₆	4 ¹¹ / ₃₂	3 ¹ / ₂	2 ¹⁵ / ₁₆	1.39
4.75	³ / ₄	⁷ / ₈	1 ¹³ / ₁₆	³ / ₄	1 ⁷ / ₃₂	3	2	5 ³ / ₃₂	4 ¹ / ₁₆	3 ¹ / ₂	2.23
6.5	⁷ / ₈	1	2 ¹ / ₁₆	⁷ / ₈	1 ¹³ / ₃₂	3 ⁹ / ₃₂	2 ⁹ / ₃₂	5 ²¹ / ₃₂	4 ¹¹ / ₁₆	4 ¹ / ₃₂	3.31
8.5	1	1 ¹ / ₈	2 ⁵ / ₁₆	³¹ / ₃₂	1 ¹¹ / ₁₆	3 ³ / ₄	2 ¹¹ / ₁₆	6 ¹⁵ / ₃₂	5 ¹³ / ₃₂	4 ²¹ / ₃₂	4.87
9.5	1 ¹ / ₈	1 ¹ / ₄	2 ⁵ / ₈	1 ³ / ₃₂	1 ²⁷ / ₃₂	4 ¹ / ₄	2 ¹⁵ / ₁₆	7 ⁵ / ₁₆	6 ¹ / ₃₂	5 ⁵ / ₃₂	6.97
12	1 ¹ / ₄	1 ³ / ₈	2 ⁷ / ₈	1 ¹ / ₄	2	4 ¹⁷ / ₃₂	3 ⁹ / ₃₂	7 ²⁹ / ₃₂	6 ¹¹ / ₁₆	5 ²⁵ / ₃₂	9.5
13.5	1 ³ / ₈	1 ¹ / ₂	3 ¹ / ₈	1 ³ / ₈	2 ¹ / ₄	5 ¹ / ₄	3 ⁵ / ₈	8 ¹⁵ / ₁₆	7 ⁵ / ₁₆	6 ³ / ₈	12.3
17	1 ¹ / ₂	1 ⁵ / ₈	3 ¹⁵ / ₃₂	1 ¹ / ₂	2 ³ / ₈	5 ³ / ₄	3 ²⁹ / ₃₂	9 ¹³ / ₁₆	8	6 ⁷ / ₈	16.38
25	1 ³ / ₄	2	4 ³ / ₃₂	1 ²⁵ / ₃₂	2 ²⁹ / ₃₂	7	4 ³¹ / ₃₂	11 ¹³ / ₁₆	9 ⁹ / ₁₆	8 ¹ / ₂	27.6
35	2	2 ¹ / ₄	4 ¹³ / ₃₂	1 ³¹ / ₃₂	3 ⁹ / ₃₂	7 ³ / ₄	5 ⁷ / ₁₆	13 ¹ / ₁₆	10 ²³ / ₃₂	9 ³ / ₈	37.9
42.5	2 ¹ / ₄	2 ⁹ / ₁₆	5 ³ / ₁₆	2 ¹ / ₄	3 ³ / ₄	8 ³ / ₄	6 ⁵ / ₁₆	14 ⁷ / ₈	12 ⁷ / ₃₂	10 ²⁵ / ₃₂	58
55	2 ¹ / ₂	2 ³ / ₄	5 ²³ / ₃₂	2 ⁹ / ₁₆	4 ¹ / ₈	10 ¹ / ₄	7 ³ / ₃₂	17 ¹ / ₁₆	13 ¹⁷ / ₃₂	12 ⁷ / ₃₂	83

CAD RFID INFO

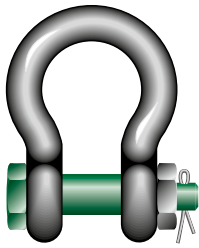




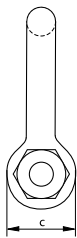
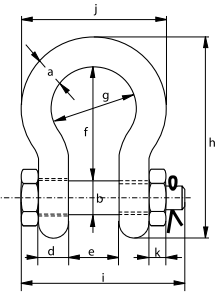
Green Pin® Bow Shackle BN

Standard bow shackle with safety bolt

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G-4163



- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade A, from 2 t and upward these shackles comply with ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 2.7-1^a * DNV 2.7-1^b * DNV 0378 CE IIA ABS PDA ABS MA
- **Article code:** scan QR code to see article codes

working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs
0.5	1/4	5/16	5/8	9/32	15/32	1 5/32	25/32	1 7/8	1 21/32	1 11/32	5/32	0.13
0.75	5/16	3/8	25/32	11/32	17/32	1 1/4	7/8	2 7/32	1 31/32	1 9/16	3/16	0.24
1	3/8	7/16	29/32	13/32	21/32	1 15/32	1 1/32	2 17/32	2 3/8	1 13/16	5/16	0.35
1.5	7/16	1/2	31/32	7/16	3/4	1 11/16	1 5/32	2 7/8	2 5/8	2	7/16	0.49
2	1/2	5/8	1 1/32	1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1/2	0.93
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	21/32	1.63
4.75	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	2.60
6.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	3.90
8.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	5.69
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	8.07
12	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	10.82
13.5	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	14.42
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	18.06
25	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	30.9
35	2	2 1/4	4 13/32	1 31/32	3 9/32	7 3/4	5 7/16	13 1/16	10 19/32	9 3/8	1 1/32	41.4
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	8 3/4	6 5/16	14 7/8	11 27/32	10 25/32	1 5/32	62
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	10 1/4	7 3/32	17 1/16	12 15/16	12 7/32	1 1/4	87
85	3	3 1/4	6 9/16	2 15/16	5	13	7 15/32	20 7/8	15	13 3/8	1 17/32	137

CAD RFID INFO



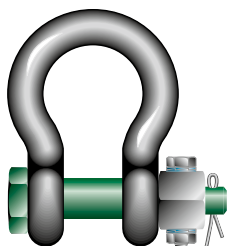
* For shackles ≥ WLL 2 t



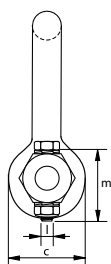
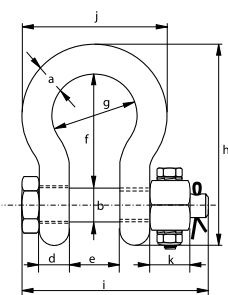
Green Pin® Bow Shackle FN

Standard bow shackle with safety bolt and fixed nut

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G-4143



- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade A, from 2 t and upward these shackles comply with ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 2.7-1^a DNV 2.7-1^b CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length bow	length bolt	width bolt	thickness nut	securing bolt thread	securing bolt length	torque	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l mm	m inch	Ft.Lbs	lbs
2	1 1/2	5/8	1 11/32	1 1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1 1/2	M6	1 3/8	6.2	0.93
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	2 1/32	M6	1 9/16	6.2	1.63
4.75	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	M6	1 25/32	6.2	2.6
6.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	M8	1 31/32	15	3.9
8.5	1	1 1/8	2 5/16	1 1/8	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	M8	2 5/32	15	5.69
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	M10	2 3/8	29	8.07
12	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	M10	2 9/16	29	10.58
13.5	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	M10	2 3/4	29	14.42
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	M8	2 15/16	15	18.06
25	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	M8	3 17/32	15	30.9
35	2	2 1/4	4 13/32	1 31/32	3 9/32	7 3/4	5 7/16	13 1/16	10 19/32	9 3/8	1 1/32	M10	3 15/16	29	43.9
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	8 3/4	6 5/16	14 7/8	11 27/32	10 25/32	1 5/32	M12	4 11/32	50	62
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	10 1/4	7 3/32	17 1/16	12 15/16	12 7/32	1 1/4	M12	4 23/32	50	87
85	3	3 1/4	6 9/16	2 15/16	5	13	7 15/32	20 7/8	15	13 3/8	1 17/32	M12	5 1/2	50	137

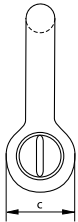
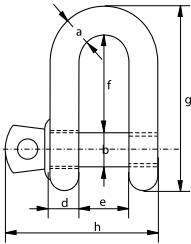
CAD RFID INFO





- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVB Class 2, grade A, from 2 t upward these shackles comply with ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 0378 CE IIA ABS PDA ABS MA
- **Article code:** scan QR code to see article codes

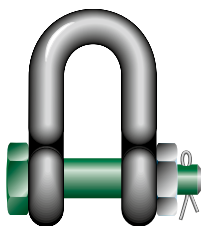
G-4151



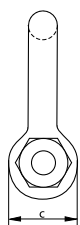
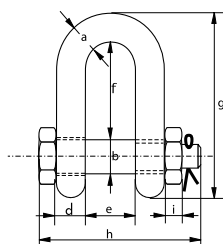
working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length	length bolt	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	lbs
0.33	³ / ₁₆	¹ / ₄	¹⁵ / ₃₂	³ / ₁₆	³ / ₈	³ / ₄	1 ⁵ / ₁₆	1 ⁵ / ₃₂	0.04
0.5	¹ / ₄	⁵ / ₁₆	⁵ / ₈	⁹ / ₃₂	¹⁵ / ₃₂	⁷ / ₈	1 ⁵ / ₈	1 ¹ / ₂	0.11
0.75	⁵ / ₁₆	³ / ₈	²⁵ / ₃₂	¹¹ / ₃₂	¹⁷ / ₃₂	1 ¹ / ₁₆	2	1 ²⁷ / ₃₂	0.2
1	³ / ₈	⁷ / ₁₆	²⁹ / ₃₂	¹³ / ₃₂	²¹ / ₃₂	1 ¹ / ₄	2 ⁵ / ₁₆	2 ¹ / ₈	0.31
1.5	⁷ / ₁₆	¹ / ₂	³¹ / ₃₂	⁷ / ₁₆	³ / ₄	1 ¹⁵ / ₃₂	2 ⁵ / ₈	2 ¹¹ / ₃₂	0.42
2	¹ / ₂	⁵ / ₈	1 ¹¹ / ₃₂	¹ / ₂	⁷ / ₈	1 ¹¹ / ₁₆	3 ⁷ / ₃₂	2 ⁷ / ₈	0.71
3.25	⁵ / ₈	³ / ₄	1 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₁₆	2	3 ¹³ / ₁₆	3 ¹ / ₂	1.19
4.75	³ / ₄	⁷ / ₈	1 ¹³ / ₁₆	³ / ₄	1 ⁷ / ₃₂	2 ⁵ / ₁₆	4 ¹³ / ₃₂	4 ¹ / ₁₆	1.92
6.5	⁷ / ₈	1	2 ¹ / ₁₆	⁷ / ₈	1 ¹³ / ₃₂	2 ⁷ / ₈	5 ⁹ / ₃₂	4 ¹¹ / ₁₆	2.95
8.5	1	1 ¹ / ₈	2 ⁵ / ₁₆	³¹ / ₃₂	1 ¹¹ / ₁₆	3 ¹¹ / ₃₂	6 ¹ / ₁₆	5 ¹³ / ₃₂	4.59
9.5	1 ¹ / ₈	1 ¹ / ₄	2 ⁵ / ₈	1 ³ / ₃₂	1 ²⁷ / ₃₂	3 ¹⁷ / ₃₂	6 ⁵ / ₈	6 ¹ / ₃₂	6.11
12	1 ¹ / ₄	1 ³ / ₈	2 ⁷ / ₈	1 ¹ / ₄	2	3 ¹¹ / ₁₆	7 ³ / ₃₂	6 ¹¹ / ₁₆	8.2
13.5	1 ³ / ₈	1 ¹ / ₂	3 ¹ / ₈	1 ³ / ₈	2 ¹ / ₄	4 ¹⁷ / ₃₂	8 ⁷ / ₃₂	7 ⁵ / ₁₆	11.99
17	1 ¹ / ₂	1 ⁵ / ₈	3 ¹⁵ / ₃₂	1 ¹ / ₂	2 ³ / ₈	5	9 ¹ / ₁₆	8	15.1
25	1 ³ / ₄	2	4 ³ / ₃₂	1 ²⁵ / ₃₂	2 ²⁹ / ₃₂	5 ⁷ / ₈	10 ²¹ / ₃₂	9 ⁹ / ₁₆	25.4
35	2	2 ¹ / ₄	4 ¹³ / ₃₂	1 ³¹ / ₃₂	3 ⁹ / ₃₂	6 ²³ / ₃₂	12 ¹ / ₁₆	10 ²³ / ₃₂	37.3
42.5	2 ¹ / ₄	2 ⁹ / ₁₆	5 ³ / ₁₆	2 ¹ / ₄	3 ³ / ₄	7 ¹⁵ / ₃₂	13 ⁵ / ₈	12 ⁷ / ₃₂	54
55	2 ¹ / ₂	2 ³ / ₄	5 ²³ / ₃₂	2 ⁹ / ₁₆	4 ¹ / ₈	8	14 ¹³ / ₁₆	13 ¹⁷ / ₃₂	72

CAD RFID INFO





G-4153



Green Pin® Dee Shackle BN

Standard dee shackle with safety bolt

- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889, ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVB Class 3, grade A
- **Finish:** hot dipped galvanized
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC* DNV 2.7-1* * DNV 2.7-1* * DNV 0378 CE IIA ABS PDA ABS MA
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length	length bolt	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
2	1/2	5/8	1 11/32	1/2	7/8	1 11/16	3 7/32	3 5/32	1/2	0.86
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2	3 13/16	3 27/32	21/32	1.48
4.75	3/4	7/8	1 13/16	3/4	1 7/32	2 5/16	4 13/32	4 17/32	3/4	2.38
6.5	7/8	1	2 1/16	7/8	1 13/32	2 7/8	5 9/32	5 1/8	7/8	3.66
8.5	1	1 1/8	2 5/16	31/32	1 11/16	3 11/32	6 1/16	5 29/32	31/32	5.42
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	3 17/32	6 5/8	6 17/32	1 1/16	7.5
12	1 1/4	1 3/8	2 7/8	1 1/4	2	3 11/16	7 3/32	7 1/4	1 3/16	9.94
13.5	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	4 17/32	8 7/32	7 3/4	1 5/16	13.45
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5	9 1/16	7 15/16	3/4	16.82
25	1 3/4	2	4 3/32	1 25/32	2 29/32	5 7/8	10 21/32	9 9/16	29/32	28.4
35	2	2 1/4	4 13/32	1 31/32	3 9/32	6 23/32	12 1/16	10 19/32	1 1/32	38.4
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	7 15/32	13 5/8	11 27/32	1 5/32	57
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	8	14 13/16	12 15/16	1 1/4	78
85	3	3 1/4	6 9/16	2 15/16	5	9 1/32	16 7/8	15	1 17/32	117

CAD RFID INFO



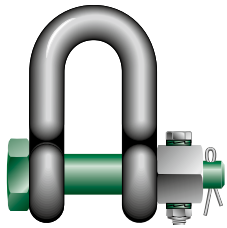
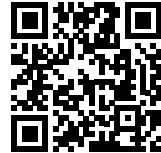
* For shackles ≥ WLL 2 t



Green Pin® Dee Shackle FN

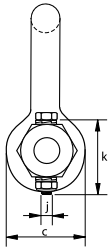
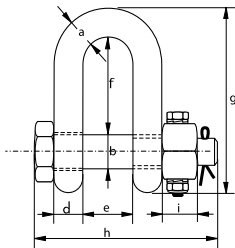
Standard dee shackle with safety bolt and fixed nut

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- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889, ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVB Class 3, grade A hot dipped galvanized
- **Finish:** -40 °F up to +392 °F
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ DNV 2.7-1² DNV 2.7-1³ CE IIA
- **Article code:** scan QR code to see article codes

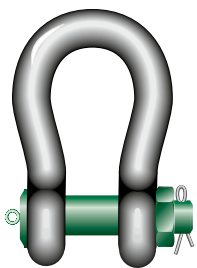
G-4133



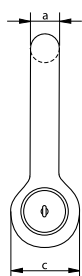
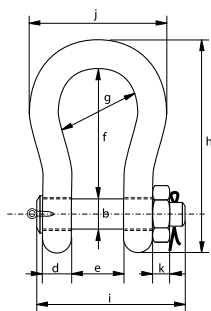
working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length bolt	thickness nut	securing bolt thread	securing bolt length	torque	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j mm	k inch	Ft.Lbs lbs
2	1/2	5/8	1 11/32	1/2	7/8	1 11/16	3 7/32	3 5/32	1/2	M6	1 3/8	6.2 0.86
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2	3 13/16	3 27/32	21/32	M6	1 9/16	6.2 1.48
4.75	3/4	7/8	1 13/16	3/4	1 7/32	2 5/16	4 13/32	4 17/32	3/4	M6	1 25/32	6.2 2.38
6.5	7/8	1	2 1/16	7/8	1 13/32	2 7/8	5 9/32	5 1/8	7/8	M8	1 31/32	15 3.66
8.5	1	1 1/8	2 5/16	31/32	1 11/16	3 11/32	6 1/16	5 29/32	31/32	M8	2 5/32	15 5.42
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	3 17/32	6 5/8	6 17/32	1 1/16	M10	2 3/8	29 7.5
12	1 1/4	1 3/8	2 7/8	1 1/4	2	3 11/16	7 3/32	7 1/4	1 3/16	M10	2 9/16	29 9.94
13.5	1 3/8	1 1/2	3 1/8	1 5/8	2 1/4	4 17/32	8 7/32	7 3/4	1 5/16	M10	2 3/4	29 13.45
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5	9 1/16	7 15/16	3/4	M8	2 15/16	15 16.82
25	1 3/4	2	4 3/32	1 25/32	2 29/32	5 7/8	10 21/32	9 9/16	29/32	M8	3 17/32	15 29.3
35	2	2 1/4	4 13/32	1 31/32	3 9/32	6 23/32	12 1/16	10 19/32	1 1/32	M10	3 15/16	29 40.8
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	7 15/32	13 5/8	11 27/32	1 5/32	M12	4 11/32	50 57
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	8	14 13/16	12 15/16	1 1/4	M12	4 23/32	50 78
85	3	3 1/4	6 9/16	2 15/16	5	9 1/32	16 7/8	15	1 17/32	M12	5 1/2	50 117

CAD RFID INFO





P-6036



Green Pin® Heavy Duty Bow Shackle BN

High load capacity bow shackle with safety bolt

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** shackle bow painted silver, pin painted green
- **Temperature range:** -40 °F up to +392 °F ≤ WLL 600 t
- **Certification:** 2.1 2.2 3.1 MTC^a CE II A Optional^{***}: MTC^b * LROS^{**} MPI^a US^a
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length bow	length bolt	width	thickness nut	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs	lbs
120	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	15 3/4	9 3/8	25 11/32	17 27/32	16 27/32	1 31/32	62	251
150	4 1/8	4 1/4	9 1/4	4 1/8	6 21/32	16 5/32	10 13/16	27	19 17/32	19 3/32	1 31/32	86	254
200	4 23/32	5 1/8	10 5/8	4 23/32	7 1/16	20 3/16	11 13/32	32 25/32	22 7/32	20 7/8	2 25/32	153	518
250	5 1/8	5 1/2	11 3/16	5 1/8	8 1/16	21 13/16	12	35 9/32	24 3/16	22 1/4	2 25/32	188	650
300	5 1/2	5 29/32	12 1/8	5 1/2	8 1/16	24 11/32	12	38 27/32	25 11/32	23 1/32	3 3/16	225	807
400	6 11/16	6 7/8	14 13/16	6 15/32	9 3/32	26 11/32	12 25/32	43 29/32	27 15/32	26 3/16	2 3/4	333	1367
500	7 3/32	7 9/32	15 21/32	6 15/32	10 3/32	28 5/16	13 25/32	46 7/8	28 21/32	27 15/16	2 3/4	381	1587
600	7 7/8	8 1/16	17 15/32	7 7/16	11 3/32	28 5/16	14 3/4	48 31/32	31 7/8	30 1/2	2 3/4	503	2150
700	8 9/32	8 15/32	17 7/8	8 1/32	12 1/8	28 9/32	15 3/4	49 23/32	34 1/4	32 9/32	2 3/4	584	2370
800	8 9/32	8 21/32	18 9/32	8 1/32	12 1/8	28 9/32	15 3/4	50	34 1/4	32 9/32	2 3/4	608	2425
900	8 21/32	9 1/16	19 3/32	8 15/32	12 29/32	28 9/32	16 17/32	51 1/32	36 7/32	33 27/32	2 3/4	703	2756
1000	9 7/16	9 7/16	20 9/32	8 15/32	13 3/4	28 9/32	16 17/32	52 19/32	37	35 7/16	2 3/4	774	3219
1250	10 1/4	10 5/8	23 1/32	9 1/16	14 17/32	30 1/4	17 23/32	57 5/16	40 11/32	38 3/16	2 3/4	1060	4277
1500	11 1/32	11 13/32	24 19/32	9 1/16	14 17/32	32 7/32	17 23/32	61 1/4	40 11/32	39 3/4	2 3/4	1226	5082

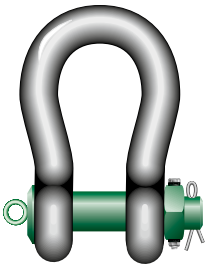
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- * Standard certification for shackles ≥ WLL 400 t
- ** For shackles ≥ WLL 120 t and ≤ WLL 600 t available with or without LROS.
For shackles ≥ WLL 700 t available with LROS
- *** With additional charges

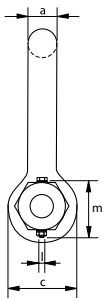
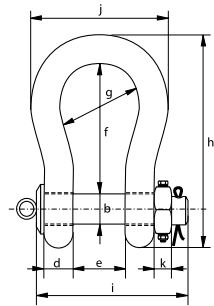


Green Pin® Heavy Duty Bow Shackle FN

High load capacity bow shackle with safety bolt and fixed nut



P-6016

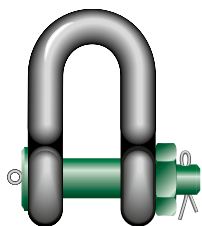


- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** shackle bow painted silver, pin painted green
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA **Optional:** MTC^b LROS MPI^a US^a
- **Article code:** scan QR code to see article codes

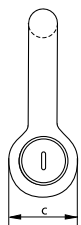
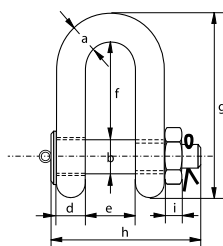
working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	securing bolt thread	securing bolt length	torque	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l mm	m inch	Ft.Lbs	lbs	lbs
120	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	15 3/4	9 3/8	25 11/32	17 27/32	16 27/32	1 31/32	M12	5 29/32	68	62	243
150	4 1/8	4 1/4	9 1/4	4 1/8	6 21/32	16 5/32	10 13/16	27	19 17/32	19 3/32	1 31/32	M12	6 5/16	68	86	342

CAD RFID INFO

* With additional charges



P-6038



Green Pin® Heavy Duty Dee Shackle BN

High load capacity dee shackle with safety bolt

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** painted
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE II A **Optional:** MTC^a LROS MPI^a US^a
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length	length bolt	thickness nut	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs	lbs
120	3 ³ / ₄	3 ³ / ₄	8 ¹ / ₃₂	3 ³ / ₄	5 ²⁵ / ₃₂	10 ²⁵ / ₃₂	20 ¹⁵ / ₃₂	17 ⁵ / ₁₆	1 ³¹ / ₃₂	62	243

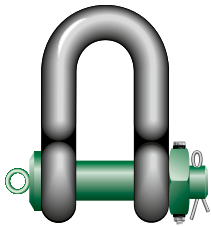
CAD RFID

* With additional charges

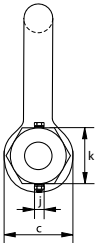
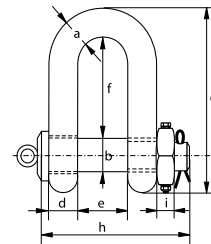


Green Pin® Heavy Duty Dee Shackle FN

High load capacity dee shackle with safety bolt and fixed nut



P-6018

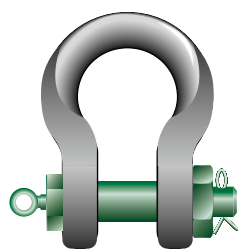


- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** painted
- **Temperature range:** -40 °F up to +392 °F ≤ WLL 600 t
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA **Optional:** MTC^a LROS MPI^a US^a
- **Article code:** scan QR code to see article codes

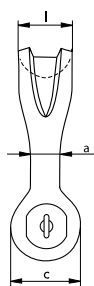
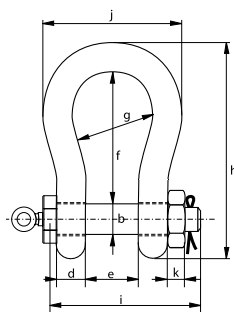
working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length	length bolt	thickness nut	securing bolt thread	securing bolt length	torque	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j mm	k inch	Ft.Lbs	lbs	lbs
120	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	10 25/32	20 15/32	17 5/16	1 31/32	M12	5 29/32	50	62	214

CAD RFID INFO

* With additional charges



P-6033



Green Pin® Sling Shackle BN

High load capacity bow shackle with safety bolt

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 ≤ WLL 600 t
- **Finish:** shackle bow painted silver, pin painted green
(WLL 7 t up to 55 t shackles are hot dipped galvanized)
- **Temperature range:** -40 °F up to +392 °F ≤ WLL 600 t
-4 °F up to +392 °F ≥ WLL 700 t
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA **Optional***:** MTC^b * LROS^{**} MPI^a US^b
- **Article code:** scan QR code to see article codes

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working load limit	diameter body	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	bearing surface	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l inch	lbs	lbs
7	7/8	7/8	1 13/16	3/4	1 1/4	3 25/32	2 17/32	6 1/32	4 17/32	4 11/32	3/4	1 5/8	2.18	4.34
12.5	1 1/8	1 1/8	2 13/32	31/32	1 23/32	4 23/32	3 7/32	7 3/4	5 15/16	5 3/4	15/16	2 1/8	4.01	9.41
18	1 3/8	1 3/8	2 23/32	1 3/16	2 1/8	5 13/16	4 1/32	9 13/32	6 23/32	7 3/32	1 5/32	2 17/32	5.18	15.1
30	1 9/16	1 21/32	3 17/32	1 3/8	2 23/32	6 1/2	4 31/32	10 11/32	8 5/32	7 7/8	1 11/32	3 1/8	9.33	27.6
40	2 5/32	2	4 9/32	1 25/32	3 5/16	7 27/32	5 1/2	13 1/32	9 29/32	9 1/4	1 1/2	3 13/16	9.52	44.8
55	2 3/8	2 1/4	4 17/32	2 5/32	3 17/32	9 7/16	6 5/16	15 5/16	11 9/16	10 5/8	1 25/32	3 15/16	10.14	67
75	2 11/16	2 3/4	4 31/32	2 1/8	4 11/32	11 13/32	7 9/32	18 5/8	12 15/32	13 1/32	1 9/16	4 23/32	25.1	99
125	3 11/32	3 5/32	6 5/32	3 11/32	5 13/32	14 13/32	8 21/32	22 15/16	16 1/4	15 11/32	1 9/16	5 29/32	41	203
150	3 11/16	3 3/4	7 1/8	3 1/2	5 25/32	15 13/32	9 31/32	25 1/2	17 17/32	17 11/16	1 31/32	6 11/16	60	309
200	4 11/32	4 1/8	7 29/32	3 15/16	6 7/32	18 15/16	11 1/32	29 7/8	18 29/32	19 11/16	1 31/32	8 1/16	81	452
250	4 31/32	4 23/32	9 1/32	4 11/32	7 1/16	21 11/32	11 13/16	33 27/32	21 1/16	21 23/32	2 3/8	9 7/16	120	582
300	5 11/32	5 9/32	9 11/16	5 9/32	7 11/16	23 21/32	13 25/32	37 1/4	23 7/32	24 13/32	2 3/4	10 7/16	175	794
400	6 5/16	6 5/16	11 5/8	5 23/32	9 3/32	22 11/16	14 9/16	39 3/16	27 1/16	27 15/16	3 5/32	12 19/32	280	1285
500	6 11/16	7 3/32	13	6 5/16	10 11/32	26 13/16	17 23/32	45 23/32	30	33 15/32	3 17/32	13 3/8	388	1900
600	7 15/32	7 7/8	14 3/16	6 11/16	11 3/8	29 3/16	19 9/32	49 3/4	32 9/16	36 5/8	3 15/16	14 9/16	516	2500
700	7 7/8	8 15/32	15 7/16	7 15/32	12 13/32	29 9/16	21 1/4	50 9/16	34 19/32	35 15/32	3 15/16	15 3/4	639	2518
800	8 19/32	9 1/16	16 17/32	7 7/8	13 15/32	33 1/2	21 13/16	56 5/32	37 3/32	37 9/32	4 11/32	16 17/32	772	3126
900	9 17/32	10 1/32	18 11/32	8 21/32	14 1/2	33 1/2	22 27/32	58 19/32	40 9/32	40 9/32	4 23/32	17 5/16	1023	3990
1000	10 1/4	10 5/8	19 9/32	9 7/16	15 23/32	33 1/2	24 3/16	60 5/16	43 7/16	43 19/32	4 23/32	18 1/8	1232	5258
1250	11 7/32	11 13/16	20 3/32	10 1/4	17 25/32	36 21/32	25 19/32	65 19/32	48 5/16	46 17/32	5 29/32	20 7/8	1724	6856
1550	11 7/32	12 19/32	21 21/32	11 1/32	19 1/32	37 13/32	26 25/32	67 5/16	51 3/16	49 11/32	5 29/32	22 1/16	2024	8058

CAD RFID

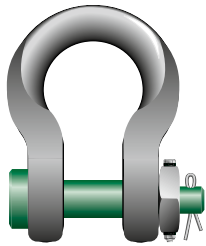
- * Standard certification for shackles ≥ WLL 400 t
- ** For shackles ≥ WLL 75 t and ≤ WLL 600 t available with or without LROS.
For shackles ≥ WLL 700 t available with LROS
- *** With additional charges



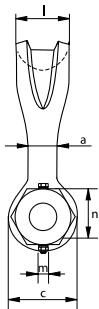
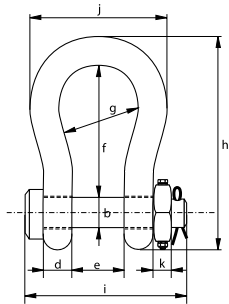
Green Pin® Sling Shackle FN

High load capacity bow shackle with safety bolt and fixed nut

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P-6013



- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 ≤ WLL 600 t
- **Finish:** shackle bow painted silver, pin painted green (WLL 7 t up to 55 t shackles are hot dipped galvanized)
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA **Optional**:** MTC^b LROS^c MPI^a US^b
- **Article code:** scan QR code to see article codes

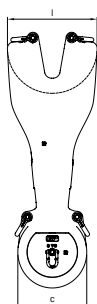
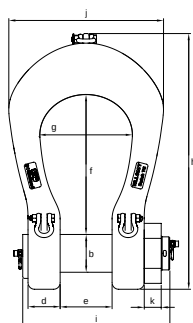
working load limit	diameter body	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	bearing surface	securing bolt thread	securing bolt length	torque	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l inch	m mm	n inch	Ft.Lbs	lbs
7	7/8	7/8	1 13/16	3/4	1 1/4	3 25/32	2 17/32	6 1/32	4 17/32	4 11/32	3/4	1 5/8	M6	1 25/32	6.2	4.34
12.5	1 1/8	1 1/8	2 13/32	31/32	1 23/32	4 23/32	3 7/32	7 3/4	5 15/16	5 3/4	15/16	2 1/8	M8	1 31/32	15	9.41
18	1 3/8	1 3/8	2 23/32	1 3/16	2 1/8	5 13/16	4 1/32	9 13/32	6 23/32	7 3/32	1 5/32	2 17/32	M10	2 9/16	29	15.1
30	1 9/16	1 21/32	3 17/32	1 3/8	2 23/32	6 1/2	4 31/32	10 31/32	8 5/32	7 7/8	1 11/32	3 1/8	M6	2 15/16	6.2	27.6
40	2 5/32	2	4 9/32	1 25/32	3 5/16	7 27/32	5 1/2	13 1/32	9 29/32	9 1/4	1 1/2	3 13/16	M8	3 17/32	15	44.8
55	2 3/8	2 1/4	4 17/32	2 5/32	3 17/32	9 7/16	6 5/16	15 5/16	11 9/16	10 5/8	1 25/32	3 15/16	M10	3 15/16	29	67
75	2 11/16	2 3/4	4 31/32	2 1/8	4 11/32	11 13/32	7 9/32	18 5/8	12 15/32	12 15/32	1 9/16	4 23/32	M12	4 23/32	50	99
125	3 11/32	3 5/32	6 5/32	3 11/32	5 13/32	14 13/32	8 21/32	22 15/16	16 1/4	15 11/32	1 9/16	5 29/32	M12	5 1/8	50	203
150	3 11/16	3 3/4	7 1/8	3 1/2	5 25/32	15 13/32	9 31/32	25 1/2	17 17/32	17 11/16	1 31/32	6 11/16	M12	5 1/2	50	309

CAD RFID INFO

* For shackles ≥ WLL 75 t and ≤ WLL 150 t available with or without LROS
 ** With additional charges



P-6043



Green Pin Power Sling® Shackle BN

High load capacity, grade 80 shackle with safety bolt

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** shackle bow painted silver, pin painted green
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC[®] DNV PL MPI[®] US[®] DNV 0377 DNV 0378 CE IIA
- **Article code:** scan QR code to see article codes

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working load limit	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length bow	length bolt	width bolt	thickness nut	bearing surface	weight pin	weight each
metric t	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l inch	lbs	lbs
125	3 3/16	6 1/2	3 1/2	5 9/32	14 3/8	8 21/32	24 7/8	15 1/32	14 9/16	1 13/32	8 1/16	38.1	214
150	3 3/4	7 5/32	3 17/32	5 21/32	15 11/32	9 27/32	27 11/16	16 1/16	16 17/32	1 21/32	9 3/4	58	295
200	4 1/8	8 1/32	3 15/16	6 1/16	18 29/32	10 7/8	33	17 9/16	18 21/32	1 27/32	11 13/32	78	432
250	4 23/32	9 3/8	4 11/32	6 27/32	21 1/4	11 13/16	36 15/16	19 13/16	20 3/16	2 3/8	12 3/8	121	602
300	5 9/32	10 1/4	4 3/4	7 7/16	23 5/8	13 25/32	40 5/8	21 21/32	23 25/32	2 3/8	13 19/32	159	818
400	6 5/16	12	5 1/2	8 13/16	24 13/32	14 9/16	44 7/32	25	25 23/32	2 3/4	15 7/16	251	1239
500	7 3/32	13 3/8	5 31/32	10 1/32	26 25/32	17 23/32	48 25/32	27 17/32	30 1/32	2 3/4	17 5/16	346	1726
600	7 7/8	14 3/8	6 11/16	11 1/32	29 3/16	19 9/32	53 5/16	30 1/4	32 7/32	2 3/4	18 11/16	467	2211
700	8 15/32	15 15/16	7 15/32	12 19/32	29 9/16	21 1/4	55 23/32	33 7/16	35 5/32	2 3/4	20 5/32	597	2813
800	9 1/16	16 15/16	7 7/8	13 21/32	33 1/2	21 13/16	60 29/32	35 19/32	36 3/32	2 3/4	21 3/32	723	3272
900	10 1/32	18 3/4	8 15/32	14 11/16	33 1/2	22 27/32	62 15/16	37 29/32	38 9/32	2 3/4	22 1/16	939	3986
1000	10 5/8	19 11/16	9 1/8	15 15/16	33 1/2	24 3/16	64 11/16	40 9/16	40 1/4	2 3/4	23 7/32	1120	4700
1250	11 13/16	22 7/16	9 21/32	17 5/32	36 5/8	25 19/32	71 3/8	43 27/32	45 1/32	2 3/4	26 3/8	1484	6374
1550	12 19/32	23 7/32	10 5/8	18 15/32	37 13/32	26 25/32	74 13/32	48 1/32	47 5/8	2 3/4	27 23/32	1823	7754

CAD RFID INFO





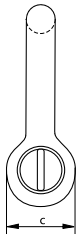
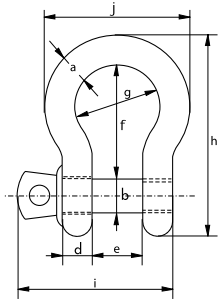
Green Pin Super® Bow Shackle SC

Grade 80 bow shackle with screw pin

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G-5261



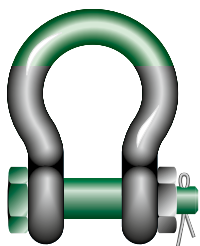
- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 2, grade B
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ CE IIA ABS PDA * ABS MA *
- **Article code:** scan QR code to see article codes

working load limit	diameter body	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	lbs
3.3	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	2 7/8	2 5/16	0.79
5	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 1/2	2 15/16	1.39
7	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 1/16	3 1/2	2.23
9.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	4 11/16	4 1/32	3.31
12.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 13/32	4 21/32	4.87
15	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 1/32	5 5/32	6.97
18	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	6 11/16	5 25/32	9.5
21	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 5/16	6 3/8	12.3
30	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	8	6 7/8	16.38
40	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	27.6
55	2	2 1/4	4 11/16	2 1/4	3 9/32	7 3/4	5 7/16	13 7/16	11 1/4	9 29/32	37.9
85*	2 1/2	2 3/4	5 23/32	2 3/4	4 1/8	10 1/4	7 3/32	17 7/32	13 15/16	12 19/32	83

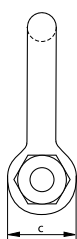
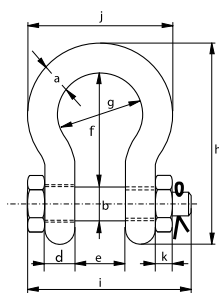
CAD RFID INFO



* Excluded from ABS Type Approval



G-5263



Green Pin Super® Bow Shackle BN

Grade 80 bow shackle with safety bolt

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade B
- **Finish:** hot dipped galvanized (150 t and upward are painted)
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC * LROS * CE IIA ABS PDA * ABS MA *
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs
3.3	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1/2	0.88
5	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	21/32	1.61
7	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	2.62
9.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	3.81
12.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	5.64
15	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	7.94
18	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	10.91
21	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	14.59
30	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	17.88
40	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	32.6
55	2	2 1/4	4 11/16	2 1/4	3 9/32	7 3/4	5 7/16	13 15/32	11 5/32	9 29/32	1 1/32	53
85	2 1/2	2 3/4	5 23/32	2 3/4	4 1/8	10 1/4	7 3/32	17 1/4	13 11/32	12 19/32	1 1/4	99
120	3	3 1/4	6 15/32	3 9/32	5	12 15/16	7 15/32	21 5/32	15 5/8	14 1/32	1 17/32	159
150*	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	15 3/4	9 3/8	25 11/32	17 27/32	16 27/32	1 31/32	247
175*	4 1/8	4 1/4	9 1/4	4 1/8	6 21/32	16 5/32	10 13/16	27	19 17/32	19 3/32	1 31/32	353

CAD RFID INFO



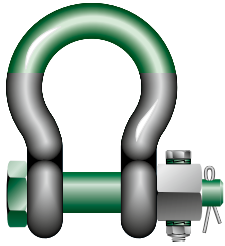
- * For shackles WLL 150 t and WLL 175 t available with and without LROS
- * With round headed bolt
- * Excluded from ABS Type Approval



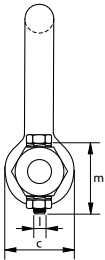
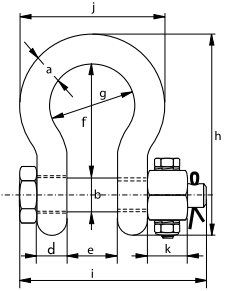
Green Pin Super® Bow Shackle FN

Grade 80 bow shackle with safety bolt and fixed nut

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G-5243



- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade B
- **Finish:** hot dipped galvanized (150 t and upward are painted)
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ LROS⁴ * CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length bow	length bolt	width	thickness nut	securing bolt thread	securing bolt length	torque	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l mm	m inch	Ft.Lbs	lbs
3.3	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1/2	M6	1 3/8	6.2	0.97
5	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	21/32	M6	1 9/16	6.2	1.74
7	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	M6	1 25/32	6.2	2.78
9.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	M8	1 31/32	15	4.14
12.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	M8	2 5/32	15	6.13
15	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	M10	2 3/8	29	8.53
18	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	M10	2 9/16	29	11.6
21	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	M10	2 3/4	29	15.3
30	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	M8	2 15/16	15	19.38
40	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	M8	3 17/32	15	33.1
55	2	2 1/4	4 11/16	2 1/4	3 9/32	7 3/4	5 7/16	13 15/32	11 5/32	9 29/32	1 1/32	M10	3 15/16	29	48.5
85	2 1/2	2 3/4	5 23/32	2 3/4	4 1/8	10 1/4	7 3/32	17 1/4	13 11/32	12 19/32	1 1/4	M12	4 23/32	50	93
120	3	3 1/4	6 15/32	3 9/32	5	13	7 15/32	21 5/32	15 5/8	14 1/32	1 17/32	M12	5 1/2	50	154
150*	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	15 3/4	9 3/8	25 11/32	17 27/32	16 27/32	1 31/32	M12	5 29/32	50	247
175*	4 1/8	4 1/4	9 1/4	4 1/8	6 21/32	16 5/32	10 13/16	27	19 17/32	19 3/32	1 31/32	M12	6 5/16	50	353

CAD RFID INFO

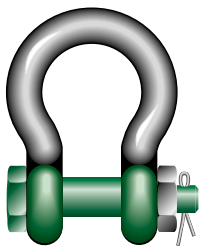
- * For shackles WLL 150 t and WLL 175 t available with and without LROS
- * With round headed bolt



Green Pin Polar® Bow Shackle BN

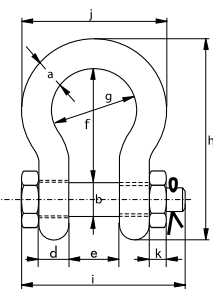
Grade 80 bow shackle with safety bolt
for use under low temperatures

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G-5163

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 7 x WLL
for shackles with WLL 55 t and 85 t the MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889, ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade A
- **Finish:** hot dipped galvanized
- **Temperature range:** -76 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 2.7-1^a DNV 2.7-1^b DNV 0378 CE IIA ABS PDA ABS MA
- **Article code:** scan QR code to see article codes



working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs
2	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1/2	0.93
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	21/32	1.63
4.75	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	2.6
6.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	3.9
8.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	5.69
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	8.07
12	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	10.82
13.5	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	14.42
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	18.06
25	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	31.3
35	2	2 1/4	4 13/32	1 31/32	3 9/32	7 3/4	5 7/16	13 1/16	10 19/32	9 3/8	1 1/32	43.9
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	8 3/4	6 5/16	14 7/8	11 27/32	10 25/32	1 5/32	62
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	10 1/4	7 3/32	17 1/16	12 15/16	12 7/32	1 1/4	87
85	3	3 1/4	6 9/16	2 15/16	5	13	7 15/32	20 7/8	15	13 3/8	1 17/32	137

CAD RFID INFO

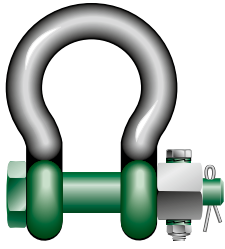




Green Pin Polar® Bow Shackle FN

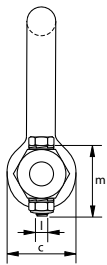
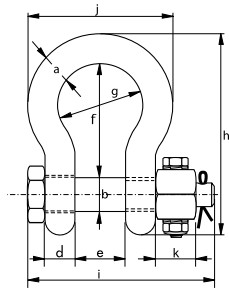
Grade 80 bow shackle with safety bolt and fixed nut
for use under low temperatures

Scan for
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details



G-5143

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 7 x WLL
for shackles with WLL 55 t and 85 t the MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889, ASME B30.26 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 3, grade A
- **Finish:** hot dipped galvanized
- **Temperature range:** -76 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a DNV 2.7-1^a DNV 2.7-1^b CE IIA
- **Article code:** scan QR code to see article codes



working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length bow	length bolt	width	thickness nut	securing bolt thread	securing bolt length	torque	weight each
metric	a	b	c	d	e	f	g	h	i	j	k	l	m	Ft.Lbs	lbs
t	inch	inch	inch	inch	inch	inch	inch	inch	inch	inch	inch	mm	inch		
2	1/2	5/8	1 11/32	1/2	7/8	2	1 1/4	3 17/32	3 5/32	2 5/16	1/2	M6	1 3/8	6.2	0.93
3.25	5/8	3/4	1 9/16	5/8	1 1/16	2 17/32	1 11/16	4 11/32	3 27/32	2 15/16	21/32	M6	1 9/16	6.2	1.63
4.75	3/4	7/8	1 13/16	3/4	1 7/32	3	2	5 3/32	4 17/32	3 1/2	3/4	M6	1 25/32	6.2	2.6
6.5	7/8	1	2 1/16	7/8	1 13/32	3 9/32	2 9/32	5 21/32	5 1/8	4 1/32	7/8	M8	1 31/32	15	3.9
8.5	1	1 1/8	2 5/16	31/32	1 11/16	3 3/4	2 11/16	6 15/32	5 29/32	4 21/32	31/32	M8	2 5/32	15	5.69
9.5	1 1/8	1 1/4	2 5/8	1 3/32	1 27/32	4 1/4	2 15/16	7 5/16	6 17/32	5 5/32	1 1/16	M10	2 3/8	29	8.07
12	1 1/4	1 3/8	2 7/8	1 1/4	2	4 17/32	3 9/32	7 29/32	7 1/4	5 25/32	1 3/16	M10	2 9/16	29	10.82
13.5	1 3/8	1 1/2	3 1/8	1 3/8	2 1/4	5 1/4	3 5/8	8 15/16	7 3/4	6 3/8	1 5/16	M10	2 3/4	29	14.42
17	1 1/2	1 5/8	3 15/32	1 1/2	2 3/8	5 3/4	3 29/32	9 13/16	7 15/16	6 7/8	3/4	M8	2 15/16	15	18.06
25	1 3/4	2	4 3/32	1 25/32	2 29/32	7	4 31/32	11 13/16	9 9/16	8 1/2	29/32	M8	3 17/32	15	31.3
35	2	2 1/4	4 13/32	1 31/32	3 9/32	7 3/4	5 7/16	13 1/16	10 19/32	9 3/8	1 1/32	M10	3 15/16	29	43.9
42.5	2 1/4	2 9/16	5 3/16	2 1/4	3 3/4	8 3/4	6 5/16	14 7/8	11 27/32	10 25/32	1 5/32	M12	4 11/32	50	62
55	2 1/2	2 3/4	5 23/32	2 9/16	4 1/8	10 1/4	7 3/32	17 1/16	12 15/16	12 7/32	1 1/4	M12	4 23/32	50	87
85	3	3 1/4	6 9/16	2 15/16	5	13	7 15/32	20 7/8	15	13 3/8	1 17/32	M12	5 1/2	50	137

CAD RFID INFO

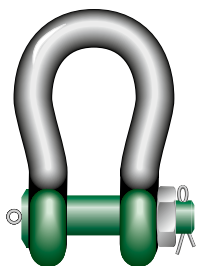




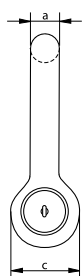
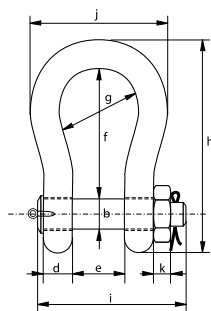
Green Pin Polar® Heavy Duty Bow Shackle BN

High load capacity, grade 80 bow shackle with safety bolt for use under low temperatures

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P-6031



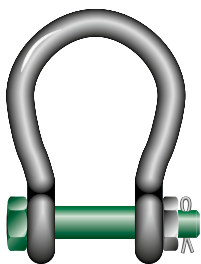
- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** shackle bow painted silver, pin painted green
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA **Optional***:** MTC^b * LROS^{**} MPI^a US^a
- **Article code:** scan QR code to see article codes



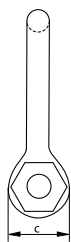
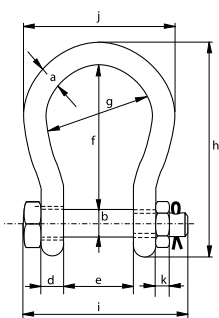
working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight pin	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs	lbs
120	3 3/4	3 3/4	8 1/32	3 3/4	5 25/32	15 3/4	9 3/8	25 11/32	17 27/32	16 27/32	1 31/32	62	243
150	4 1/8	4 1/4	9 1/4	4 1/8	6 21/32	16 5/32	10 13/16	27	19 17/32	19 3/32	1 31/32	86	353
200	4 23/32	5 1/8	10 5/8	4 23/32	7 1/16	20 3/16	11 13/32	32 25/32	22 7/32	20 7/8	2 25/32	153	518
250	5 1/8	5 1/2	11 3/16	5 1/8	8 1/16	21 13/16	12	35 9/32	24 3/16	22 1/4	2 25/32	188	628
300	5 1/2	5 29/32	12 1/8	5 1/2	8 1/16	24 11/32	12	38 27/32	25 11/32	23 1/32	3 3/16	225	811
400	6 11/16	6 7/8	14 13/16	6 15/32	9 3/32	26 5/16	12 25/32	43 27/32	27 5/32	26 3/16	2 3/4	333	1235
500	7 3/32	7 9/32	15 21/32	6 15/32	10 3/32	28 9/32	13 25/32	46 27/32	28 11/32	27 15/16	2 3/4	381	1579
600	7 7/8	8 1/16	17 15/32	7 7/16	11 3/32	28 9/32	14 3/4	48 15/16	31 7/8	30 1/2	2 3/4	503	2150
700	8 9/32	8 15/32	17 7/8	8 1/32	12 1/8	28 9/32	15 3/4	49 23/32	34 1/4	32 9/32	2 3/4	584	2370
800	8 9/32	8 21/32	18 9/32	8 1/32	12 1/8	28 9/32	15 3/4	50	34 1/4	32 9/32	2 3/4	608	2425
900	8 21/32	9 1/16	19 3/32	8 15/32	12 29/32	28 9/32	16 17/32	51 1/32	36 7/32	33 27/32	2 3/4	703	2756
1000	9 7/16	9 7/16	20 9/32	8 15/32	13 3/4	28 9/32	16 17/32	52 19/32	37	35 7/16	2 3/4	774	3219
1250	10 1/4	10 5/8	23 1/32	9 1/16	14 17/32	30 1/4	17 23/32	57 5/16	40 11/32	38 3/16	2 3/4	1060	4277
1500	11 1/32	11 13/32	24 19/32	9 1/16	14 17/32	32 7/32	17 23/32	61 1/4	40 11/32	39 3/4	2 3/4	1226	5082

CAD RFID

- * Standard certification for shackles $\geq$ WLL 400 t
- ** For shackles between WLL 120 t and WLL 600 t available with and without LROS.
For shackles WLL 700 t and up available with LROS
- *** With additional charges



G-4263



Green Pin BigMouth® Bow Shackle BN

Grade 80 bow shackle with safety bolt and wider shackle mouth

- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE II A
- **Article code:** scan QR code to see article codes

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working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs
4.75	7/8	1	2 1/16	7/8	2 15/32	4 13/32	3 15/32	6 13/16	6 3/16	5 3/16	7/8	4.59
6.5	1	1 1/8	2 5/16	31/32	2 15/16	5 5/16	4 1/8	8 1/32	7 7/32	6 3/32	31/32	6.92
8.5	1 1/8	1 1/4	2 19/32	1 3/32	3 7/32	5 13/16	4 17/32	8 27/32	8 1/16	6 23/32	1 1/16	9.61
9.5	1 1/4	1 3/8	2 27/32	1 1/4	3 17/32	6 3/8	4 31/32	9 3/4	8 13/16	7 15/32	1 3/16	13.12
12	1 3/8	1 1/2	3 1/8	1 3/8	3 15/16	7 3/32	5 1/2	10 25/32	9 21/32	8 9/32	1 5/16	17.35
16	1 1/2	1 5/8	3 15/32	1 1/2	4 3/16	8 1/2	6 1/4	12 9/16	9 3/4	9 1/4	3/4	22.5
25	1 3/4	2	4 1/16	1 25/32	5	9 3/4	6 7/8	14 9/16	11 21/32	10 7/16	29/32	36.8
30	2	2 1/4	4 21/32	1 31/32	5 3/4	10 3/4	8 5/32	16 3/16	13 1/16	12 3/32	1 1/32	55
55	2 1/2	2 3/4	5 23/32	2 9/16	6 1/2	12 3/8	8 3/8	19 3/16	15 5/16	13 1/2	1 1/4	99
75	3 1/4	3 1/4	6 15/32	3 9/32	7 1/4	13	10	21 5/32	17 29/32	16 17/32	1 17/32	154

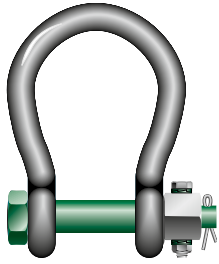
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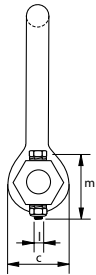
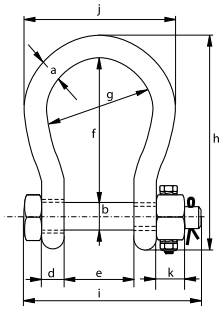
Green Pin BigMouth® Bow Shackle FN

Grade 80 bow shackle with safety bolt, fixed nut and wider shackle mouth

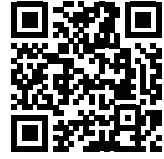
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G-4243



- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIA
- **Article code:** scan QR code to see article codes



working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	securing bolt thread	securing bolt length	torque	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	l mm	m inch	Ft.Lbs	lbs
4.75	7/8	1	2 1/16	7/8	2 15/32	4 13/32	3 15/32	6 13/16	6 3/16	5 3/16	7/8	M8	1 31/32	15	4.59
6.5	1	1 1/8	2 5/16	31/32	2 15/16	5 5/16	4 1/8	8 1/32	7 7/32	6 3/32	31/32	M8	2 5/32	15	6.92
8.5	1 1/8	1 1/4	2 19/32	1 3/32	3 7/32	5 13/16	4 17/32	8 27/32	8 1/16	6 23/32	1 1/16	M10	2 3/8	29	9.61
9.5	1 1/4	1 3/8	2 27/32	1 1/4	3 17/32	6 3/8	4 31/32	9 3/4	8 13/16	7 15/32	1 3/16	M10	2 9/16	29	13.12
12	1 3/8	1 1/2	3 1/8	1 3/8	3 15/16	7 3/32	5 1/2	10 25/32	9 21/32	8 9/32	1 5/16	M10	2 3/4	29	17.35
16	1 1/2	1 5/8	3 15/32	1 1/2	4 3/16	8 1/2	6 1/4	12 9/16	9 3/4	9 1/4	3/4	M8	2 15/16	15	22.5
25	1 3/4	2	4 1/16	1 25/32	5	9 3/4	6 7/8	14 9/16	11 21/32	10 7/16	29/32	M8	3 17/32	15	36.8
30	2	2 1/4	4 21/32	1 31/32	5 3/4	10 3/4	8 5/32	16 3/16	13 1/16	12 3/32	1 1/32	M10	3 15/16	29	55
55	2 1/2	2 3/4	5 23/32	2 9/16	6 1/2	12 3/8	8 3/8	19 3/16	15 5/16	13 1/2	1 1/4	M12	4 23/32	50	99
75	3 1/4	3 1/4	6 15/32	3 9/32	7 1/4	13	10	21 5/32	17 29/32	16 17/32	1 17/32	M12	5 1/2	50	154

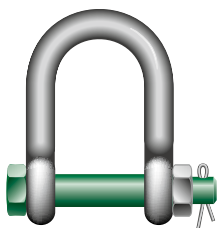
CAD RFID INFO



Green Pin BigMouth® Dee Shackle BN

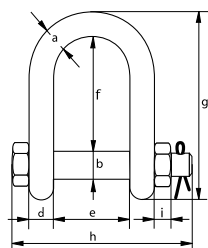
Dee shackle with a longer inside length, wider mouth and safety bolt

Scan for additional product details



- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** hot dipped galvanized
- **Temperature range:** -4 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a CE IIIA
- **Article code:** scan QR code to see article codes

G-4553



working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	length	length bolt	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	lbs
4.6	$\frac{3}{4}$	$\frac{7}{8}$	$1\frac{13}{16}$	$\frac{3}{4}$	$2\frac{3}{4}$	$4\frac{9}{16}$	$6\frac{21}{32}$	$6\frac{1}{16}$	$\frac{3}{4}$	3.31
8.6	1	$1\frac{1}{8}$	$2\frac{5}{16}$	$\frac{31}{32}$	$3\frac{9}{32}$	$5\frac{1}{2}$	$8\frac{3}{16}$	$7\frac{15}{32}$	$\frac{31}{32}$	6.9
15.5	$1\frac{1}{2}$	$1\frac{5}{8}$	$3\frac{15}{32}$	$1\frac{1}{2}$	$4\frac{17}{32}$	7	$11\frac{1}{16}$	$10\frac{1}{8}$	$\frac{3}{4}$	20.8

CAD

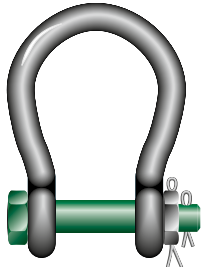




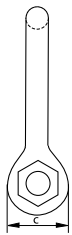
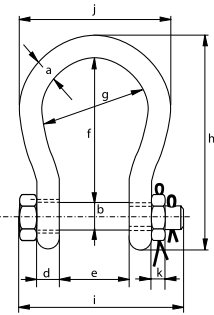
Green Pin BigMouth® Towing Shackle BN

Grade 80 towing bow shackle with safety bolt and wider shackle mouth

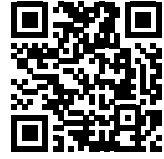
Scan for additional product details



G-4463



- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Finish:** hot dipped galvanized
- **Certification:** 2.1 2.2 3.1 CE IIA
- **Article code:** scan QR code to see article codes
- **Note:** for towing only, not for lifting applications



working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	thickness nut	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	k inch	lbs
22	1 1/2	1 5/8	3 15/32	1 1/2	4 3/16	8 1/2	6 1/4	12 9/16	9 3/4	9 1/4	3/4	22.5
30	1 3/4	2	4 1/16	1 25/32	5	9 3/4	6 7/8	14 9/16	11 21/32	10 7/16	15/16	36.8
40	2	2 1/4	4 21/32	1 31/32	5 3/4	10 3/4	8 5/32	16 3/16	13 5/16	12 3/32	1 1/16	55
55	2 1/2	2 3/4	5 23/32	2 9/16	6 1/2	12 3/8	8 3/8	19 3/16	15 5/16	13 1/2	1 5/16	99
100	3 1/4	3 1/4	6 15/32	3 9/32	7 1/4	13	10	21 1/4	17 29/32	16 17/32	1 9/16	154

CAD



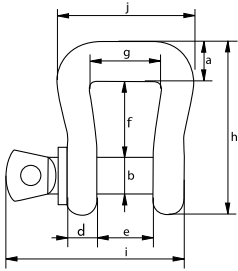
Green Pin® Web Sling Shackle SC

Shackle for synthetic web slings with screw collar pin

Scan for additional product details



P-5461



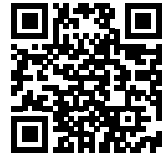
- **Material:** bow and pin alloy steel, grade 80, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** EN1677-1 and ASME B30.26
- **Finish:** painted green
- **Temperature range:** -40 °C up to 200 °C
- **Certification:** 2.1 2.2 3.1 CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width inside	length	length bolt	width	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	lbs
3.25	²⁵ / ₃₂	³ / ₄	1 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₁₆	1 ¹ / ₂	1 ³ / ₈	3 ⁷ / ₁₆	3 ¹ / ₂	2 ¹¹ / ₁₆	1.46
4.75	¹⁵ / ₁₆	⁷ / ₈	1 ¹³ / ₁₆	³ / ₄	1 ⁷ / ₃₂	1 ⁷ / ₈	1 ¹³ / ₁₆	4 ³ / ₁₆	4 ¹ / ₁₆	3 ¹¹ / ₃₂	2.43
6.5	1 ¹ / ₁₆	1	2 ¹ / ₁₆	⁷ / ₈	1 ¹³ / ₃₂	2 ²⁷ / ₃₂	2 ⁷ / ₁₆	5 ¹³ / ₃₂	4 ¹¹ / ₁₆	4 ⁹ / ₃₂	3.95
8.5	1 ¹ / ₄	1 ¹ / ₈	2 ⁵ / ₁₆	³¹ / ₃₂	1 ¹¹ / ₁₆	3 ⁹ / ₃₂	3 ¹ / ₈	6 ⁷ / ₃₂	5 ¹³ / ₃₂	5 ¹ / ₄	6.04

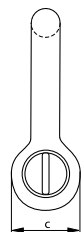
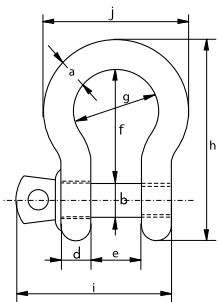
CAD

Green Pin® Theatre Shackle SC

Matte black bow shackle with screw collar pin



G-4161T



- **Material:** bow and pin high tensile steel, grade 60, quenched and tempered
- **Safety factor:** MBL equals 6 x WLL
- **Standard:** ISO 2415, EN 13889 and meets performance requirements of US Fed. Spec. RR-C-271 Type IVA Class 2, grade A, from 2 t and upward these shackles comply with ASME B30.26
- **Finish:** black
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC³ CE IIA
- **Article code:** scan QR code to see article codes

working load limit	diameter bow	diameter pin	diameter eye	width eye	width inside	length inside	width bow	length	length bolt	width	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	g inch	h inch	i inch	j inch	lbs
0.33	³ / ₁₆	¹ / ₄	¹⁵ / ₃₂	³ / ₁₆	³ / ₈	⁷ / ₈	⁵ / ₈	1 ¹³ / ₃₂	1 ⁵ / ₃₂	1 ¹ / ₃₂	0.04
0.5	¹ / ₄	⁵ / ₁₆	⁵ / ₈	⁹ / ₃₂	¹⁵ / ₃₂	1 ⁵ / ₃₂	²⁵ / ₃₂	1 ⁷ / ₈	1 ¹ / ₂	1 ¹¹ / ₃₂	0.11
0.75	⁵ / ₁₆	³ / ₈	²⁵ / ₃₂	¹¹ / ₃₂	¹⁷ / ₃₂	1 ¹ / ₄	⁷ / ₈	2 ⁷ / ₃₂	1 ²⁷ / ₃₂	1 ⁹ / ₁₆	0.22
1	³ / ₈	⁷ / ₁₆	²⁹ / ₃₂	¹³ / ₃₂	²¹ / ₃₂	1 ¹⁵ / ₃₂	1 ¹ / ₃₂	2 ¹⁷ / ₃₂	2 ¹ / ₈	1 ¹³ / ₁₆	0.31
1.5	⁷ / ₁₆	¹ / ₂	³¹ / ₃₂	⁷ / ₁₆	³ / ₄	1 ¹¹ / ₁₆	1 ⁵ / ₃₂	2 ⁷ / ₈	2 ¹¹ / ₃₂	2	0.42
2	¹ / ₂	⁵ / ₈	1 ¹¹ / ₃₂	¹ / ₂	⁷ / ₈	2	1 ¹ / ₄	3 ¹⁷ / ₃₂	2 ⁷ / ₈	2 ⁵ / ₁₆	0.79
3.25	⁵ / ₈	³ / ₄	1 ⁹ / ₁₆	⁵ / ₈	1 ¹ / ₁₆	2 ¹⁷ / ₃₂	1 ¹¹ / ₁₆	4 ¹¹ / ₃₂	3 ¹ / ₂	2 ¹⁵ / ₁₆	1.39
4.75	³ / ₄	⁷ / ₈	1 ¹³ / ₁₆	³ / ₄	1 ⁷ / ₃₂	3	2	5 ³ / ₃₂	4 ¹ / ₁₆	3 ¹ / ₂	2.23
6.5	⁷ / ₈	1	2 ¹ / ₁₆	⁷ / ₈	1 ¹³ / ₃₂	3 ⁹ / ₃₂	2 ⁹ / ₃₂	5 ²¹ / ₃₂	4 ¹¹ / ₁₆	4 ¹ / ₃₂	3.31
8.5	1	1 ¹ / ₈	2 ⁵ / ₁₆	³¹ / ₃₂	1 ¹¹ / ₁₆	3 ³ / ₄	2 ¹¹ / ₁₆	6 ¹⁵ / ₃₂	5 ¹³ / ₃₂	4 ²¹ / ₃₂	4.87



CM[®] **CHAIN & RIGGING ATTACHMENTS**

SHACKLES

**CHAIN & SYNTHETIC
SLING COMPONENTS**

RIGGING HARDWARE

CLAMPS

**TRANSPORTATION
CHAIN & ATTACHMENTS**

SPECIALTY PRODUCTS



C O L U M B U S M c K I N N O N C O R P O R A T I O N





SHACKLES OVERVIEW

When it comes to shackles, Columbus McKinnon prides itself on providing the strongest and most reliable products on the market. We carry a full line of anchor and chain shackles, manufactured in the U.S.A. through our state-of-the-art forging process.

CM shackles are available in three materials, including carbon, super strong and alloy. Our innovative Super Strong Shackles are unique in the industry, featuring strength ratings up to 50 percent stronger than comparable sized carbon shackles and a 6:1 design factor for ultimate safety.

CM shackles are available in a three styles: **Screw Pin; Bolt, Nut & Cotter; and Round Pin.** Learn more about the uses and benefits of each shackle style below.

BOLT, NUT & COTTER SHACKLES

Of all shackle types, Bolt, Nut and Cotter Shackles provide the most secure pin arrangement, resisting axial and torsional loading. This type of shackle should be used in semi-permanent applications where the pin is removed infrequently or where cyclical loading occurs. This is the preferred type of shackle in areas that are difficult to reach or inspect. Recommended for overhead lifting, bolt, nut and cotter shackles are available in capacities up to 50 tons.



SCREW PIN SHACKLES

Screw Pin Shackles allow for quick and easy removal of the screw pin, which makes this style ideal for applications where the shackle is removed frequently. While the threaded pin can resist axial forces, it should not be cyclically loaded and is unreliable and vulnerable to backing out in applications where the pin is subjected to a torque or twisting action. Recommended for overhead lifting, screw pin shackles are available in capacities up to 43 tons. Screw pins should be moused in some applications.



ROUND PIN SHACKLES

Round Pin Shackles allow for easy removal by simply removing the cotter that holds the pin in place. These shackles perform well where the pin is subjected to a torque or twisting action, but they should not be subjected to an axial load. Round pin shackles are available in capacities up to 43 tons and are **not recommended for overhead lifting**.



3 TYPES OF SHACKLE MATERIAL

MATERIAL	STYLE	WLL (TONS)	SIZES (IN.)	STYLES	DESIGN FACTOR	FINISHES
CARBON Carbon/Government rated chain shackles are available through Special Applications	Anchor	1/3 to 35 ton	3/16" to 2"	Bolt, Nut & Cotter; Screw Pin; Round Pin	6:1	Orange Powder Coated, Galvanized
	Chain	1/2 to 35 ton	1/4" to 2"			
SUPER STRONG 17 to 50% stronger than comparable-sized Carbon	Anchor	1/2 to 35 ton	3/16" to 2"	Bolt, Nut & Cotter; Screw Pin; Round Pin	6:1**	Orange Powder Coated, Self Colored, Galvanized
	Chain	3/4 to 35 ton	1/4" to 2"			
ALLOY (U.S.) ~50% stronger than comparable-sized Carbon and ~25% stronger than Super Strong	Anchor	2 to 50 ton	3/8" to 2"	Bolt, Nut & Cotter; Screw Pin; Round Pin	5:1	Orange Powder Coated, Self Colored, Galvanized

** Round pin shackles have a 5:1 design factor.



UNDERSTANDING & CALCULATING BEARING STRESS

Bearing stress is an important, but commonly overlooked, calculation to ensure safe sling use. This calculation will determine how much stress a sling is seeing in the working area of the shackle.

Per WSTDA, the recommended bearing stress value is 7,000 lbs./sq. in. or less. This calculation should be done any time a lift is being conducted with a synthetic strap and connecting hardware.

To calculate bearing stress, follow the calculations below:

DETERMINE EFFECTIVE WIDTH:

This is the width the sling can use inside of a shackle.

Effective Width = Shackle Width (catalog width from the manufacturer) x .75

DETERMINE LOAD BEARING AREA:

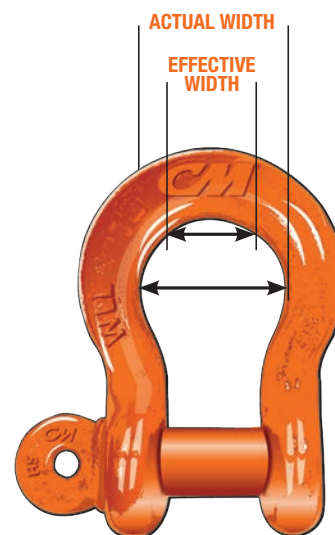
This is the area the sling is in contact with.

Load Bearing Area = Hardware Diameter x Effective Width

DETERMINE BEARING STRESS AT HARDWARE CONNECTION:

Should be 7,000 lbs./sq. in. or less to prevent damage to the sling.

Bearing Stress = Sling Load (IN POUNDS) ÷ Load Bearing Area



SLING INSPECTION,
USE & CARE

Source: WSTDA 4.7.1.1 – 4.7.1.3





SHACKLES

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Long Reach Shackles	57

Columbus McKinnon is proud to offer some of the strongest and most reliable shackles on the market. Manufactured in the U.S.A. through our state-of-the-art forging process, CM shackles are available in a variety of different styles and materials for virtually any rigging application.

DESIGN & DEVELOPMENT

SPECIFICATIONS

When manufacturing our shackles, Columbus McKinnon utilizes state-of-the-art forging equipment. The forging process is closely monitored to ensure consistent quality and the heat treatment process is computer-controlled and monitored to ensure that maximum performance parameters are met.

Each lot of product is checked to verify that the desired hardness range has been obtained.

All CM shackles are made from special bar quality material and comply with ASTM A322, ASTM A576 or ASTM A921. Galvanized shackles meet ASTM A153 or ASTM B695. Pins and bolts meet SAE J429 and ASTM A354.

CM shackles meet or exceed the performance requirements of the specs listed below:

- ASME B30.26 ■ ANSI B18
- EN 13889 ■ ISO 2415

CM also offers shackles that meet U.S. Government Specification RR-C-271.

Every CM shackle is marked with an alpha-numeric trace code. For full information on CM shackle identification markings, see the Shackle Identification box to the right.

ENGINEERING & TESTING REQUIREMENTS

Columbus McKinnon has the capability to apply fracture mechanics, predict product life expectancy and conduct a multi-axial fatigue analysis to solve engineering problems related to safety-critical applications.

CM products having strength requirements are sample tested to ensure hardness, ductility and requisite loading parameters. All testing and measuring equipment is calibrated on a periodic basis. CM testing equipment is calibrated to National Institute for Standards and Testing (NIST) requirements. Columbus McKinnon is also ISO 9001:2015 certified.

Certifications for all shackles are available online. RR-C-271 certification is available if requested at time of order.

Columbus McKinnon can also provide the following information if requested at time of order:

- Material certification
- Magnetic particle inspection
- Proof, ultimate, charpy, deformation and fatigue testing

CUSTOMIZATION

CM shackles are always designed to meet internal, customer, contractual and regulatory requirements. Columbus McKinnon has the capability to develop original product designs based on unique customer applications. The CM Engineering department has CAD stations to facilitate design and development activities. New product design and tooling is subject to computerized Finite Element Analysis (FEA) and all drawings are filed electronically.

Custom products, or specials, designed to meet customer specifications require customer approval before the design is finalized.

In addition to product design, Columbus McKinnon also performs tooling and machine design to manufacture and process these products.

► SHACKLE IDENTIFICATION

CM shackles and other rigging products can be identified by their unique markings.

We have taken extra efforts to enhance our shackle identification markings and our products now feature some of the largest and most user-friendly forged identification markings on the market. This innovation improves operator safety, reduces replacement costs and allows for easier identification of CM products in the field.

Every shackle is forged with the CM logo, its body or diameter size in imperial and/or metric units, trace code, USA, "Forged" and its specified strength requirements/working load limit (WLL). Most CM products also carry an alpha-numeric traceability code. Implemented in July 1980, this trace code system enables us to identify and track products once they ship from our plant, as well as determine:

- ▲ Date the product was forged
- ▲ Type and chemistry of steel
- ▲ Heat treating parameters
- ▲ In-process hardness testing results
- ▲ Strength data testing

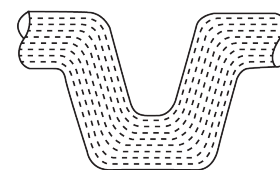
Design and markings meet or exceed ASME B30.26

FORGING VERSUS CASTING

Forging and casting are two very different manufacturing methods. When something is cast the material is heated above its melting temperature and poured into a mold where it solidifies. When something is forged it is physically forced into shape while remaining in a solid state – although it is frequently heated.

Forged shackles are generally better than cast. Forgings normally have less porosity, finer grain structure, higher tensile strength, better fatigue life and strength, and greater ductility than cast shackles. Why is this the case? When you melt metal to cast it, the grain size is free to expand. When it cools back to a solid, the grain structure is coarser and more random, decreasing its strength. Interior voids are also possible. The diagrams on the right illustrate the difference in grain flow between a forging and a casting.

For these reasons, CM utilizes a best-in-class forging process to ensure our shackles are strong, durable and reliable. All of our forged shackles are made right here in America at our Chattanooga, Tennessee facility.



FORGING
Uniform grain flow gives material higher strength



CASTING
Random grain flow with larger grain structure makes material weaker than forged products

STANDARD & SPECIALTY SHACKLES

CM offers a full line of forged chain and anchor shackles for standard and specialty applications that are made right here in America. Chain shackles are best used for straight-line pulls, while anchor shackles have a more generous loop that allows them to be side loaded or used for multiple connections.

STANDARD:

SCREW PIN SHACKLES

Screw Pin Shackles allow for quick and easy removal of the screw pin, which makes this style ideal for applications where the shackle is removed frequently. While the threaded pin can resist axial forces, it should not be cyclically loaded and is unreliable and vulnerable to backing out in applications where the pin is subjected to a torque or twisting action. In some applications, it is recommended to mouse the screw pin to prevent the pin from unscrewing. **Recommended for overhead lifting**, screw pin shackles are available in Super Strong, carbon and alloy steel with capacities up to 43 tons. Shackles meet ASME B30.26. They also meet the performance and dimensional requirements of RR-C-271.



SHACKLES

BOLT, NUT & COTTER SHACKLES

Of all shackle types, Bolt, Nut and Cotter Shackles provide the most secure pin arrangement, resisting axial and torsional loading. This type of shackle should be used in semi-permanent applications where the pin is removed infrequently.

Recommended for overhead lifting, bolt, nut and cotter shackles are available in Super Strong, carbon and alloy steel with capacities up to 50 tons. Shackles meet ASME B30.26. They also meet the performance and dimensional requirements of RR-C-271.



ROUND PIN SHACKLES

Round Pin Shackles allow for easy removal by simply removing the cotter that holds the pin in place. These shackles perform well where the pin is subjected to a torque or twisting action, but are not best for use where the pin is subject to an axial load. Round pin shackles are **not recommended for overhead lifting or side loaded**. They feature a forged, heat-treated steel body with forged, heat-treated alloy steel pin and are available in Super Strong, carbon and alloy steel with capacities up to 43 tons. Shackles meet the performance and dimensional requirements of RR-C-271.



SPECIALTY:

WEB SLING SHACKLES

Designed primarily for use with a synthetic slings up to 6" in width. Available in capacities up to 12 tons. Body is made of heat-treated carbon steel or alloy steel.

NOTE: Shackles cannot be point loaded. The load should be evenly distributed over the entire pin to achieve full working load limit.

LONG REACH SHACKLES

Made of alloy steel, CM is one of the only manufacturers of long reach shackles. These shackles are ideal for use in construction applications where a longer reach is needed to attach to pick points, and can also be used as a bail for lifting thicker products.

NOTE: Shackles cannot be side loaded. The load should be evenly distributed over the entire pin to achieve full working load limit.



SHACKLE USE, CARE & INSPECTION

Improper use or care of shackles can result in bodily injury or property damage. Always observe the following guidelines when using shackles.

- Do not exceed the working load limit.
- Do not shock load.
- If the shackle is side loaded, the WLL must be reduced in accordance with the manufacturer's recommendation or a qualified person. Shackles are designed and rated for in-line applied tension. You can attach multiple slings in the body of a shackle without reducing the capacity provided that the shackle is symmetrically loaded and the included angle does not exceed 120 degrees. (See Side Loading and Symmetrical Loading sections).
- Do not replace pin or bolt with other than original equipment.
- Inspect before use for wear, deformation and pin engagement as outlined in ASME B30.26. (See full inspection guidelines below).

Care should be exercised so that the shackle is not abused during use. When using shackles, it is important to:

- If necessary, use spacers on the shackle pin to assure that the shackle is not loaded at an angle. Load line of action should be through the center line of the shackle body and the middle of the shackle pin.
- The shackle should be protected with zinc plating or a galvanized finish if used in harsh environments.
- The shackle should not be subjected to high or low temperatures that could affect thermal treatment and the strength of the shackle. (Note: Per ASME B30.26 shackles are rated for temperatures between -40°F to 400°F.

SCREW PIN TIGHTENING

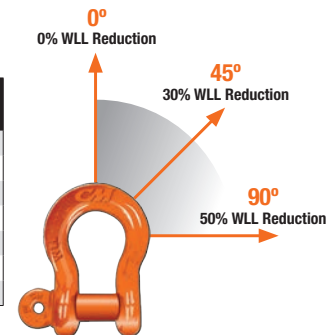
When tightening screw pins, it is important that shackle screw pin threads and the tapped threads in the shackle head are clean and free of burrs and damage. These conditions can cause an under-tightening of the shackle screw pin. The shackle screw pin should be tightly fitted into the shackle's leg opening until the threads engage and the shoulder of the screw pin makes contact with shackle body.

SIDE LOADING

When side loading a shackle with a single sling, the rated WLL will be reduced in accordance with the manufacturer's recommendation or a qualified person. ASME B30.26 also recommends reducing the capacity of a shackle when it is side loaded. (See figure below.) Note that only anchor shackles 3/16" to 2" may be side loaded. Chain or long-reach shackles should not be side loaded.

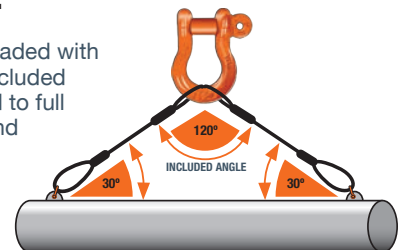
ANCHOR SHACKLES SIZES 3/16" TO 2"

Angles in Degrees	Working Load Limit Reduction
0° to 10°	0%
11° to 20°	15%
21° to 30°	25%
31° to 45°	30%
46° to 55°	40%
56° to 70°	45%
71° to 90°	50%



SYMMETRICAL LOADING

Shackles symmetrically loaded with two legs at a maximum included angle of 120° can be used to full working load limit. Side and symmetrical loading data applies to screw pin and bolt nut cotter anchor shackles as shown to the right.



SHACKLE INSPECTION

Shackles should be visually inspected before each use in line with ASME B30.26 regulations. Shackles should be discarded if any of the following conditions are apparent:

- Any parts are worn more than 10% of the original dimensions
- Load bearing components are bent, twisted, distorted, stretched, elongated, cracked or broken
- Excessive pitting, corrosion, nicks or gouges
- Indication of heat damage
- Missing or illegible manufacturer's name or trademark, working load limit or size
- Load pins have bent or visibly damaged threads
- Cotter pins or hairpin retainers are damaged

CHARPY IMPACT TEST

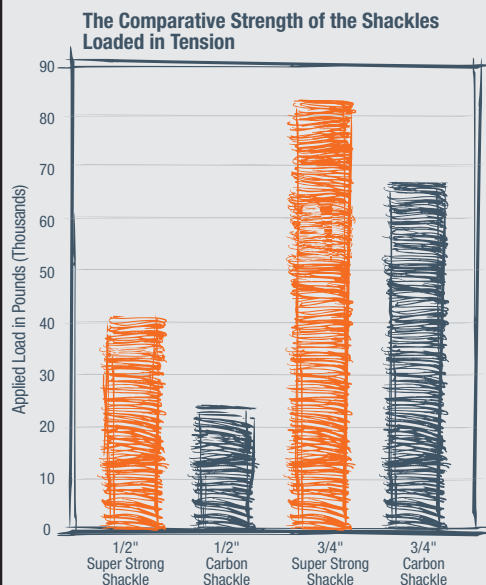
The Charpy V-Notch Test was developed during World War 2 to test the penetration resistance of steel armor. It has since evolved into a method to test for toughness of steel in critical structures such as buildings or bridges.

In this test, a bar is mounted horizontally with the notch facing away from an impact weight suspended on a pendulum. When the weight is released, it swings down and breaks through the bar. An indicator measures how far the pendulum continues to swing after breaking the bar. The momentum of the pendulum is then the measure of the resistance of the material to breaking or penetration.

CM Super Strong shackles, with the lower hardness values, will consistently pull more than a competitor's carbon shackles of the same diameter. CM Super Strong shackles were designed to improve overall load strength and ductility without an increase in shackle diameter.

CM alloy shackles will meet the Charpy Impact Test requirements. Results of this testing show that CM Super Strong shackles greatly exceed the minimum strength requirements.

RESULTS OF COMPARISON TESTING CM SUPER STRONG SHACKLES VERSUS STANDARD CARBON SHACKLES



"Clearly the CM Big Orange® shackles exhibited superior strength and more ductility than the carbon steel shackles of the same nominal section size. While all of the shackles performed above their ratings, the CM Big Orange shackle performance was superior.

The CM Big Orange® shock test results indicated severe deformation occurred but no fracture was present. The carbon steel parts fractured in two tests and were severely cracked in a third test. These results indicate that the CM Big Orange shackle assembly is stronger and more ductile than the carbon steel shackle of the same size. For these reasons, the CM Big Orange shackle provides more extensive deformation prior to fracture. In conclusion, this test demonstrates the superiority of the CM Big Orange shackles when compared to the carbon steel shackles under the shock loaded conditions."

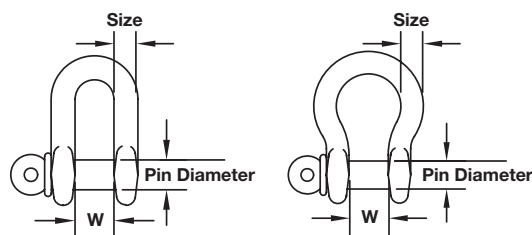
Verified by John Bloodsworth, P.E.
Q.C. Metallurgical Laboratory, Inc.

* CM Big Orange® shackles are now referred to as CM Super Strong shackles.

SHACKLE SPECIFICATIONS

DIMENSIONS & WORKING LOAD LIMITS

CM shackles are available in different dimensions with varying working load limits depending on the material they are made of. See the charts below for sizes and working load limits of our alloy, carbon and super strong shackles.



CARBON

Size (in.)	WLL (tons)	WLL (lbs.)	Pin Dia. (in.)	W dim. (in.)
3/16	1/3	667	0.25	0.38
1/4	1/2	1,000	0.31	0.47
5/16	3/4	1,500	0.38	0.53
3/8	1	2,000	0.44	0.66
7/16	1-1/2	3,000	0.50	0.72
1/2	2	4,000	0.63	0.84
5/8	3-1/4	6,500	0.75	1.06
3/4	4-3/4	9,500	0.88	1.28
7/8	6-1/2	13,000	1.00	1.44
1	8-1/2	17,000	1.13	1.72
1-1/8	9-1/2	19,000	1.25	1.84
1-1/4	12	24,000	1.38	2.03
1-3/8	13-1/2	27,000	1.50	2.25
1-1/2	17	34,000	1.63	2.41
1-5/8	20	40,000	1.75	2.66
1-3/4	25	50,000	2.00	2.94
2	35	70,000	2.25	3.28

SUPER STRONG

Size (in.)	WLL (tons)	WLL (lbs.)	Pin Dia. (in.)	W dim. (in.)
3/16	1/2	1,000	0.25	0.38
1/4	3/4	1,500	0.31	0.47
5/16	1	2,000	0.38	0.53
3/8	1-1/2	3,000	0.44	0.66
7/16	2	4,000	0.50	0.72
1/2	3	6,000	0.63	0.84
5/8	4-1/2	9,000	0.75	1.06
3/4	6-1/2	13,000	0.88	1.28
7/8	8-1/2	17,000	1.00	1.44
1	10	20,000	1.13	1.72
1-1/8	12	24,000	1.25	1.84
1-1/4	14	28,000	1.38	2.03
1-3/8	17	34,000	1.50	2.25
1-1/2	20	40,000	1.63	2.41
1-5/8	24	48,000	1.75	2.66
1-3/4	30	60,000	2.00	2.94
2	35	70,000	2.25	3.28

ALLOY

Size (in.)	WLL (tons)	WLL (lbs.)	Pin Dia. (in.)	W dim. (in.)
3/8	2	4,000	0.44	0.66
7/16	2.6	5,200	0.50	0.72
1/2	3.3	6,600	0.63	0.84
5/8	5	10,000	0.75	1.06
3/4	7	14,000	0.88	1.28
7/8	9.5	19,000	1.00	1.44
1	12.5	25,000	1.13	1.72
1-1/8	15	30,000	1.25	1.84
1-1/4	18	36,000	1.38	2.03
1-3/8	21	42,000	1.50	2.25
1-1/2*	25	50,000	1.63	2.41
1-1/2**	30	60,000	1.63	2.41
1-5/8*	29	58,000	1.75	2.66
1-5/8**	35	70,000	1.75	2.66
1-3/4*	34	68,000	2.00	2.94
1-3/4**	40	80,000	2.00	2.94
2*	43	86,000	2.25	3.28
2**	50	100,000	2.25	3.28

* Screw Pin & Round Pin style only

** Bolt, Nut & Cotter style only

SUPER STRONG ANCHOR SHACKLES

WORKING LOAD LIMIT: 1/2 TO 35 TONS



CM Super Strong Shackles are carbon-type shackles with strength ratings that are 17 to 50% stronger than comparable-sized carbon shackles. As a result, these shackles are designed with a 6:1 design factor. Anchor shackles can be side loaded or used for multiple connections.

BENEFITS & FEATURES

- Manufactured from technically advanced domestic (U.S.A.) microalloy steel with optimal hardness for strength and ductility
- Shackles show major deformation before failure
- Working load limit and traceability codes shown as permanent markings on body
- All shackles have alloy quenched and tempered pins
- Available in sizes 3/16" to 2"
- Available finishes include powder coated, galvanized or self-colored
- Shackles meet dimensional requirements and exceed performance requirements of RR-C-271
- Special testing and certification is available if requested at the time of the order
- Note: Screw pin and bolt/nut/cotter shackles have a 6:1 design factor. All round pin shackles have a 5:1 design factor.
- Screw pin & bolt/nut/cotter shackles meet ASME B30.26



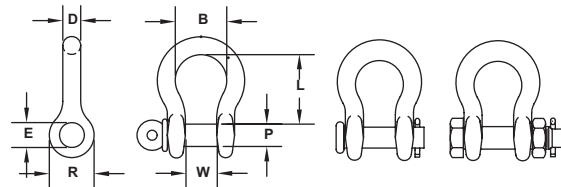
Screw Pin



Round Pin



Bolt, Nut & Cotter



STYLES: Screw Pin, Round Pin, Bolt/Nut/Cotter
FINISHES: Self Colored, Galvanized, Orange Powder Coated

Size D (in.)	Working Load Limit (Ton)	Std. Pkg.	Weight (lbs.)	Product Code									Dimensions (in.)					
				Screw Pin			Round Pin			Bolt, Nut & Cotter			P	E	W	R	L	B (min.)
				Self Colored	Galvanized	Orange Powder Coated	Self Colored	Galvanized	Orange Powder Coated	Self Colored	Galvanized	Orange Powder Coated						
3/16	1/2	50	0.06	—	M645G	—	—	M345G	—	—	—	—	0.25	0.29	0.38	0.57	0.88	0.58
1/4	3/4	50	0.12	M646	M646G	M646P	M346	M346G	M346P	M846	M846G	M846P	0.31	0.36	0.47	0.75	1.13	0.75
5/16	1	50	0.20	M647	M647G	M647P	M347	M347G	M347P	M847	M847G	M847P	0.38	0.45	0.53	0.84	1.25	0.81
3/8	1-1/2	50	0.30	M648	M648G	M648P	M348	M348G	M348P	M848	M848G	M848P	0.44	0.52	0.66	1.00	1.40	1.00
7/16	2	50	0.50	M649	M649G	M649P	M349	M349G	M349P	M849	M849G	M849P	0.50	0.58	0.72	1.15	1.69	1.19
1/2	3	50	0.75	M650	M650G	M650P	M350	M350G	M350P	M850	M850G	M850P	0.63	0.70	0.84	1.34	1.94	1.38
5/8	4-1/2	25	1.30	M651	M651G	M651P	M351	M351G	M351P	M851	M851G	M851P	0.75	0.83	1.06	1.66	2.41	1.63
3/4	6-1/2	10	2.30	M652	M652G	M652P	M352	M352G	M352P	M852	M852G	M852P	0.88	0.95	1.28	1.94	2.84	1.89
7/8	8-1/2	10	3.50	M653	M653G	M653P	M353	M353G	M353P	M853	M853G	M853P	1.00	1.09	1.44	2.14	3.31	2.06
1	10	5	5.00	M654	M654G	M654P	M354	M354G	M354P	M854	M854G	M854P	1.13	1.22	1.72	2.44	3.75	2.52
1-1/8	12	Bulk	7.00	M655	M655G	M655P	M355	M355G	M355P	M855	M855G	M855P	1.25	1.36	1.84	2.66	4.02	2.69
1-1/4	14	Bulk	9.50	M656	M656G	M656P	M356	M356G	M356P	M856	M856G	M856P	1.38	1.52	2.03	3.15	4.63	2.88
1-3/8	17	Bulk	12.50	M666	M666G	M666P	M366	M366G	M366P	M866	M866G	M866P	1.50	1.65	2.25	3.25	5.19	3.25
1-1/2	20	Bulk	17.20	M657	M657G	M657P	M357	M357G	M357P	M857	M857G	M857P	1.63	1.77	2.41	3.50	5.63	3.50
1-5/8	24	Bulk	23.50	M685	M685G	M685P	M385	M385G	M385P	M885	M885G	M885P	1.75	1.88	2.66	3.91	6.13	4.13
1-3/4	30	Bulk	27.70	M677	M677G	M677P	M377	M377G	M377P	M877	M877G	M877P	2.00	2.13	2.94	4.06	6.97	4.75
2	35	Bulk	39.00	M658	M658G	M658P	M358	M358G	M358P	M858	M858G	M858P	2.25	2.38	3.28	4.51	7.44	5.50

SUPER STRONG CHAIN SHACKLES

WORKING LOAD LIMIT: 3/4 TO 35 TONS



CM Super Strong Shackles are carbon-type shackles with strength ratings that are 17 to 50% stronger than comparable-sized carbon shackles. As a result, these shackles are designed with a 6:1 design factor. Chain shackles are best suited for straight-line pulls.

BENEFITS & FEATURES

- Manufactured from technically advanced domestic (U.S.A.) microalloy steel with optimal hardness for strength and ductility
- Shackles show major deformation before failure
- Working load limit and traceability codes shown as permanent markings on body
- All shackles have alloy quenched and tempered pins
- Available in sizes 1/4" to 2"
- Available finishes: galvanized
- Carbon industrial/government rated chain shackles available through special quote
- Shackles meet dimensional requirements and exceed performance requirements of RR-C-271
- Special testing and certification is available if requested at the time of the order
- Note: Screw pin and bolt/nut/cotter shackles have a 6:1 design factor. Round pin shackles have a 5:1 design factor.
- Screw pin & bolt/nut/cotter shackles meet ASME B30.26



Round Pin

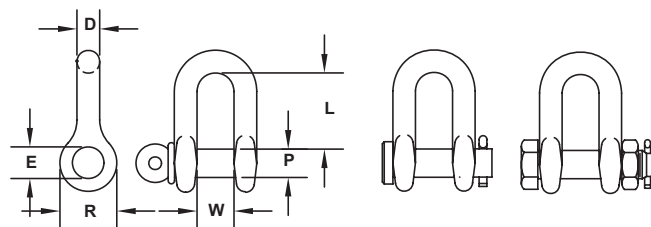


Screw Pin



Bolt, Nut & Cotter

STYLES: Screw Pin, Round Pin, Bolt/Nut/Cotter
FINISHES: Galvanized



Size D (in.)	Working Load Limit (Ton)	Std. Pkg.	Weight (lbs.)	Product Code			Dimensions (in.)				
				Screw Pin	Round Pin	Bolt, Nut & Cotter	P	E	W	R	L
				Galva- nized	Galva- nized	Galva- nized					
1/4	3/4	50	0.12	M746G	M546G	M946G	0.31	0.36	0.47	0.75	0.88
5/16	1	50	0.20	M747G	M547G	M947G	0.38	0.46	0.56	0.84	1.03
3/8	1-1/2	50	0.30	M748G	M548G	M948G	0.44	0.52	0.66	0.99	1.25
1/2	3	50	0.75	M750G	M550G	M950G	0.63	0.70	0.84	1.25	1.69
5/8	4-1/2	25	1.30	M751G	M551G	M951G	0.75	0.83	1.09	1.58	2.00
3/4	6-1/2	10	2.30	M752G	M552G	M952G	0.88	0.95	1.25	1.89	2.38
7/8	8-1/2	10	3.50	M753G	M553G	M953G	1.00	1.09	1.44	2.14	2.88
1	10	5	5.00	M754G	M554G	M954G	1.13	1.22	1.72	2.41	3.19
1-1/8	12	Bulk	7.00	M755G	M555G	M955G	1.25	1.34	1.81	2.69	3.56
1-1/4	14	Bulk	9.50	M756G	M556G	M956G	1.38	1.50	2.03	3.13	3.94
1-3/8	17	Bulk	12.50	M766G	M566G	M966G	1.50	1.63	2.25	3.32	4.44
1-1/2	20	Bulk	17.20	M757G	M557G	M957G	1.63	1.78	2.38	3.57	4.88
1-5/8	24	Bulk	23.50	M785G	M585G	M985G	1.75	1.88	2.63	3.94	5.25
1-3/4	30	Bulk	27.70	M777G	M577G	M977G	2.00	2.13	2.88	4.06	5.75
2	35	Bulk	39.00	M758G	M558G	M958G	2.25	2.38	3.28	4.53	6.75

CARBON ANCHOR SHACKLES (INDUSTRIAL/GOVERNMENT-RATED)



WORKING LOAD LIMIT: 1/3 TO 35 TONS

CM Industrial/Government-Rated Carbon Shackles are designed with a 6:1 design factor. Anchor shackles can be side loaded or used for multiple connections.

BENEFITS & FEATURES

- Manufactured from technically advanced domestic (U.S.A.) microalloy steel with optimal hardness for strength and ductility
- All shackles have alloy quenched and tempered pins
- Working load limit and traceability codes shown as permanent markings on body
- Available in sizes 3/16" to 2"
- Available finishes include powder coated, self-colored or galvanized per ASTM A153
- All bolt, nut & cotter shackles have thread-protected ends
- Shackles meet dimensional, marking and performance requirements of RR-C-271
- Standard industry tolerances apply
- Design factor 6:1
- Screw pin & bolt/nut/cotter shackles meet ASME B30.26



Screw Pin



Round Pin

Bolt, Nut & Cotter

► DID YOU KNOW?

CM INDUSTRIAL/GOVERNMENT VS. CM SUPER STRONG SHACKLES

An Industrial/Government shackle is a Super Strong shackle de-rated to meet, not exceed, the Federal Specification RR-C-271.

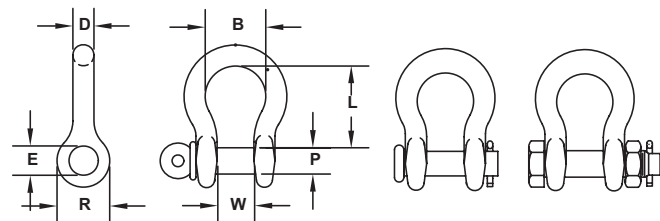
That means it has the same dimensions and performance characteristics as a Super Strong shackle but is marked with specifications to meet government requirements.

Example:

- 1" Super Strong shackle will be marked 10 Ton WLL

For more information, visit us at WWW.CMCO.COM

STYLES: Screw Pin, Round Pin, Bolt/Nut/Cotter
FINISHES: Galvanized, Orange Powder Coated



Size D (in.)	Working Load Limit (Ton)	Std. Pkg.	Weight (lbs.)	Product Code				Dimensions (in.)					
				Screw Pin		Round Pin	Bolt, Nut & Cotter						
				Galvanized	Orange Powder Coated	Galvanized	Galvanized	P	E	W	R	L	B (min.)
3/16	1/3	50	0.06	MC645G	—	MC345G	—	0.25	0.29	0.38	0.57	0.88	0.58
1/4	1/2	50	0.12	MC646G	MC646P	MC346G	MC846G	0.31	0.36	0.47	0.75	1.13	0.75
5/16	3/4	50	0.20	MC647G	MC647P	MC347G	MC847G	0.38	0.45	0.53	0.84	1.25	0.81
3/8	1	50	0.30	MC648G	MC648P	MC348G	MC848G	0.44	0.52	0.66	1.00	1.40	1.00
7/16	1-1/2	50	0.50	MC649G	MC649P	MC349G	MC849G	0.50	0.58	0.72	1.15	1.69	1.19
1/2	2	50	0.75	MC650G	MC650P	MC350G	MC850G	0.63	0.70	0.84	1.34	1.94	1.38
5/8	3-1/4	25	1.30	MC651G	MC651P	MC351G	MC851G	0.75	0.83	1.06	1.66	2.41	1.63
3/4	4-3/4	10	2.25	MC652G	MC652P	MC352G	MC852G	0.88	0.95	1.28	1.94	2.84	1.89
7/8	6-1/2	10	3.50	MC653G	MC653P	MC353G	MC853G	1.00	1.09	1.44	2.14	3.31	2.06
1	8-1/2	5	5.00	MC654G	MC654P	MC354G	MC854G	1.13	1.22	1.72	2.44	3.75	2.52
1-1/8	9-1/2	Bulk	7.00	MC655G	—	MC355G	MC855G	1.25	1.36	1.84	2.66	4.02	2.69
1-1/4	12	Bulk	9.00	MC656G	—	MC356G	MC856G	1.38	1.52	2.03	3.15	4.63	2.88
1-3/8	13-1/2	Bulk	12.50	MC666G	—	MC366G	MC866G	1.50	1.65	2.25	3.25	5.19	3.25
1-1/2	17	Bulk	17.20	MC657G	—	MC357G	MC857G	1.63	1.77	2.41	3.50	5.63	3.50
1-5/8	20	Bulk	23.50	MC685G	—	MC385G	MC885G	1.75	1.88	2.66	3.91	6.97	4.75
1-3/4	25	Bulk	227.70	MC677G	—	MC377G	MC877G	2.00	2.13	2.94	4.51	7.44	5.50
2	35	Bulk	39.00	M658G	M658P	M358GG	—	2.25	2.38	3.28	4.51	7.44	5.50

ALLOY ANCHOR SHACKLES

WORKING LOAD LIMIT: 2 TO 50 TONS



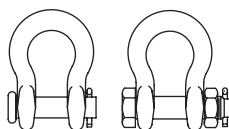
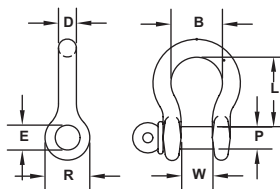
CM Alloy Shackles are designed with a 5:1 design factor and have a strength rating approximately 50% higher than a comparable-sized carbon shackle and about 25% stronger than super strong shackles. Anchor shackles can be side loaded or used for multiple connections.

BENEFITS & FEATURES

- 3/8" to 2" sizes made in the U.S.A. from domestic (U.S.A.) Special Bar Quality (SBQ) steel having fine grain, reduced sulfur and phosphorus.
- Shackles show major deformation before failure
- Working load limit and traceability codes shown as permanent markings on body
- All shackles have alloy quenched and tempered pins and bodies
- Available in sizes 3/8" to 2"
- Available finishes include powder htxxhb b coated, galvanized or self-colored
- Shackles meet dimensional, marking and performance requirements of RR-C-271
- Special testing and certification is available if requested at the time of the order
- Design factor 5:1
- Screw pin & bolt/nut/cotter shackles meet ASME B30.26



Screw Pin



Round Pin



Bolt, Nut & Cotter

STYLES: Screw Pin, Round Pin, Bolt/Nut/Cotter
FINISHES: Self Colored, Galvanized, Orange Powder Coated

Size D (in.)	Working Load Limit (Ton)	Std. Pkg.	Weight (lbs.)	Product Code									Dimensions (in.)					
				Screw Pin			Round Pin			Bolt, Nut & Cotter			P	E	W	R	L	B min
				Self Colored	Galva- nized	Orange Powder Coated	Self Colored	Galva- nized	Orange Powder Coated	Self Colored	Galva- nized	Orange Powder Coated						
3/8	2	50	0.30	M648A	M648AG	M648AP	—	—	—	M848A	M848AG	M848AP	0.44	0.52	0.66	1.00	1.40	1.00
7/16	2.6	50	0.50	M649A	M649AG	M649AP	—	—	—	M849A	M849AG	M849AP	0.50	0.58	0.72	1.15	1.69	1.19
1/2	3.3	50	0.75	M650A	M650AG	M650AP	M350A	M350AG	M350AP	M850A	M850AG	M850AP	0.63	0.70	0.84	1.34	1.94	1.38
5/8	5	25	1.30	M651A	M651AG	M651AP	M351A	M351AG	M351AP	M851A	M851AG	M851AP	0.75	0.83	1.06	1.66	2.41	1.63
3/4	7	10	2.30	M652A	M652AG	M652AP	M352A	M352AG	M352AP	M852A	M852AG	M852AP	0.88	0.95	1.28	1.94	2.84	1.89
7/8	9.5	10	3.50	M653A	M653AG	M653AP	M353A	M353AG	M353AP	M853A	M853AG	M853AP	1.00	1.09	1.44	2.14	3.31	2.06
1	12.5	5	5.00	M654A	M654AG	M654AP	M354A	M354AG	M354AP	M854A	M854AG	M854AP	1.13	1.22	1.72	2.44	3.75	2.52
1-1/8	15	Bulk	7.00	M655A	M655AG	M655AP	M355A	M355AG	M355AP	M855A	M855AG	M855AP	1.25	1.36	1.84	2.66	4.02	2.69
1-1/4	18	Bulk	9.50	M656A	M656AG	M656AP	M356A	M356AG	M356AP	M856A	M856AG	M856AP	1.38	1.52	2.03	3.15	4.63	2.88
1-3/8	21	Bulk	12.50	M666A	M666AG	M666AP	M366A	M366AG	M366AP	M866A	M866AG	M866AP	1.50	1.65	2.25	3.25	5.19	3.25
1-1/2	25	Bulk	17.20	M657A	M657AG	M657AP	M357A	M357AG	M357AP	—	—	—	1.63	1.77	2.41	3.50	5.63	3.50
1-1/2	30	Bulk	17.20	—	—	—	—	—	—	M857A	M857AG	M857AP	1.63	1.77	2.41	3.50	5.63	3.50
1-5/8	29	Bulk	23.50	M685A	M685AG	M685AP	M385A	M385AG	M385AP	—	—	—	1.75	1.88	2.66	3.91	6.13	4.13
1-5/8	35	Bulk	23.50	—	—	—	—	—	—	M885A	M885AG	M885AP	1.75	1.88	2.66	3.91	6.13	4.13
1-3/4	34	Bulk	27.70	M677A	M677AG	M677AP	M377A	M377AG	M377AP	—	—	—	2.00	2.13	2.94	4.06	6.97	4.75
1-3/4	40	Bulk	27.70	—	—	—	—	—	—	M877A	M877AG	M877AP	2.00	2.13	2.94	4.06	6.97	4.75
2	43	Bulk	39.00	M658A	M658AG	M658AP	M358A	M358AG	M358AP	—	—	—	2.25	2.38	3.28	4.51	7.44	5.50
2	50	Bulk	39.00	—	—	—	—	—	—	M858A	M858AG	M858AP	2.25	2.38	3.28	4.51	7.44	5.50

CARBON WEB SLING SHACKLE

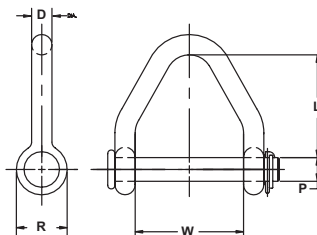
WORKING LOAD LIMIT: 8,000 TO 23,500 LBS.



Web sling shackles are designed to connect synthetic slings to other lifting hardware.

BENEFITS & FEATURES

- Design factor 4:1
- Web Sling Shackles can be used on web slings from 2 to 6 inches in width
- Shackle body: carbon steel, heat treated
- Shackle pin: alloy steel, heat treated
- Finish: hot dip galvanized
- Zinc-plated linch pin comes standard. Cotter or hairpin available on special order.
- Do not point load. The load should be evenly distributed over the entire pin to achieve full working load limit.



Product Code	Pin Number	Linch Pin Number	Working Load Limit (lbs.)	Dimensions (in.)					Weight (lbs.)
				P	D	L	W	R	
M702	2X702	35480	8,000	0.75	0.63	2.25	2.00	1.63	1.70
M703	2X703	35480	13,000	0.88	0.75	3.25	3.00	1.88	2.86
M704	2X704	35480	11,000	0.88	0.75	3.75	4.00	1.88	3.15
M705	2X705	5511	18,000	1.00	0.88	4.25	5.00	2.13	4.75
M706	2X706	5511	18,000	1.13	1.00	4.75	6.00	2.38	6.75
M706H	2X706H	5511	23,500	1.25	1.13	4.75	6.00	2.63	9.80

ALLOY WEB SLING SHACKLE

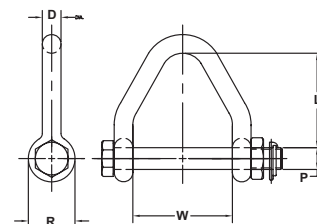
WORKING LOAD LIMIT: 13,500 TO 22,500 LBS.



Web sling shackles are designed to connect synthetic slings to lifting hardware.

BENEFITS & FEATURES

- Design factor 6:1
- Web Sling Shackles can be used on web slings from 3 to 6 inches in width
- Utilize a bolt and nut with linchpin to secure the assembly in place
- All shackles are galvanized for longer life
- Marked with working load limit (WLL) and size
- Do not point load. The load should be evenly distributed over the entire pin to achieve full working load limit



Product Code	Pin Number	Linch Pin Number	Working Load Limit (lbs.)	Dimensions (in.)					Weight (lbs.)
				P	D	L	W	R	
M703A	2X8703A	35480	13,500	0.88	0.75	3.25	3.00	1.88	3.01
M704A	2X8704A	35480	14,500	0.88	0.75	3.75	4.00	1.88	3.16
M705A	2X8705A	5511	19,000	1.00	1.00	4.25	5.00	2.38	6.04
M706A	2X8706A	5511	22,500	1.13	1.13	4.75	6.00	2.63	9.02

LONG-REACH SHACKLE

WORKING LOAD LIMIT: 7,000 TO 50,000 LBS.

As one of the only manufacturers of long-reach shackles, we designed these shackles for use in construction applications where a longer reach is needed to attach to pick points.

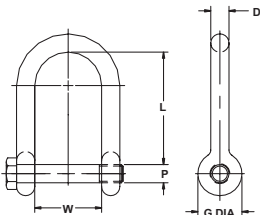
BENEFITS & FEATURES

- Design factor of 5:1
- Meets the requirements of ASME B30.26, but must not be side loaded
- Alloy steel
- WLL forged on body
- Offered in self-colored or durable orange powder coated finish
- Do not point load. The load should be evenly distributed over the entire pin to achieve full working load limit.
- Do not side load long-reach shackles
- 80% of bolt/pin must be covered to obtain full working load limit

STYLES: Screw Pin, Bolt/Nut/Cotter
FINISHES: Self Colored, Orange Powder Coated



Screw Pin



Bolt, Nut & Cotter

SHACKLES

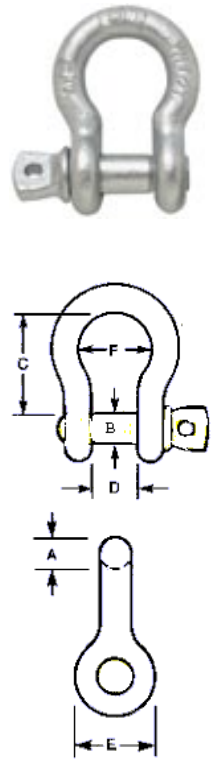
Size (in.)	Working Load Limit (lbs.)	Screw Pin			Bolt, Nut & Cotter			Dimensions (in.)				
		Product Code		Weight (lbs.)	Product Code		Weight (lbs.)					
		Self Colored	Painted		Self Colored	Painted		P	D	L	W	G
5/8	7,000	M7151	M7151P	1.80	M9151	M9151P	1.95	0.75	0.63	4.00	2.25	1.57
3/4	10,000	M7152	M7152P	2.72	M9152	M9152P	3.21	0.88	0.75	4.53	2.75	1.81
1	19,000	M7154	M7154P	5.86	M9154	M9154P	6.31	1.00	1.00	5.50	3.25	2.38
1-1/4	28,000	M7156	M7156P	11.90	M9156	M9156P	12.90	1.38	1.25	6.19	3.88	3.06
1-1/2	34,000	M7157	M7157P	19.60	M9157	M9157P	20.70	1.625	1.50	7.00	4.50	3.50
1-3/4	50,000	M7177	M7177P	30.70	M9177	M9177P	33.30	2.00	1.75	8.00	5.25	4.00



SCREW PIN ANCHOR SHACKLES - IMPORTED

 Never Exceed Working Load Limit

Size (A)	Working Load Limit in Tons	Pin Diam. in Inches (B)	Approx. Weight Ea. in Lbs.	Dimensions in Inches			
				C	D	E	F
3/16 in.	0.33	0.25	0.05	0.88	0.38	0.56	0.63
1/4 in.	0.50	0.31	0.13	1.06	0.47	0.66	0.75
5/16 in.	0.75	0.38	0.21	1.25	0.53	0.81	0.84
3/8 in.	1.00	0.44	0.33	1.44	0.63	0.97	1
7/16 in.	1.50	0.50	0.47	1.69	0.75	1.06	1.13
1/2 in.	2.00	0.63	0.76	1.88	0.81	1.31	1.25
5/8 in.	3.25	0.75	1.44	2.44	1.06	1.56	1.69
3/4 in.	4.75	0.88	2.3	2.88	1.19	1.88	1.94
7/8 in.	6.50	1.00	3.5	3.38	1.44	2.13	2.19
1 in.	8.50	1.13	5	3.81	1.69	2.38	2.63
1 1/8 in.	9.50	1.25	7	4.25	1.75	2.63	2.88
1 1/4 in.	12.00	1.38	9.5	4.69	2	3	3.13
1 3/8 in.	13.50	1.50	13	5.31	2.13	3.25	3.5
1 1/2 in.	17.00	1.63	16.5	5.69	2.31	3.5	3.75
1 3/4 in.	25.00	2.00	29	7.06	2.88	4.25	4.88
2 in.	35.00	2.25	43	7.75	3.31	4.88	5.44
2 1/4 in.	45.00	2.50	59.4	9.25	3.63	5.25	5.5
2 1/2 in.	55.00	2.75	86	10.75	4.13	5.88	7.25
3 in.	85.00	3.25	119.5	13.75	4.88	6.75	7.63

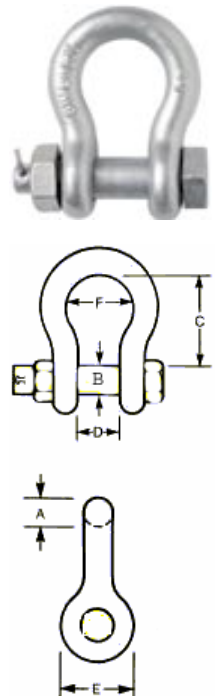


Do not use screw pin shackles if the pin can roll under load and unscrew.

BOLT TYPE SAFETY ANCHOR SHACKLES - IMPORTED

 Never Exceed Working Load Limit

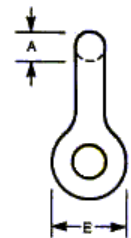
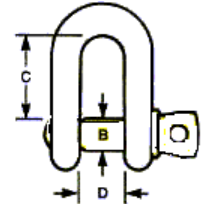
Size (A)	Working Load Limit in Tons	Bolt Diam. in Inches (B)	Approx. Weight Ea. in Lbs.	Dimensions in Inches			
				C	D	E	F
1/2 in.	2.00	0.63	0.9	1.88	0.81	1.31	1.25
5/8 in.	3.25	0.75	1.6	2.44	1.06	1.56	1.69
3/4 in.	4.75	0.88	2.8	2.88	1.19	1.88	1.94
7/8 in.	6.50	1.00	4.2	3.38	1.44	2.13	2.19
1 in.	8.50	1.13	6.4	3.81	1.69	2.38	2.63
1 1/8 in.	9.50	1.25	8.3	4.25	1.75	2.63	2.88
1 1/4 in.	12.00	1.38	11.7	4.69	2	3	3.13
1 3/8 in.	13.50	1.50	15.6	5.31	2.13	3.25	3.5
1 1/2 in.	17.00	1.63	19.7	5.69	2.31	3.5	3.75
1 3/4 in.	25.00	2.00	32.6	7.06	2.88	4.25	4.88
2 in.	35.00	2.25	48.3	7.75	3.31	4.88	5.44
2 1/4 in.	45.00	2.50	67	9.25	3.63	5.25	5.5
2 1/2 in.	55.00	2.75	94	10.75	4.13	5.88	7.25
3 in.	85.00	3.25	145	13.75	4.88	6.75	7.63



SCREW PIN CHAIN SHACKLES - IMPORTED

 Never Exceed Working Load Limit

Size (A)	Working Load Limit in Tons	Pin Diam. in Inches (B)	Approx. Weight Each in Pounds	Dimensions in Inches		
				C	D	E
1/4 in.	1/2	5/16	0.09	0.88	0.5	0.63
5/16 in.	3/4	3/8	0.17	1	0.53	0.78
3/8 in.	1	7/16	0.32	1.28	0.69	0.97
1/2 in.	2	5/8	0.8	1.72	0.84	1.28
5/8 in.	3 1/4	3/4	1.3	2.03	1.06	1.5
3/4 in.	4 3/4	7/8	2.2	2.41	1.25	1.81
7/8 in.	6 1/2	1	3.2	2.91	1.72	2.13
1 in.	8 1/2	1 1/8	5.1	3.25	1.75	2.38
1 1/8 in.	9 1/2	1 1/4	6.5	3.56	1.81	2.63
1 1/4 in.	12	1 3/8	9	3.97	2	2.91
1 3/8 in.	13 1/2	1 1/2	12.2	4.56	2.19	3.25
1 1/2 in.	17	1 5/8	17	5	2.34	3.5
1 3/4 in.	25	2	25.5	6	2.88	4.38
2 in.	35	2 1/4	37	7	3.31	4.91
2 1/2 in.	55	2 3/4	75	8	4.13	5.88
3 in.	85	3 1/4	108	8.25	5.25	6.75



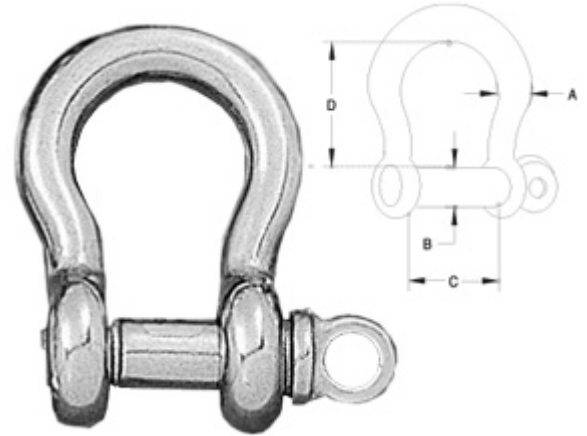
Do not use screw pin shackles if the pin can roll under load and unscrew.

SCREW PIN ANCHOR SHACKLES - TYPE 316 SS

Precision cast. Load rated. Meets Fed. Spec. RR-C-271D, Type IV, Class 2.

 Never Exceed Working Load Limit

Size (A)	B	C	D	W.L.L. (lbs.)	Wt. Ea. (lbs.)
3/16"	1/4"	3/8"	7/8"	650	0.04
1/4"	5/16"	15/32"	1-1/16"	1000	0.1
5/16"	3/8"	1/2"	1-1/4"	1500	0.18
3/8"	1/2"	5/8"	1-7/16"	2000	0.28
1/2"	5/8"	7/8"	1-7/8"	4000	0.68
5/8"	3/4"	1-1/16"	2-5/16"	6000	1.26
3/4"	7/8"	1-3/16"	2-7/8"	8000	2.1
7/8"	1"	1-1/2"	3-7/16"	10000	3.3
1"	1-1/8"	1-5/8"	3-3/4"	12000	4.6



Do not use screw pin shackles if the pin can roll under load and unscrew.

SAFETY ANCHOR SHACKLES - TYPE 316 SS

Precision cast. Load rated. Meets Fed. Spec. RR-C-271D, Type IV, Class 3.

 Never Exceed Working Load Limit

Size (A)	B	C	D	W.L.L. (lbs.)	Wt. Ea. (lbs.)
1/4"	5/16"	15/32"	1-1/16"	1000	0.12
5/16"	3/8"	1/2"	1-1/4"	1500	0.2
3/8"	1/2"	5/8"	1-7/16"	2000	0.3
1/2"	5/8"	7/8"	1-7/8"	4000	0.7
5/8"	3/4"	1-1/16"	2-5/16"	6000	1.3
3/4"	7/8"	1-3/16"	2-7/8"	8000	2.3
7/8"	1"	1-1/2"	3-7/16"	10000	3.4
1"	1-1/8"	1-5/8"	3-3/4"	12000	4.8

